Tengo

Independent expert review

Product, technology, privacy, legal, risk, data, design, accessibility, behaviour, language, commercial and judging perspectives

Subject: Tengo, a money app for people with irregular income, entered in the FNB App of the Year hackathon (Financial inclusion and money confidence).

State reviewed: commit d114497, reviewed 2026-10-20.

Nature of the review: a structured review by thirteen simulated professional perspectives, run as independent agents, with a synthesis. No reviewer is or claims to be FNB staff. Nothing here is legal or financial advice. All app data is fictional.

Files: docs/review/ (REPORT.pdf, findings.csv, ROADMAP.md, risk-register.md, judge-qa.md, model.xlsx, backtest/, reviewer files).

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1 Executive summary

What this review could not establish. Real user behaviour or comprehension (no user sessions took place); forecast accuracy on real bank data (the backtest uses synthetic data written to the same assumptions as the engine); legal sign-off (the legal chapter is a simulated perspective, not legal advice); FNB's internal criteria, systems, data or pricing (all bank figures are public); real costs at scale; translation quality (no translations exist yet). Every external figure carries a source and access date or is labelled Assumption.

Verdict. Tengo is a well-built, honest prototype of a distinct idea: give people with irregular income the date their money runs out, beside the date their next money usually lands. The build is fast, accessible and carefully explained. It is not yet ready for real users or a bank pilot: the Home screen contradicts itself, the engine misses the moments that matter most for irregular earners (income that stops, payday, large bills just after payday), some outputs read as permissions or could cause harm, and nothing has been tested with real people or real data. The 13 reviewers average **5.0 of 10** (range 4 to 6); the simulated judge panel scores **39 of 70**. They raised **151 findings** (48 High, 66 Medium, 37 Low).

Reviewer	Score
Product lead	5
Technology	6
Security and privacy	5
Legal	5
Credit risk	4
Data scientist	5
UI and UX	6
Accessibility	6
Behavioural	5
Localisation	4
Commercial	4
Investor	4
Judge panel	6

Judging criterion	Score
Product	6
Creativity	7
Usability	6
Sustainability	4
Impact	5
Presentation	5
Intelligence	6
Total	**39 of 70**

Top five strengths

1. A distinct, date-first answer for people without a payday, with next money inferred from irregular deposits (PROD, COMM, judge Creativity 7).
2. A transparent, deterministic engine with 97.6% unit coverage that beats a naive baseline on synthetic data (MAE 2.1 against 3.4 days).
3. An accessible, polished interface: Lighthouse accessibility 100, one axe WCAG 2.2 violation across 15 pages, both themes.
4. Honest labelling and a privacy-first demo: nothing leaves the device and simulated controls are marked as such.
5. Fast on a throttled phone: Home in 1.5 s cold, 142 KB of compressed JavaScript.

Top five risks

1. Home shows Short beside "R310 safe to spend" and a chart saying Enough (R1).
2. Income that has stopped still counts as coming, so the app can show Enough with high confidence (R3).
3. "What helps most" can suggest paying a car instalment late; "safe" reads as a promise (R4, R2).
4. No users, and validation is circular on synthetic data (R8).
5. No backend, auth or encryption: real data must not be touched yet (R12).

Top three fixes

1. **One rule drives Home** ("Tight until", daily budget not advised when Short, chart labelled "if money lands as usual"): about 4 hours, the largest single score lift (JUDGE-01).
2. **Remove harmful outputs and teach the engine when a pattern breaks** (no late credit or insurance, renamed safe figures, paused-income and no-income rules): about 2 days (JUDGE-03, JUDGE-04).
3. **Make every claim true** (languages "English today", privacy screens as demo statements) and run a six-person comprehension test: 1 hour plus 2 days (JUDGE-02, JUDGE-07).

2 Scope, method and limits

This is an independent, read-only review of Tengo as it stood at commit `d114497` on 2026-10-20, following `REVIEW.md`. No application code, tests or configuration were changed; everything written lives in `docs/review/`.

2.1 What was reviewed

- The codebase (6 747 lines of application TypeScript, 642 lines of unit tests), README, BRIEF.md and the README "Redesign" section. `REVIEW.md` refers to `docs/BRIEF.md` and `docs/REDESIGN.md`; those files do not exist, so the root `BRIEF.md` and the README sections were used.
- The production build served locally, driven with Playwright. 264 screenshots: every page, both themes, at 390, 768 and 1280 px, for all three personas (`docs/review/screens/`).
- Build, unit tests, lint, i18n, fonts and contrast checks; the full Playwright suite (221 tests); unit coverage; npm audit; bundle sizes; licence summary; Lighthouse mobile; a throttled timing of the in-app Home page.
- A persona walkthrough and a fact sheet of what the product does today, shared with every reviewer (Section 3).

2.2 Panel method

Thirteen reviewers were run as independent agents. Twelve worked in parallel without reading each other's files, so they could not anchor on one another. The judge panel ran last and was told to use the others' evidence. Each received the same shared instructions: cite a file and line, a screenshot, a command output or a source URL for every finding; label anything unsourced as Assumption with a range; do not claim to be FNB staff; name at least three weaknesses; follow the copy rules. Two reviewers (data scientist, commercial) were interrupted by a usage limit; the data scientist's file was already complete and the commercial reviewer was resumed with its context to finish.

The synthesis checked every reviewer file for required sections, findings tables and copy rules, merged all findings into `findings.csv`, and verified the most consequential claims independently (Section 8.2).

2.3 Check results

Check	Result
npm run build	pass
npm run test (contrast, then 66 unit tests)	pass, 116 of 116 contrast pairs
npm run lint (ESLint and dash check)	pass
npm run i18n:check	pass, 11 locales with identical keys
npm run fonts:check	pass
Playwright (screens, flows, phone, axe)	221 passed
Unit coverage	engine 97.6% of statements; all of src 20.0%
npm audit	12 advisories (2 critical, 5 high, 5 moderate); 2 moderate ship to users

2.4 Limits

Everything in the "could not establish" box in Section 1 applies. In addition: the backtest is only as good as the generator; prices and market figures move, so every one carries an access date; Lighthouse cannot reach in-app pages because it clears storage, so the in-app timing used a separate Playwright measurement with matching throttling; and LibreOffice was not available, so the financial model's 1 474 formulas were verified with the Python `formulas` engine (zero errors) and the workbook is set to recalculate fully when opened in Excel.

3 What the product does today

Prepared from the code at commit d114497 on 2026-10-20. Every line here is checked against the source, the check outputs in this folder, or the persona snapshot. It describes what exists, not what the pitch says.

Form

- A single page web app: React 18, TypeScript (strict), Vite, Tailwind, React Router 6, i18next, Zustand. No backend. Source: `package.json`, `src/App.tsx`.
- All data lives in the browser's localStorage under two keys: `tengo.session.v1` (language, theme, active persona, sidebar) and `tengo.data.v1` (records, consent, audit log, review requests, deletion receipts). Source: `src/store/session.ts:31`, `src/data/localRepository.ts:15`.
- Data access goes through a `TengoRepository` interface with one implementation, `LocalRepository`, which adds a 140 ms artificial delay. Source: `src/data/repository.ts`, `src/data/index.ts`.
- Installable as a PWA with a same-origin service worker for offline use. Source: `public/sw.js`, `src/main.tsx`.
- 6 747 lines of application TypeScript and 642 lines of unit tests. Source: `wc -l` over `src`.

Data

- No real data. Three fictional personas (Thabo, e-hailing driver, Tshwane; Lerato, hairdresser, Mamelodi; Ayanda, spaza owner, Tembisa) with 90 days of deterministic synthetic transactions ending yesterday. Each persona's balance is chosen by the engine so it starts in Tight, Enough and Short respectively. Source: `src/data/seed/generate.ts`, `src/data/seed/personas.ts`.
- Seed data regenerates every day unless the user imports or edits. Source: `src/data/localRepository.ts getRecord`.
- A user can replace a persona's history with a CSV bank statement, parsed in the browser (PapaParse). It detects columns, skips bad rows, requires 14 days, warns under 28, and shifts statements that ended more than 3 days ago so they end yesterday. PDF import is shown as "coming next" and does not exist. Source: `src/lib/csv.ts`, `src/pages/Import.tsx`.
- No bank connection, no open banking, no login, no accounts.

Forecast engine (`src/engine/`)

- Splits history into income, recurring bills (same description, amounts within 20 percent, 24 to 36 days apart, at least 2 times) and everyday spend. Source: `classify.ts:16`.
- Daily spend: median, 80th and 20th percentile of rolling 7 day spend over the last 60 days, divided by 7. Low confidence (under 28 days) widens the spread by 1.5. Source: `forecast.ts:20,26`.
- Next money: deposits of at least 10 percent of typical monthly income. Weekday pattern when 60 percent land on one weekday (latest date is one day after typical), otherwise gaps between deposits (20th, 50th, 80th percentile). No pattern means a 30 day plan. Source: `forecast.ts:24,28`.
- Runway: balance run down daily at each rate minus bills, ignoring future income. The date is the day the normal path reaches zero. Horizon 180 days. Source: `forecast.ts:18`.
- State: Enough if the heavy path reaches the latest next money; Tight if the normal path reaches the typical next money; otherwise Short.
- Safe to spend today: (balance minus bills before the latest next money) divided by days to it, floored at zero, rounded down to R5.
- Also: a 30 day timeline that does include expected income, a what-if, a loan check with "safe to lend", "What helps most" (extra money, daily cut or bill to move), and a keyword-based spending breakdown. Source: `forecast.ts`, `help.ts`, `spending.ts`.

Persona snapshot on 2026-10-20 (from `persona-snapshot.json`)

Persona	State	Date shown	Range	Safe to spend	Next money	Usual daily spend
Thabo	Tight	27 Oct	25 to 28 Oct	R295	26 Oct (weekday)	R295
Lerato	Enough	1 Nov	26 Oct to 1 Nov	R330	24 Oct (weekday)	R120
Ayanda	Short	23 Oct	22 to 25 Oct	R310	25 Oct (interval)	R623

Screens

Landing, persona picker, consent, Home (date hero, range band, state, stats, 30 day chart, What helps most, recent activity, Why this date drawer), Plan (what-if, loan check), Money (history, filters, search, corrections, recurring bills, spending breakdown), Import, Inbox (simulated WhatsApp or SMS preview in 11 languages), Privacy centre (consent toggles, data minimisation, see and export data, audit log, human review request, sample breach notice, retention, cross-border card, placeholder Information Officer, delete with receipt), Me (persona, language, theme, about, reset). Command menu on Ctrl or Cmd plus K. Source: `src/App.tsx`, `src/pages/`.

Languages

11 locale files with identical keys (577 keys). All 10 non-English files currently contain English text. A Google Cloud Translation script exists but has not been run (no API key). Source: `npm run i18n:check`, `src/i18n/status.json`, `scripts/translate.mjs`.

What is simulated, not real

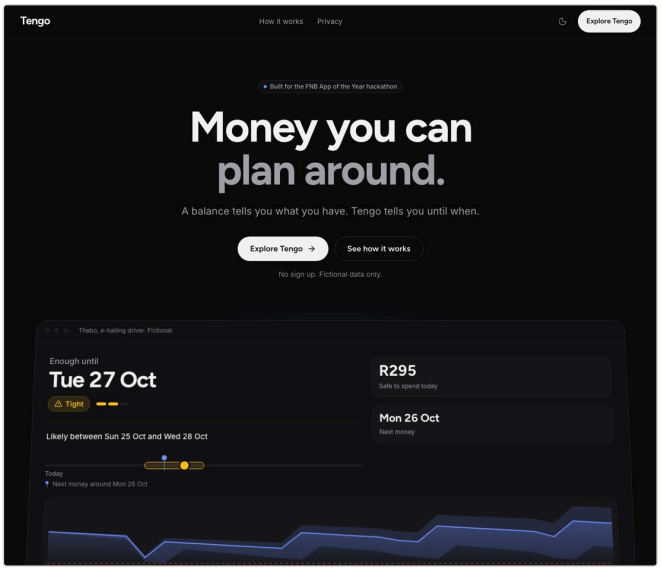
Bank data, messaging (nothing is sent), privacy controls (labelled "Simulated privacy controls. Not a compliance certification."), the Information Officer contact (placeholder), the human review (stored locally, nobody receives it), translations, the logo and icons (placeholders), encryption claims (described for a real deployment only).

Quality signals (see `checks.md`, `audit.md`, `playwright.txt`, `coverage.txt`, `lighthouse/`)

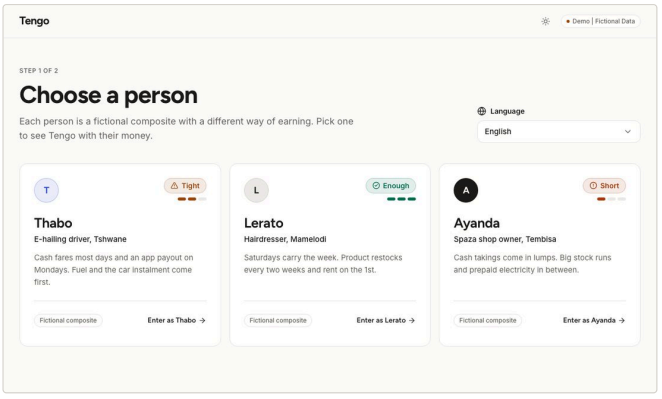
- Build, 66 unit tests, lint, i18n check, fonts check and 116 contrast pairs all pass.
- Playwright: screenshot, flow, phone and axe accessibility suites (see `playwright.txt` for the count and result).
- Unit coverage: 97.6 percent of statements in `src/engine`, 20.0 percent across all of `src` (UI is covered by Playwright only).
- npm audit: 12 issues in all dependencies (2 critical, 5 high, 5 moderate), 2 moderate in production dependencies (react-router).

- Bundle: one 696 KB JavaScript file (142 KB gzip) that includes all 11 locale files (337 KB of JSON).
- Lighthouse mobile (simulated slow 4G, 4x CPU): landing 94 performance, 100 accessibility, 100 best practices, 91 SEO; picker 96, 100, 100, 91.

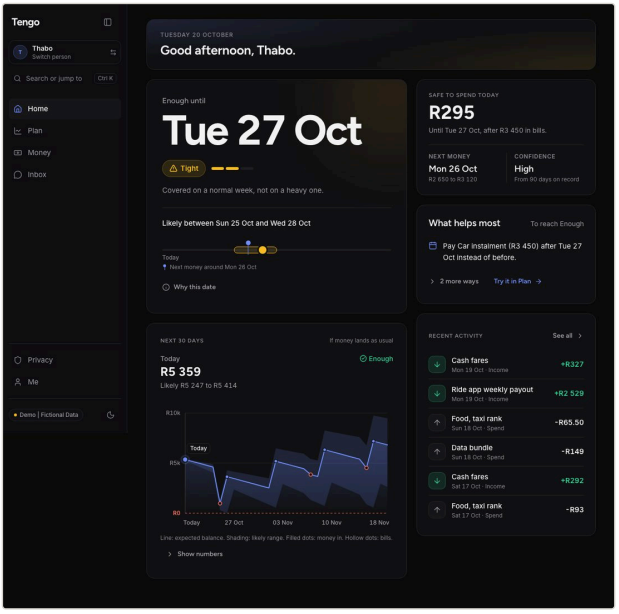
Screens



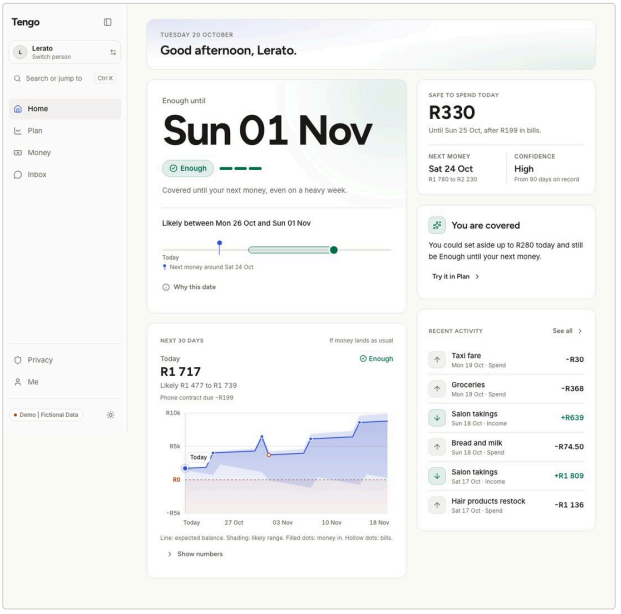
Landing, dark, 1280 px



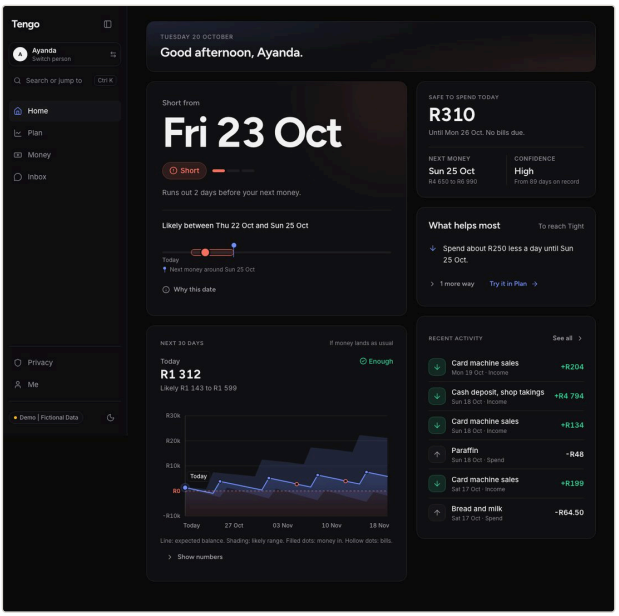
Persona picker, light, 1280 px



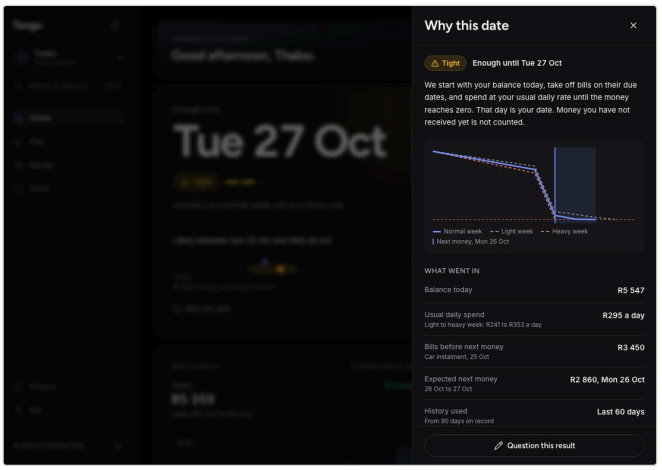
Home for Thabo (Tight), dark



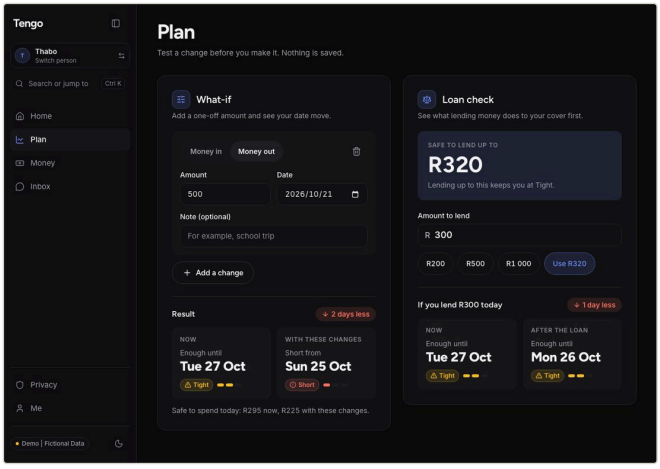
Home for Lerato (Enough), light



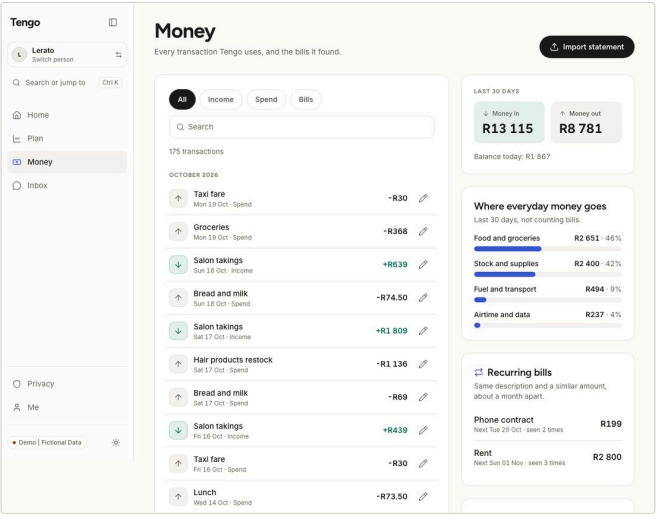
Home for Ayanda (Short), dark



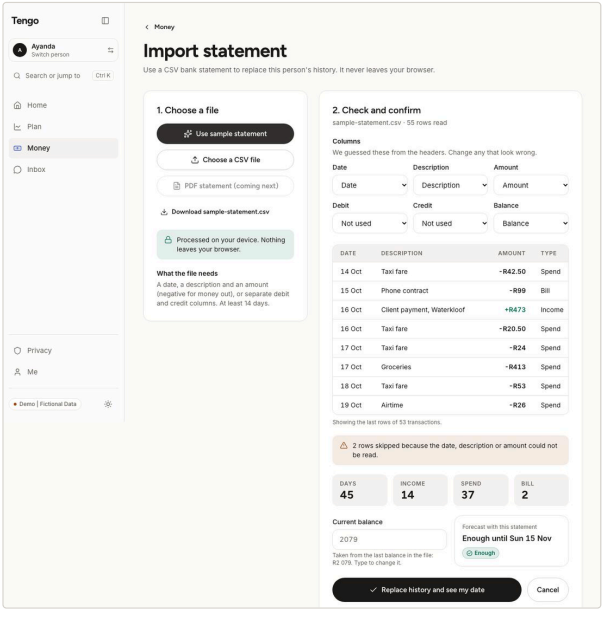
Why this date drawer



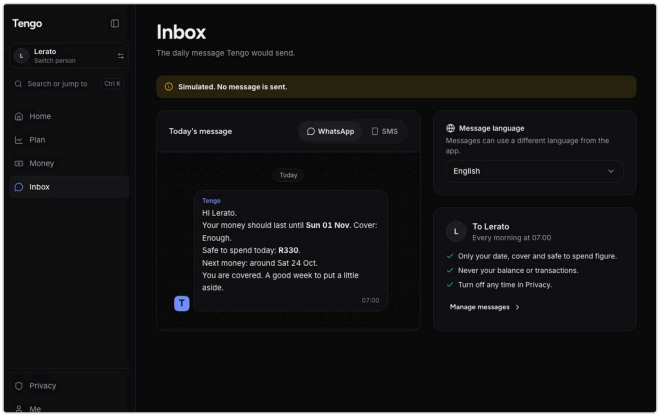
Plan: what-if and loan check



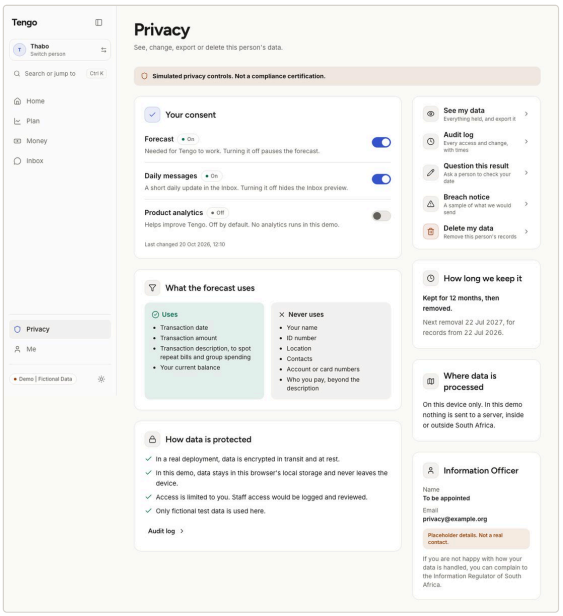
Money: history, bills and spending



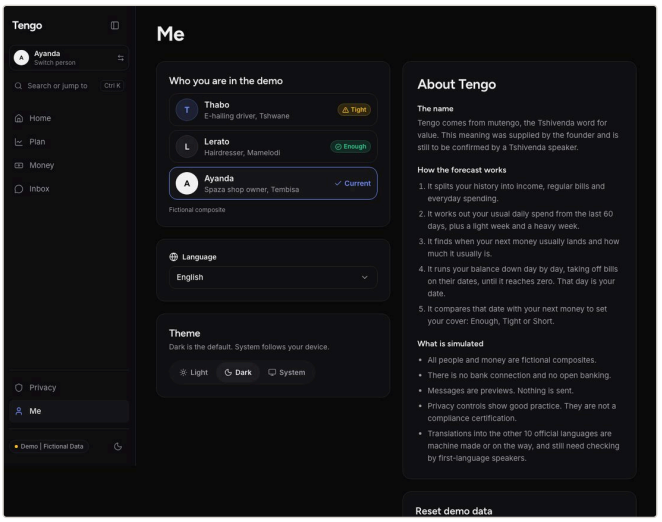
Import with the sample statement



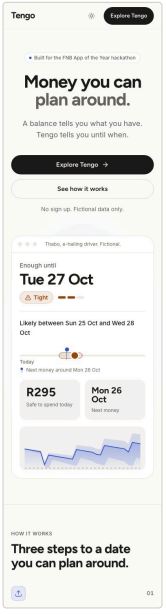
Inbox message preview



Privacy centre



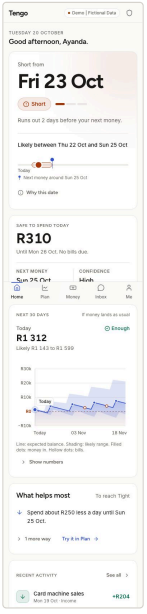
Me



Landing, phone



Consent, phone



Home, phone

Walked on 2026-10-20 against the production build (`npx vite preview`, port 4180) for Thabo, Lerato and Ayanda: landing, picker, consent, Home, Plan, Money, import with the sample CSV, Inbox, Privacy, Me. Screens are in `docs/review/screens/` named `<persona>-<page>-<theme>-<width>.png`. These are observations for the panel to weigh, not conclusions.

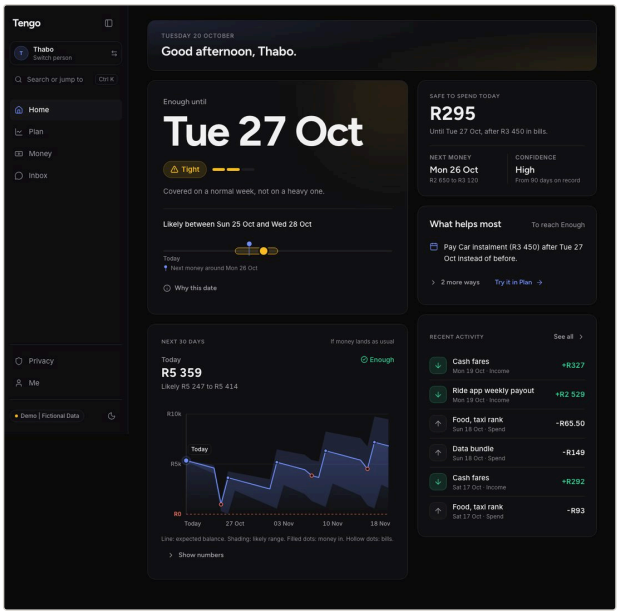
Flow and friction points

1. **Three steps before any value.** Landing, then the picker, then a consent screen with five notice cards and three switches, then Home. On a 390px phone the consent screen is several screens tall before "Agree and continue". Evidence: `thabo-consent-dark-390.png`.
2. **Short but "safe to spend R310".** Ayanda is shown as Short from 23 Oct, two days before her next money, yet the stats card says R310 is safe to spend today, while her usual daily spend is R623. The two figures come from different rules (runway versus balance spread to the latest next money) and are not reconciled on screen. Evidence: `persona-snapshot.json` (ayanda), `ayanda-home-dark-1280.png`.
3. **Headline and chart can disagree.** For Ayanda the hero says Short, while the 30 day chart's readout for today says Enough, because the chart assumes income lands as usual and the headline does not. It is labelled "If money lands as usual" in small text. Evidence: `ayanda-home-light-390.png`.
4. **Pre-filled what-if.** Plan opens with a R500 expense already entered, so the result panel shows a change ("days less") before the user has done anything. Evidence: `thabo-plan-dark-1280.png`.
5. **Languages show English.** Choosing any of the 10 other languages keeps English text everywhere, with a small note. A judge choosing isiZulu sees no change. Evidence: `src/i18n/status.json` (all "pending"), picker screenshots.
6. **Starting state on the picker is fixed.** Persona cards always show the designed starting state (Tight, Enough, Short) even after an import or edit changes it. Evidence: `src/pages/Picker.tsx` uses `p.startState`.
7. **Edited seed data freezes.** Once a seed transaction is corrected, the record stops refreshing daily, so on later days the history no longer ends yesterday and the 60 day spend window drifts. Evidence: `src/data/localRepository.ts` `getRecord` (only `source === 'seed'` refreshes).
8. **Privacy is one tap further on phones.** The bottom bar has Home, Plan, Money, Inbox and Me; Privacy is the shield icon in the top bar. Evidence: `src/layouts/nav.ts`, any `*-390.png`.

9. **Sample import changes the person.** "Use sample statement" replaces the active persona's history with a different, fictional domestic worker's statement, so Thabo becomes someone else with Thabo's name. Evidence: `src/lib/sampleStatement.ts`, `thabo-import-sample-*.png`.
10. **What helps most can suggest moving a car instalment.** For Thabo, the bill-move idea names the car instalment, which in real life may carry penalties. Evidence: `persona-snapshot.json` (thabo help.billToMove).
11. **Human review goes nowhere.** "Question this result" stores a request locally and confirms "Request received" with a reference. It is labelled simulated, but a user could believe a person will reply. Evidence: `src/pages/privacy/Review.tsx`, `thabo-privacy-review-*.png`.
12. **Things that worked smoothly.** Persona switch from the sidebar, Ctrl K menu, theme switching with no flash, CSV sample import with skipped-row count, export of JSON and CSV, delete with receipt, offline reload (Playwright flow test).

4.1 Retail banking product lead

Reviewer file: `docs/review/product-lead.md`. Findings are listed in full in Appendix A.



Thabo, Home, dark, 1280 px: the date-first answer a bank would embed.

Simulated perspective. I am not FNB staff and know nothing of FNB's internal policy, systems, roadmap or pricing. Everything about banks below comes from public pages cited in Sources.

Verdict

Tengo has one idea a bank app does not clearly offer today: a date-first answer ("Enough until Tue 27 Oct") built on how irregular income actually arrives, with a range and a plain state. That is a real gap next to what is public: FNB's nav» Money does categorised spend, smart budgets and alerts and reports five million users; Standard Bank's Future Payments add-on already forecasts debits 14 and 30 days ahead; Standard Bank's Budget Manager pre-fills a budget; Capitec auto-categorises spend; Absa publicises a Savings Coach. None of those, as publicly described, predicts when income lands or turns that into a run-out date. But as a bank product it is not pilot-ready. The engine ignores the things a bank knows best and that matter most to this segment (overdraft and credit limits, scheduled debit orders, multiple accounts, pending card holds), the headline mixes signals ("Enough until" next to a Tight chip), the Plan page uses credit-adjacent language ("Safe to lend up to") that a bank's compliance team would not ship as is, and there is no USSD or low-data path for the customers it claims to serve. The best route is not a separate app but a single "until when" card inside an existing money-management area, fed by internal transaction data, piloted with a narrow segment whose income is paid into the account (e-hailing and platform earners) before anyone whose income is mostly cash.

What works

- **A crisp, distinct core answer.** The Home hero leads with one date, a state word, an icon and a three-segment gauge, then a range and "Next money around". Evidence: `docs/review/screens/thabo-home-dark-1280.png`, `docs/review/screens/ayanda-home-dark-390.png`, `src/components/forecast/DateHero.tsx:9`.
- **Income cadence is modelled, not assumed monthly.** Weekday patterns for daily earners and gap percentiles for lumpy earners give an earliest, typical and latest next money. That is the piece salaried-oriented bank budgeting tools do not publicly describe. Evidence: `docs/review/prep/fact-sheet.md` (Forecast engine), `src/engine/forecast.ts:103`, `docs/review/prep/persona-snapshot.json` (`pattern`: weekday versus interval).
- **Transparent and explainable.** The "Why this date" drawer lists every input, which suits a bank's need to explain automated outputs to customers and complaints teams. Evidence: `docs/review/screens/thabo-why-drawer-dark-1280.png`, README "How the forecast works, in plain words".
- **Integration-friendly seams.** All data access sits behind `TengoRepository` and messages behind `MessageWriter`, so a bank could swap in an internal data feed and its own notification service without touching UI. Evidence: `src/data/repository.ts`, `src/data/index.ts`, README "What is simulated".
- **Engine is small, pure and tested.** 97.6 percent statement coverage in `src/engine`, which lowers the cost of a bank team re-implementing or validating it server side. Evidence: `docs/review/prep/fact-sheet.md` (Quality signals), `docs/review/prep/coverage.txt`.
- **Daily message is privacy-aware by design.** The preview promises only the date, cover and safe-to-spend figure, never balance or transactions. Evidence: `docs/review/screens/thabo-inbox-dark-390.png`.

What to improve

PROD-01. The engine ignores the bank data that defines this customer's real position. - **Severity:** High - **Evidence:** `zeroDate` treats zero as the end of money (`src/engine/forecast.ts:196` to `210`, "if (bal <= 0) return date"). There is no input for an overdraft or credit facility limit, scheduled debit orders or future-dated payments, pending card authorisations, or more than one account. Bills are inferred from history only (same description, within 20 percent, 24 to 36 days apart, at least twice: `src/engine/classify.ts:16`), so a new debit order or an annual premium is invisible. FNB publicly offers app-based overdraft applications (fnb.co.za overdraft page), and Standard Bank's Future Payments already predicts upcoming debits from recurring payments. A bank customer with a R2 000 overdraft who is at R0 is not "out of money" in the bank's view, and a customer with a debit order next week that has only run once is at risk Tengo cannot see. - **Recommendation:** Define a bank data contract for the engine: available balance versus ledger balance, facility limit and whether to count it (default off, shown separately as "then you would be using your overdraft"), scheduled debit orders and future-dated payments as known bills, and an account set. Keep inferred bills as a fallback. - **Effort:** M

PROD-02. Overlap with existing bank tools is not addressed, so the proposition reads as "another budgeting screen". - **Severity:** High - **Evidence:** FNB nav» Money: auto-categorised spend, smart budgets with alerts, credit status, saving goals, five million users and 1.1 million smart budgets (ITWeb, March 2025). Standard Bank: Budget Manager, Money Movements and Future Payments with 14 and 30 day predicted debits (Standard Bank pages). Capitec: automatic spend categorisation (Capitec blog). Tengo's Money page repeats categorisation and a recurring bills list (`docs/review/screens/lerato-money-dark-1280.png`), and the spending breakdown is keyword-based (`src/engine/spending.ts`), which is weaker than a bank's merchant-coded categories. Nothing in the app or README

positions it against these. - **Recommendation:** Narrow the bank pitch to the one thing that is new: an income-aware run-out date and safe-to-spend figure for irregular earners. Drop or de-emphasise categorisation in the bank version and show how the date card would sit beside an existing budgeting tool. Add a one-page "what a bank already has versus what Tengo adds" slide. - **Effort:** S

PROD-03. Headline signals conflict, which in a bank app means complaints and call-centre volume. - **Severity:** High - **Evidence:** Thabo shows "Enough until Tue 27 Oct" with a Tight chip: the label is "Enough until" for every non-Short state (`src/components/forecast/DateHero.tsx:9`, `docs/review/screens/thabo-home-dark-1280.png`). Ayanda is Short from Fri 23 Oct yet "Safe to spend today R310", while her usual daily spend is R623, and the 30 day chart readout for today says Enough (`docs/review/screens/ayanda-home-dark-390.png`, `docs/review/prep/persona-snapshot.json`). The same "Enough until" plus Tight pairing appears in Plan (`docs/review/screens/thabo-plan-dark-1280.png`, `src/pages/Plan.tsx:44`). - **Recommendation:** Use a neutral label ("Lasts until" or "Covered until") for all states, and make the safe-to-spend figure obey the state (for Short, show the gap or the cut needed rather than a positive spend figure). Make the chart readout say "if money lands as usual" next to the state, not in small print. - **Effort:** S

PROD-04. Credit-adjacent and advice-like copy would not pass a bank's brand and compliance review as written. - **Severity:** Medium - **Evidence:** Loan check headline "Safe to lend up to R320" (`src/i18n/locales/en.json:523`, `docs/review/screens/thabo-plan-dark-1280.png`). "What helps most" suggests paying the car instalment after its due date (`docs/review/screens/thabo-home-dark-1280.png`, persona snapshot `help.billToMove`: "Car instalment"), which for a bank that may hold that very loan is a direct conflict. "You could set aside up to" (`en.json:702`) edges towards a savings recommendation. Whether any of this is regulated advice is for the legal reviewer; my point is that a bank's own compliance and brand teams would block it by default. - **Recommendation:** Reword to descriptive outcomes ("If you lend R300, your date moves to Mon 26 Oct"), remove "safe" from lending, never suggest delaying a credit repayment, and add a short "This is an estimate, not advice" line on Plan. Prepare the copy set for a compliance review before any pilot. - **Effort:** S

PROD-05. No channel fit for the lowest-data customers it claims to serve. - **Severity:** Medium - **Evidence:** Tengo is a web PWA only (`docs/review/prep/fact-sheet.md`, Form). The daily message is previewed as WhatsApp or SMS but nothing is sent (`docs/review/screens/thabo-inbox-dark-390.png`), and the SMS text is the same six-line message, which is long for a single SMS. FNB already serves cellphone banking over USSD on *120*321# with balances and mini statements (FNB cellphone banking page). Sending a safe-to-spend figure over WhatsApp also puts financial information on a third-party channel, which a bank would need to approve. - **Recommendation:** Design three bank channel formats: an in-app card (primary), a one-line USSD answer ("Lasts until Tue 27 Oct. Tight. Safe today R295") on an existing USSD menu, and a short push or SMS under 160 characters. Treat WhatsApp as optional and subject to the bank's channel policy. - **Effort:** M

PROD-06. No realistic external data path, so a pilot depends entirely on a bank sponsor. - **Severity:** Medium - **Evidence:** The only real data entry is CSV import (`src/lib/csv.ts`, `src/pages/Import.tsx`); PDF is "coming next" and there is no bank connection (`en.json:472`). Public commentary says South Africa's open finance framework had no formal developments in 2025 and adoption by the FSCA is "still on the cards" for 2026 (Moonstone). Asking an irregular earner to export and upload a CSV every week is not a viable retention loop. - **Recommendation:** Make the pilot ask explicit: internal transaction data, run server side by the bank, with an internal sponsor in the digital money-management or financial inclusion team. Pilot prerequisites to list in the pitch: a data contract (PROD-01), a server-side engine with backtest results on real anonymised data, a copy set cleared by compliance (PROD-04), a channel plan (PROD-05), success metrics (for example arrears and unpaid debit orders in the pilot group versus a control), and a kill criterion. - **Effort:** L

PROD-07. Segment fit is asserted, not shown, and the cash-heavy persona is the weakest fit for bank data. - **Severity:** Medium - **Evidence:** All three personas are synthetic composites (`src/data/seed/personas.ts`); there is no user research in the repo. Ayanda (spaza owner) earns "lumpy cash takings" (BRIEF.md section 11); cash spent before it is deposited never appears in a bank feed, so the forecast sees only net deposits. Thabo's "Ride app weekly payout" lands in the account (`docs/review/screens/thabo-home-dark-1280.png`), which is the case bank data handles best. - **Recommendation:** Lead the bank pilot with platform and gig earners whose income is paid into the account, then hairdressers and small traders who deposit regularly. Treat cash-first traders as a later phase that needs a cash-entry feature. Run 5 to 8 interviews per segment before a pilot. - **Effort:** M

PROD-08. Standalone onboarding and placeholders do not fit inside a bank app. - **Severity:** Low - **Evidence:** Three steps before value (landing, picker, consent with five cards and three switches) (`docs/review/prep/walkthrough.md` item 1, `docs/review/screens/thabo-consent-dark-390.png`). All 10 non-English locales are English text (`src/i18n/status.json`). Logo and icons are placeholders (README "Brand and type"). - **Recommendation:** For the bank version, design an in-app opt-in of one screen that reuses the bank's existing consent and terms, inherit the bank's design system, and get first-language review of at least isiZulu, Sesotho and Afrikaans before any customer sees them. - **Effort:** M

Top three risks

1. **Wrong date, real harm, bank's name on it.** Inside a bank app, a confident "Enough until" that misses an overdraft, a new debit order or a late payout becomes the bank's error, with complaints and reputational cost (PROD-01, PROD-03).
2. **Absorbed as a feature.** Banks already have budgeting, alerts and, at Standard Bank, a 14 and 30 day payment forecast. A bank could add income cadence to an existing tool without Tengo; the defensible part is small unless backed by proven accuracy on real data (PROD-02).
3. **No data, no pilot.** With open finance not yet adopted and CSV as the only input, Tengo has no route to real users except a bank sponsor, and the current copy and channel gaps would slow that sponsor's internal approvals (PROD-04, PROD-05, PROD-06).

Score

5/10. A genuinely distinct date-first, income-aware answer with a clean explainable engine, but it ignores bank-side data like overdrafts and debit orders, mixes signals in the headline and is not yet shaped for a bank's compliance, channels or segments.

Sources

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- FNB nav» Money page (Money Coach; direct fetch blocked by a browser check, content seen in search index only): https://www.fnb.co.za/ways-to-bank/digital/navMoney.html (accessed 2026-10-20)
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- Standard Bank, "Do more with our banking app" (Budget Manager, Future Payments 14 to 30 days, Money Movements, My360): https://www.standardbank.co.za/southafrica/personal/learn/do-more-with-our-banking-app-part-1 (accessed 2026-10-20)
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- Capitec blog, "What if there was a simpler way to budget?" (automatic categorisation; direct fetch returned 403, content from search index): https://www.capitecbank.co.za/blog/articles/best-way-to-bank/what-if-there-was-a-simpler-way-to-budget/ (accessed 2026-10-20)
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4.2 Head of technology and solution architect

Reviewer file: `docs/review/technology.md`. Findings are listed in full in Appendix A.

Verdict

As a hackathon build, Tengo is well made. It has a pure, deterministic forecast engine with 97.6 percent statement coverage, clean layering, strict TypeScript, good scores on throttled mobile, and a small, permissively licensed set of dependencies. As the base for a real product it is a well-built client-only prototype, and the README's claim that it can be swapped to a backend without touching UI code is optimistic. The most serious technical defect is about correctness, not scale. Once a user imports or edits data, the record stops ageing. The engine then treats every elapsed day as zero spend against an unchanged balance, so the state drifts towards Enough while confidence stays "high". My benchmark shows Ayanda moving from Short to Enough after 30 days with no new data. The other gaps are expected in a demo but must close before a pilot. There is no observability at all. The repository contract is coarse and cannot be scoped to real users. localStorage writes are unguarded. The service worker caching is naive. The code is never split into smaller chunks.

What works

- **The engine is pure, deterministic and well tested.** `src/engine/forecast.ts` and `classify.ts` are side-effect-free functions with named constants (`forecast.ts:18–28`, `classify.ts:16–22`). Statement coverage is 97.55 percent for `src/engine` and 98.59 percent for `forecast.ts` (`docs/review/prep/coverage.txt`). There are 66 unit tests across 9 files and all pass (`docs/review/prep/checks.md`).
- **Date arithmetic is safe across time zones.** Calendar days are converted to UTC day numbers (`src/engine/dates.ts:6–9`), so a device in a different time zone cannot shift the forecast date.
- **The engine is fast enough at realistic volumes.** `forecast()` runs in 0.36 ms on 285 transactions, 0.83 to 1.02 ms on about 2 000, and 2.23 ms on 10 080 (laptop, `docs/review/tools/technology/engine-bench.out.txt`, script `engine-bench.ts`).
- **Layering is clean at the edges.** Zustand holds only UI preferences (`src/store/session.ts:18`). All persona data goes through one `TengoRepository` interface (`src/data/repository.ts:75`), and a single line chooses the implementation (`src/data/index.ts:5`).
- **It performs well on throttled mobile.** Lighthouse mobile (slow 4G at 150 ms RTT, 4x CPU) gives landing 94 and picker 96 for performance, with TBT of 160 ms and 10 ms, LCP of 2.4 s and 2.3 s, CLS 0, and 218 KiB total (`docs/review/prep/lighthouse/*.report.json`). Home under the same throttling shows the hero in 1 516 ms cold and 627 ms warm, transferring 149 KB (`docs/review/prep/perf-home.json`).
- **Privacy is tested in code.** A Playwright test asserts no third-party requests (`tests/flows.spec.ts:117`), and the service worker ignores other origins (`public/sw.js:24`).
- **Dependencies are few and permissively licensed.** There are 10 production dependencies, licensed MIT (18), OFL-1.1 (2), ISC (1) and Apache-2.0 (1), with no copyleft (`docs/review/prep/audit.md`).
- **There is a quality toolchain.** Lint, a dash checker, i18n key parity, font coverage and 116 contrast pairs are all scripted and pass (`docs/review/prep/checks.md`). A pre-paint script stops the theme from flashing (`index.html:11–22`).

What to improve

TECH-01 Imported or edited data never ages, so the forecast drifts towards Enough

- **Severity:** High
- **Evidence:** `src/data/localRepository.ts:87` refreshes only `source === 'seed'`, so imported and edited records stay fixed. `src/engine/forecast.ts:36–41` measures history up to `today`, not up to the last transaction, and `forecast.ts:66` counts days with no transactions as zero spend. The balance is never reduced for days that have passed. Running the real engine on the seeded records viewed later with no new data (`docs/review/tools/technology/engine-bench.out.txt`): Thabo moves from Tight on 27 Oct to Enough on 13 Nov after 7 days. Ayanda moves from Short to Tight after 7 days and to Enough after 30, with central spend falling from R623 to R79 a day. Confidence stays "high" throughout. The import page's rebase (`src/lib/csv.ts:160–162`) fixes this only at the moment of import. The walkthrough (item 7) noted the freeze but not that it inflates the state.
- **Recommendation:** Store an `asOf` date with the balance. Run the engine with `today = asOf + 1`, or subtract the estimated spend for the elapsed days, and measure the spend window up to the last transaction. Show "Your data is N days old". Lower the confidence after 3 days and hide the state after about 14 days until the user refreshes. Add a unit test for staleness.
- **Effort:** S

TECH-02 No observability: errors, performance and forecast quality are invisible

- **Severity:** High
- **Evidence:** The only error handling is `console.error` (`src/components/ErrorBoundary.tsx:17`). A search for Sentry, `onerror` or `unhandledrejection` finds nothing. There is one boundary, at the root (`src/App.tsx:22`), so any render error replaces the whole app, which contradicts its own comment "one broken screen never blanks the whole app" (`ErrorBoundary.tsx:8`). Promise failures in writes are not caught at all: `src/pages/Import.tsx:55–60` has no try/catch, and `void` calls in `src/data/PersonaProvider.tsx:80` drop their errors. Nothing measures web vitals, and nothing records forecast outcomes against what actually happened, so accuracy can never be tracked in production.
- **Recommendation:** Add error reporting with personal data scrubbed (no amounts, descriptions or names), route-level error boundaries, a global `unhandledrejection` handler, web-vitals sampling, and a privacy-safe forecast accuracy event (predicted state and date against the observed outcome, bucketed). Define SLOs before a pilot.
- **Effort:** M

TECH-03 The repository contract is not ready for a real backend

- **Severity:** High
- **Evidence:** Keys are a closed union of three personas (`src/data/seed/personas.ts:3`), with no user, tenant or auth context in any method (`src/data/repository.ts:75–95`). Every read and write parses and rewrites one JSON blob holding all personas (`src/data/localRepository.ts:57–69`). `getRecord` returns the full history and `replaceHistory` sends it all back: about 183 KB of JSON for 2 000 transactions (`engine-bench.out.txt`). There is no paging, no incremental sync and no version or ETag for concurrent edits. Each write makes 2 or 3 sequential calls: `setConsent` then `appendAudit` then `getAudit` (`PersonaProvider.tsx:112–117`). The "audit log" is written by the client, can be edited by the user and is cut to 200 entries (`localRepository.ts:16,146`). Pages also bypass the provider and call the repository directly (`src/pages/Me.tsx:26,32`, `src/pages/Picker.tsx:35,40`, `src/components/CommandPalette.tsx:92`).
- **Recommendation:** Redesign the contract for a hosted service: user-scoped methods that take an auth context, paged and incremental transaction sync (cursor or `since`), optimistic concurrency, write endpoints that return the audit entry they created, and a server-side append-only audit trail with no client delete. Move the engine into a shared package so the server can produce authoritative forecasts and alerts while the client keeps offline what-ifs.
- **Effort:** L

TECH-04 localStorage writes are unguarded and the storage is fragile

- **Severity:** Medium
- **Evidence:** `write()` calls `setItem` with no try/catch (`src/data/localRepository.ts:67–69`). A `QuotaExceededError` therefore rejects `replaceHistory`. `Import.tsx:57–58` then leaves `saving` set to true and never navigates. MDN gives a limit of 5 MiB per origin and says Safari deletes script-written data after seven days without interaction (source below). 10 080 transactions take 897 KB of JSON, which is about 1.8 MB in browsers that store UTF-16 (`engine-bench.out.txt`). There is no `storage` event listener, so two open tabs overwrite each other's blob (last writer wins). Corrupt data is silently replaced with an empty store (`localRepository.ts:61–64`), which loses consent and audit records without telling the user.
- **Recommendation:** Wrap writes and show a clear error. Move records to IndexedDB, request `navigator.storage.persist()`, listen for `storage` events or use a BroadcastChannel, and keep a backup copy before overwriting corrupt data. Before any pilot, add server sync so clearing the browser does not lose a user's history.
- **Effort:** M

TECH-05 The service worker caching strategy is naive

- **Severity:** Medium
- **Evidence:** The worker is registered on `load` (`src/main.tsx:16–19`), so the first visit's JS and CSS are fetched before it controls the page. Offline works only after a second load, and the test reloads before going offline (`tests/flows.spec.ts:131–140`). Every same-origin GET that is not a navigation is cache-first under a fixed name `tengo-v2` (`public/sw.js:4,40–52`). As a result `manifest.webmanifest` and the icons never refresh, and old hashed bundles pile up forever because the cache name never changes with the build. Navigations write any response, including a 404 or 5xx, over the cached `/` (`sw.js:27–35`). `vercel.json` sends every unmatched path to `index.html`, so a missing old asset comes back as HTML with status 200 and gets cached under a `.js` URL.
- **Recommendation:** Generate a precache manifest from the build (Workbox or vite-plugin-pwa) and precache it at install. Version the cache by build hash and clean up old entries. Cache only `res.ok` responses with the expected content type. Use stale-while-revalidate for files without a hash. Exclude `/assets/*` from the SPA rewrite.
- **Effort:** S

TECH-06 One 696 KB bundle with every locale loaded at start

- **Severity:** Medium
- **Evidence:** There is a single `index-*.js` of 695 602 bytes raw, 141 864 gzip (`docs/review/prep/audit.md`), and Vite warns that a chunk is over 500 kB (`checks.md`). Locales are imported eagerly (`src/i18n/index.ts:5`), as are all pages (`src/App.tsx:8–18`) and the seed generator. Lighthouse estimates 41 to 47 KiB of unused JS. Today the 11 locale files cost little because 10 are copies of English: 12.8 KB gzip together. Compressed one at a time they total 104 KB gzip (measured with `gzip -c` per file on `src/i18n/locales`). **Assumption:** real translations compress about like separate files, so translating adds roughly 80 to 100 KB gzip to the first load, about 60 to 70 percent more JS on 2G or 3G for users who pay for data.
- **Recommendation:** Load English plus the chosen language on demand and let the service worker prefetch the rest when idle, so switching language offline still works. Lazy-load routes (Plan, Money, Import, Privacy, Inbox). Keep the seed generator out of the main chunk. Set a bundle budget in CI.
- **Effort:** S

TECH-07 Test strategy leaves the UI, data layer and failure paths unproven

- **Severity:** Medium
- **Evidence:** Statement coverage is 19.95 percent for all of `src`, with 0 percent for every component and page, and 0 percent for `PersonaProvider.tsx` (`coverage.txt`). 136 of the 221 Playwright tests are screenshot runs whose only assertion is no horizontal overflow, with no visual baseline (`tests/screens.spec.ts:96–98`). Playwright uses Chromium only (`playwright.config.ts:13`), so iOS Safari, where storage eviction applies, is untested. There is no CI configuration (no `.github` folder). Nothing tests quota errors, stale data (TECH-01), multiple tabs, time zones or service worker updates. `localRepository.test.ts` has 5 tests.
- **Recommendation:** Add CI running build, unit tests, lint, audit and Playwright on every push. Add a WebKit project and a low-end Android emulation profile. Use `toHaveScreenshot` baselines for the key screens. Add component tests for `PersonaProvider` and Import. Add engine property tests (balance monotonicity, state ordering, staleness).
- **Effort:** M

TECH-08 Known advisories in shipped and build dependencies

- **Severity:** Medium
- **Evidence:** `react-router-dom` 6.30 is affected by GHSA-wrjc-x8rr-h8h6, an open redirect through `<Link>` and `useNavigate` (moderate, patched in 7.18.0), and by GHSA-337j-9hxr-rhxg, which concerns SSR hydration (`audit.md`). The app passes only internal constant paths or an encoded receipt id to `navigate` (`src/pages/privacy/Delete.tsx:21`, `CommandPalette.tsx:82`), so practical exposure is low, and Tengo has no SSR. The dev toolchain has 2 critical advisories (tinypool, through vitest 3) and 5 high (braces, through tailwind 3 and chokidar), all dev-only but a risk on build machines.
- **Recommendation:** Upgrade to react-router 7.18 or later in library mode, and to vitest 5 and tailwind 4 (or pin fixed transitive versions with npm `overrides`). Add `npm audit --omit=dev` as a CI gate and use Renovate or Dependabot.
- **Effort:** M

TECH-09 Engine runs on the main thread and is repeated many times per render

- **Severity:** Low
- **Evidence:** `helpPlan` runs `forecast` inside a binary search over R10 steps up to R200 000 (`src/engine/help.ts:6,29–39`), then again for each candidate bill. It is called on every Home render where the record changes (`src/components/forecast/HelpCard.tsx:17`). Measured on a laptop: 4.2 ms at 285 transactions, 10 to 12 ms at about 2 000, 40 ms at 10 080 (`engine-bench.out.txt`). `loanCheck` and `whatIf` run again on every keystroke (`src/pages/Plan.tsx:65–73,190`). **Assumption:** a low-end Android is 4 to 10 times slower than the laptop, giving 40 to 120 ms at 2 000 transactions and 160 to 400 ms at 10 000. That is noticeable input lag on Home and Plan.
- **Recommendation:** Debounce the Plan inputs. Cache the base forecast by record version. Solve `extraNeeded` in closed form from the runway (it is linear in balance) instead of searching. Move heavy work to a Web Worker if histories grow.
- **Effort:** S

TECH-10 The fake 140 ms latency ships to production and is applied unevenly

- **Severity:** Low
- **Evidence:** `new LocalRepository(window.localStorage, 140)` in production (`src/data/index.ts:5`). It adds 140 ms to every cold Home load (`perf-home.json`, hero at 1 516 ms). It applies to `getRecord`, `getConsent`, `getAudit` and `deletePersona`, but not to `setConsent`, `appendAudit`, `getReviewRequests` or `getDeletionReceipts` (`localRepository.ts:129–163,182`). The loading states therefore do not model a real network.
- **Recommendation:** Make the latency a development-only flag, and test loading and error states with Playwright route delays and failures instead.
- **Effort:** S

TECH-11 "Today" is fixed when the app opens

- **Severity:** Low
- **Evidence:** `const [today] = useState(() => localToday())` (`src/data/PersonaProvider.tsx:36`). An installed PWA left open overnight keeps showing yesterday's forecast and does not trigger the daily seed refresh.
- **Recommendation:** Recompute `today` on `visibilitychange` and at local midnight, and reload when it changes.
- **Effort:** S

TECH-12 No security or caching headers in the hosting config

- **Severity:** Low
- **Evidence:** `vercel.json` contains only an SPA rewrite: no Content-Security-Policy, no `frame-ancestors`, no Permissions-Policy, and no `immutable` cache header for hashed `/assets/*`. For a money app that will later hold bank data, the CSP is the main defence against injected scripts reading localStorage.
- **Recommendation:** Add a strict CSP (`default-src 'self'`, with a hash for the inline theme script at `index.html:11`), `frame-ancestors 'none'`, `Referrer-Policy`, `Permissions-Policy`, long immutable caching for `/assets/*` and `no-cache` for `/sw.js`.
- **Effort:** S

What breaks first at 100 000 users

Serving the app is not the problem: it is a static site on a CDN. What breaks first, in order:

1. **Trust, through stale and lost data.** Users who imported once see drifting forecasts (TECH-01). Safari users who stay away for a week lose everything (TECH-04).
2. **The team's ability to see any of it.** No telemetry means the first signal is a support ticket or a social media post (TECH-02).
3. **The repository when a backend arrives.** Full-history payloads and 2 or 3 calls per write (TECH-03) multiply load. **Assumption:** with 30 percent daily active users each making about 6 data calls a day, that is about 180 000 calls a day, which is trivial for a managed Postgres service. The real costs are bank-feed ingestion, key management and the audit store, none of which exist yet.

Top three risks

1. **The forecast misleads users as data ages (TECH-01).** A frozen import shows Enough to someone who is Short, with high confidence. That is a product safety risk, not only a bug.
2. **Running blind (TECH-02).** Without error, performance or accuracy telemetry, a pilot cannot show the forecast works or catch regressions on low-end phones.
3. **Underestimating the backend (TECH-03, TECH-04).** The repository swap is presented as easy. In practice auth, user scoping, sync, a server-side audit trail and data durability are most of the work for a pilot.

Score

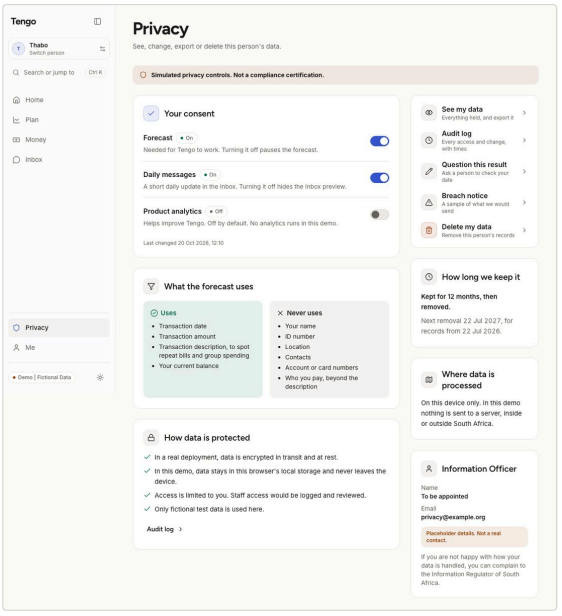
6 out of 10. The engine and front end are clean, fast and well tested for a demo, but data freshness, observability and the backend contract all need real work before a pilot.

Sources

- MDN, "Storage quotas and eviction criteria": https://developer.mozilla.org/en-US/docs/Web/API/Storage_API/Storage_quotas_and_eviction_criteria (accessed 2026-10-20)
- GitHub Advisory GHSA-wrjc-x8rr-h8h6, React Router open redirect: https://github.com/advisories/GHSA-wrjc-x8rr-h8h6 (accessed 2026-10-20)

4.3 Security and privacy lead (POPIA)

Reviewer file: `docs/review/security-privacy.md`. Findings are listed in full in Appendix A.



Privacy centre, light, 1280 px.

Verdict

Tengo's privacy centre is a thoughtful demonstration of data subject rights for a hackathon: consent, access, export, correction, human review, a sample breach notice and deletion all exist as working screens, nothing leaves the browser, and a test checks there are no third party requests. As a model of what a real deployment would do, it is weaker than it looks. Several on-screen statements are written as facts but are not implemented: the "Kept for 12 months, then removed" retention rule is only a computed date and never deletes anything, the "Never uses account numbers or who you pay" panel is contradicted by storing statement descriptions verbatim (the bundled sample already holds "Client payment, Mrs K"), and green ticks sit next to encryption and access claims. Deletion removes the persona slot but leaves a receipt that proves nothing, and the audit log stores transaction descriptions, including the value a user later corrected. None of this harms anyone today because the data is fictional and local. It would matter the moment a real statement is imported, and nothing yet exists for authentication, server-side storage, key management, operator contracts, cross-border transfer, breach handling or a registered Information Officer. This is a simulated professional view based on the code and public sources, not legal advice.

What works

- Everything is local and same-origin. The service worker only handles same-origin GETs (`public/sw.js:23`) and a Playwright test asserts no third party requests (`tests/flows.spec.ts:117`). The README states no analytics scripts (`README.md:104`).
- Clear "simulated" labelling on the consent screen and privacy centre: "Simulated privacy controls. Not a compliance certification." (`src/pages/Consent.tsx:64-66`, `src/pages/privacy/Overview.tsx:55-57`, screenshot `docs/review/screens/thabo-consent-light-390.png`). The Information Officer card is flagged "Placeholder details. Not a real contact." (`Overview.tsx:157`).
- Rights are mapped to real POPIA concepts: access and export (`src/pages/privacy/MyData.tsx:27-48`), correction routed to Money (`privacy.correctHint`), a human review route that echoes the representation right for automated decisions in section 71 (`src/pages/privacy/Review.tsx`), complaint to the Information Regulator (`privacy.complaints`), and a breach notice template with the right sections (`src/pages/privacy/Breach.tsx`, stamped "Sample").
- Withdrawing core consent pauses the forecast screens while leaving Privacy and Me open, so a person can still see and delete data (`src/layouts/AppGate.tsx:9,35`). Analytics is off by default (`Consent.tsx:28`).
- CSV import drops every column except date, description, amount and balance (`src/lib/csv.ts:130-152`) and the file is read in the browser, never uploaded (`src/pages/Import.tsx:99`). That is genuine minimisation for column-level data.
- Deletion is a two-step confirm with an explicit checkbox (`src/pages/privacy/Delete.tsx:46-66`) and does remove transactions, balance, consent, audit and review requests for that persona (`src/data/localRepository.ts:165-180`, confirmed by `docs/review/tools/security-privacy/deletion-probe.ts`).
- The repository interface (`src/data/repository.ts`) gives one seam where server-side access control, encryption and audit can be added without touching UI code.
- No `dangerouslySetInnerHTML`, `innerHTML` or `eval` in `src` (grep, 2026-10-20). React escaping covers the i18n `escapeValue: false` setting (`src/i18n/index.ts:14`) because no translated string reaches an HTML sink.

What to improve

SEC-01 Retention rule is displayed as fact but never enforced

Severity: High **Evidence:** The privacy centre says "Kept for 12 months, then removed." and "Next removal {{date}}" (`src/i18n/locales/en.json:234`, `docs/review/screens/thabo-privacy-dark-1280.png`), the consent notice says "12 months, then it is removed" (`consent.notice.how.body`), and the README says retention "works against local storage" (`README.md:94`). `src/pages/privacy/retention.ts:6-11` only computes a date; the only caller is the display (`Overview.tsx:40`). Probe: a record dated 2024-01-05 is still held on 2026-10-20 (`docs/review/tools/security-privacy/deletion-probe.ts`, output "2024 record still held after 12+ months: true"). Seed data hides this because it is 90 days old and regenerated daily. POPIA section 14(4) requires destruction or de-identification as soon as reasonably practicable once authority to keep ends, and 14(5) requires that destruction prevents reconstruction. **Recommendation:** Either implement a purge (drop transactions older than the window on every read and log it) or change the copy to "In a real service, records older than 12 months would be removed". In production, run a scheduled server job, cover backups and operator copies, and record the run. **Effort:** S

SEC-02 On-screen wording reads as real security and compliance claims

Severity: High **Evidence:** "How data is protected" shows green check icons (`Overview.tsx:105–109`) beside "In a real deployment, data is encrypted in transit and at rest." and "Access is limited to you. Staff access would be logged and reviewed." (`en.json:205,207`). There is no real deployment, so the tick asserts a control that does not exist. Delete says "I understand this removes the data for good" (`en.json:330`) and "A deletion receipt is kept" (`en.json:327`). The human review says a person would "reply within 5 working days" (`en.json:288`). The breach sample says "told the Information Regulator" (`privacy.breach.doing.body`). One banner at the top is easy to crop out of a screenshot or pitch slide (`docs/review/screens/thabo-privacy-dark-1280.png`). **Recommendation:** Replace ticks with a neutral "planned" icon, write every future control in the conditional ("would be"), put "Demo" on each card that describes a future control, and keep the banner on every privacy sub-page. Do not quote these screens as evidence of compliance in submission material. **Effort:** S

SEC-03 "Never uses names, account numbers or who you pay" is false for real statements

Severity: High **Evidence:** The panel lists "Your name", "Account or card numbers" and "Who you pay, beyond the description" under "Never uses" (`en.json:188–201`). Import keeps the description verbatim (`src/lib/csv.ts:133,163`) and it is stored and exported unchanged (`MyData.tsx:44`). Bank statement descriptions routinely hold payee names, references and partial account numbers (**Assumption**, from typical South African statement layouts; likely in most debit-order and transfer rows). The project's own sample contains third party names: "Client payment, Mrs K" and "Client payment, Waterkloof" (`public/sample-statement.csv:2,7`). That is personal information of third parties, who receive no notice. Financial history is personal information under the POPIA section 1 definition, item (b). **Recommendation:** Scrub descriptions on import (mask digit runs of 6 or more, South African ID number patterns, email and phone numbers), store a normalised merchant token instead of free text where possible, and change the copy to "Descriptions may contain names. We mask numbers." Add unit tests for the scrubber. **Effort:** M

SEC-04 No authentication, unencrypted financial data in localStorage, no security headers

Severity: High (for any real data), Low (for the demo) **Evidence:** All records, balance, consent, audit and reviews sit in plaintext JSON under `tengo.data.v1` (`localRepository.ts:15,67–69`; `src/data/index.ts:5`). Anyone with the phone, any script on the origin, or a malicious browser extension can read it. Persona switching has no lock (`src/pages/Picker.tsx:40–41`), which on a shared phone means one person can open another's data. `vercel.json` sets only a rewrite: no Content-Security-Policy, HSTS, `X-Content-Type-Options`, `Referrer-Policy` or `Permissions-Policy`. The inline theme script in `index.html:11–22` means a strict CSP would need a hash. `react-router` 6.x carries a moderate open-redirect advisory (GHSA-wrjc-x8rr-h8h6, `docs/review/prep/audit.md:45–47`). POPIA section 19 requires appropriate, reasonable technical and organisational measures with regard to generally accepted practice. **Recommendation:** For real data: server-side storage with authentication (bank SSO or phone OTP plus device binding), an app PIN or biometric lock, short-lived tokens in memory or HttpOnly cookies rather than localStorage, and at most a minimal encrypted cache on device (WebCrypto key, non-extractable). Ship CSP with a script hash, HSTS and the other headers in `vercel.json`. Upgrade react-router. **Effort:** L

SEC-05 Deletion is local only and the receipt proves nothing

Severity: Medium **Evidence:** `deletePersona` deletes the slot then appends a receipt with `personaId`, timestamp and counts (`localRepository.ts:165–180`). Probe output after delete: `{"personas":{},"receipts":[{"id":"DEL_MVGIYDW5GV114F","personaId":"thabo",...}]}`. The receipt id comes from `Math.random` and `Date.now` (`localRepository.ts:37–40`), is unsigned, is stored on the same device whose data was "deleted", and survives forever until "Reset" (`resetAll`, line 186). The id is placed in the URL (`Delete.tsx:21`), so it lands in browser history. Review request count is not recorded in the receipt. Things deletion cannot reach: downloaded JSON and CSV exports (`MyData.tsx:39,45`), the original statement file, and in a real build any server copies, backups, logs and operator copies. Reopening the persona regenerates a fresh seed record immediately (probe: "source: seed tx: 220"), which can look like the deletion failed. **Recommendation:** In production, deletion is a server workflow: delete primary rows, tombstone for backup restore, purge from operators with confirmation, expire backups within a stated window, then issue a receipt signed by the server that holds no personal data beyond a pseudonymous id. Tell the user plainly what cannot be deleted (their own downloads). Keep the deletion record itself under a documented retention period. In the demo, drop `personaId` from the URL and say "receipt on this device only". **Effort:** M (demo copy S, real workflow L)

SEC-06 Consent is bundled, partly pre-ticked, and not provable

Severity: Medium **Evidence:** Core processing is a disabled toggle labelled "Required" (`Consent.tsx:86`), so "consent" for it is not voluntary in the sense of the POPIA section 1 definition ("voluntary, specific and informed"). Contract (section 11(1)(b)) is the more honest lawful basis for the core forecast. Daily messages default to on (`Consent.tsx:27`, screenshot `thabo-consent-light-390.png`). The `ConsentRecord` is overwritten on each change (`localRepository.ts:129–135`) and holds only booleans and a time: no notice version, language, channel or text shown. The only history is the audit log, capped at 200 entries (`localRepository.ts:16,146`) and erased on delete. Section 11(2)(a) puts the burden of proving consent on the responsible party. If messages ever carry offers, section 69 requires prior opt-in for electronic direct marketing. **Recommendation:** Base core processing on contract and say so; keep consent for messages and analytics, both off by default. Store append-only consent events with notice version hash, locale, UI surface and timestamp on the server, retained beyond profile deletion as proof under a documented period. Separate service messages from any marketing. **Effort:** M

SEC-07 Audit log holds personal data, keeps pre-correction values, and is not tamper-evident

Severity: Medium **Evidence:** Editing a transaction logs `{ name: tx.description }` (`src/components/TransactionEditor.tsx:41`), so the original description survives in the log after the user corrects it (probe output `{"name":"PAYMENT TO J SMITH","fields":"category"}`). This undermines correction under section 24. Translated labels are stored as params (`Overview.tsx:47`). The log is client-writable, trimmed to the last 200 entries (`localRepository.ts:146`), and every page view adds `session.opened` and `data.viewed` (`PersonaProvider.tsx:77–81`, `MyData.tsx:20–24`), so a busy user loses older evidence quickly. **Recommendation:** Log event type, actor id, pseudonymous record id and changed field names only, never descriptions or amounts. In production write audit events server-side to append-only storage with hash chaining or WORM retention, separate from the user profile, and apply the same "no personal data in logs" rule to application logs, error tracking and analytics. **Effort:** M

SEC-08 Collection notice misses items section 18 requires

Severity: Medium **Evidence:** The consent notice covers what, why, who, where and how long (`Consent.tsx:14–20`, `en.json:146–157`). Section 18(1) also needs the responsible party's name and address, whether supply is voluntary or mandatory and the consequences, recipients or categories of recipients, the right to object, the right to complain to the Regulator with contact details, and any intended cross-border transfer. The complaint line exists only on the privacy page (`privacy.complaints`). The Information Officer is "To be appointed" with `privacy@example.org` (`Overview.tsx:150–154`); section 55(2) and the Regulator's guidance note say officers take up duties only after registration on the eServices portal. **Recommendation:** Add the missing items to a short layered notice with a link to a full privacy policy. Name the legal entity. Appoint and register an Information Officer before any pilot. Publish a PAIA manual. **Effort:** S (copy), M (entity, officer, manual)

SEC-09 Service worker would cache personal data once an API exists

Severity: Medium **Evidence:** Every same-origin GET that is not a navigation is served cache-first and cached on success (`public/sw.js:38–49`). Today that is only hashed assets. If the planned hosted repository (`README.md:95`) uses a same-origin API path or a proxy, account and transaction responses would be written to Cache Storage, survive logout and deletion, and be served stale to the next person on a shared phone. Navigations are cached as `/` (`sw.js:30`), harmless now, risky if HTML ever embeds user data. **Recommendation:** Allow-list cacheable paths (hashed `/assets/*`, icons, manifest) and make everything else network-only. Clear all caches and storage on logout and on deletion. Set `Cache-Control: no-store` on API responses. **Effort:** S

SEC-10 No plan for operators and cross-border transfer

Severity: Medium **Evidence:** The cross-border card says "nothing is sent to a server, inside or outside South Africa" (`en.json:238`). The README invites swapping in Supabase (`README.md:95`) and an LLM-backed `MessageWriter` (`README.md:96`), and the Inbox previews WhatsApp and SMS delivery. Each would be an operator

under POPIA section 1, requiring a written contract obliging section 19 safeguards and immediate breach notice to the responsible party (section 21). A database region, LLM provider or messaging platform outside South Africa triggers section 72: transfer only with adequate protection by law, binding rules or agreement, consent, or contractual necessity. Message text containing a cover date and amount is financial personal information; sending it to an LLM or WhatsApp sends it to that provider. **Recommendation:** Keep a register of operators with region, data categories and contract status. Prefer a South African hosting region. For an LLM, send only templated, de-identified fields (no names, no descriptions) and contract for zero retention and no training. For WhatsApp or SMS, keep content minimal ("Your Tengo update is ready") to limit what lock screens and the platform see. Update the section 18 notice with transfers. **Effort:** M

SEC-11 Children and vulnerable users are not considered

Severity: Medium **Evidence:** There is no age question or gate anywhere (grep for "age" checks in `src/pages`, none). POPIA defines a child as under 18 (section 1) and section 34 prohibits processing a child's personal information unless a section 35 exception applies, such as prior consent from a competent person. The target users (informal earners, shared phones per the brief) include people at risk of financial control by a partner or family member; the app has no lock, shows full history on open, and the planned daily messages could reveal a balance on a shared or watched phone. **Recommendation:** Require 18 or older at onboarding (or a bank-verified adult account), add an app lock and a discreet notification mode, and run a short risk assessment for financial abuse and coercion with an inclusion partner. **Effort:** M

SEC-12 No breach response capability behind the sample notice

Severity: Medium **Evidence:** `Breach.tsx` shows a template only. There is no incident runbook, detection, contact list or log source. Since 1 April 2025 section 22 notifications to the Regulator must go through the eServices portal, and commentary reports no low-risk exemption. An operator must tell the responsible party immediately (section 21(2)). **Recommendation:** Before a pilot: an incident response plan with roles, a decision log, the SCN1 portal process, a data subject notice in all supported languages, operator notification clauses, and a tabletop exercise. **Effort:** M

SEC-13 Withdrawn consent does not stop processing

Severity: Low **Evidence:** With core consent off, `AppGate` hides forecast pages (`AppGate.tsx:35`) but `PersonaProvider` still computes the full forecast on every load (`PersonaProvider.tsx:83–86`) and data stays held. Section 11(2)(b) allows withdrawal at any time. **Recommendation:** Skip the forecast computation when `consent.core` is false, and offer deletion on the paused screen. On a server, stop scheduled jobs and messaging for that user. **Effort:** S

SEC-14 Export and import edge cases

Severity: Low **Evidence:** `toCsv` quotes only commas, quotes and newlines (`src/lib/csv.ts:189–199`); a description beginning with `=`, `+`, `-` or `@` from an imported statement is written as a live spreadsheet formula (CSV injection). The JSON export includes the full audit log and free-text review details (`MyData.tsx:28–38`) as an unencrypted file in Downloads, with a note saying "Not real personal information", which would be false on real data. Import has no size limit (`Import.tsx:99`) and `write` does not catch `QuotaExceededError` (`localRepository.ts:67–69`), so a large file fails silently mid-flow. **Recommendation:** Prefix risky leading characters with a single quote in exported cells. Warn before export that the file is unprotected. Cap import size (for example 2 MB) and handle storage errors with a message. **Effort:** S

Threat model for a real deployment

Area	Threat	What must be built
Authentication	Account takeover via SIM swap or shared phone; persona-style switching without a lock	Bank SSO or OTP plus device binding, app PIN or biometric, session timeout, step-up for export and delete
Data in transit	Interception, downgrade	TLS 1.2 or higher, HSTS preload, certificate monitoring, no data in URLs (fix `?deleted=` pattern)
Data at rest	Database or backup leak, stolen phone	Encrypted database and backups with KMS-managed keys, field-level encryption for descriptions and balance, minimal device cache, encrypted if kept
Row-level security	One user reading another's rows through a client SDK	RLS policies keyed to the authenticated user id on every table, deny by default, tests that try cross-user reads, no service-role key in the client
Key management	Key leak in code or CI	Keys in a managed KMS or secret store, rotation schedule, separate keys per environment, no keys in the bundle
Logging	Personal data in app logs, error tracker, analytics	Structured logs with ids only, scrubbers in the error SDK, retention limits, access to logs logged
Consent records	Cannot prove consent	Append-only consent events with notice version and locale (SEC-06)
Retention and deletion	Data kept forever; deletion misses backups and operators	Scheduled purge, backup expiry window, operator purge confirmation, signed receipt (SEC-01, SEC-05)
Breach response	Late or missed section 22 notice	Runbook, eServices SCN1 process, operator clauses (SEC-12)
Operators	Processor misuse or breach	Written contracts under section 21, register, due diligence (SEC-10)
Cross-border	Transfer without basis to cloud, LLM or messaging provider	Section 72 assessment, South African region where possible, de-identified LLM inputs (SEC-10)
Automated decisions	A forecast or loan check that substantially affects the user without recourse	Real staffed human review with logic explanation, as section 71 contemplates
Children and vulnerable users	Minors, coerced users	Age gate, discreet mode, lock (SEC-11)
Client side	XSS stealing tokens or data, stale caches	CSP, no tokens in localStorage, service worker allow-list (SEC-04, SEC-09)

Must exist before real user data is touched

1. A named legal entity as responsible party, a registered Information Officer, a privacy policy and a PAIA manual.
2. A documented lawful basis per purpose (contract for the forecast, consent for messages and analytics) and a section 18 notice with every required item.
3. Authentication with device binding and an app lock; no personal data in localStorage beyond an encrypted, minimal cache.
4. A hosted database with encryption at rest, row-level security tested against cross-user reads, KMS-managed keys and backups with a stated expiry.
5. Description scrubbing on import (ID numbers, account numbers, phone numbers, emails) and an updated "what we use" panel that tells the truth.
6. An enforced retention job and a deletion workflow that reaches backups and operators, with a server-signed receipt.
7. Append-only consent and audit records with no transaction content in them.
8. Security headers and CSP, service worker allow-list, dependency upgrades with no known high or critical issues in production dependencies.
9. Written operator agreements and a section 72 assessment for every provider (database, LLM, WhatsApp or SMS, hosting, error tracking).
10. An incident response plan, tested once, using the Regulator's eServices portal process.
11. An age gate and a vulnerable-user risk assessment.

12. A staffed human review process with a real response time before the screen promises one.
13. An independent security review or penetration test, and a privacy impact assessment signed off by counsel.

Top three risks

1. **A demo statement mistaken for a compliance fact.** Ticks beside encryption and access claims, a retention rule that does not run, and "Never uses account numbers" could be quoted by judges, a bank partner or users as real controls (SEC-01, SEC-02, SEC-03).
2. **Real statements imported into an unprotected browser store.** The import path works with real bank CSVs today; their descriptions carry third party names and references and sit in plaintext localStorage on possibly shared phones, with no lock (SEC-03, SEC-04).
3. **Bolting on Supabase, an LLM or WhatsApp without the privacy plumbing.** The README makes these swaps sound like one-file changes; each adds an operator, likely a cross-border transfer and new caching and logging paths (SEC-09, SEC-10, SEC-12).

Score

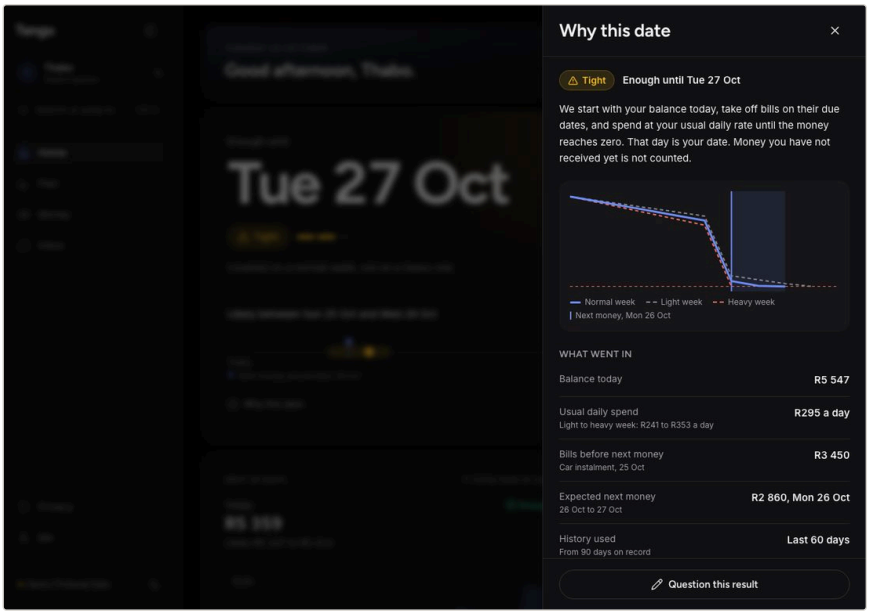
5 out of 10. Strong privacy UX for a demo and nothing leaves the device, but several screens assert controls that do not exist and none of the real-deployment foundations are in place.

Sources

- Protection of Personal Information Act 4 of 2013, official text, gov.za: https://www.gov.za/documents/protection-personal-information-act (accessed 2026-10-20)
- POPIA consolidated text, Law Library: https://lawlibrary.org.za/akn/za/act/2013/4 (accessed 2026-10-20)
- POPIA section 1 definitions (child, operator, consent, personal information): https://popia.co.za/section-1-definitions/ (accessed 2026-10-20)
- POPIA section 11: https://popia.co.za/section-11-consent-justification-and-objection/ (accessed 2026-10-20)
- POPIA section 14: https://popia.co.za/section-14-retention-and-restriction-of-records/ (accessed 2026-10-20)
- POPIA section 18: https://popia.co.za/section-18-notification-to-data-subject-when-collecting-personal-information/ (accessed 2026-10-20)
- POPIA section 19: https://popia.co.za/section-19-security-measures-on-integrity-and-confidentiality-of-personal-information/ (accessed 2026-10-20)
- POPIA section 21: https://popia.co.za/section-21-security-measures-regarding-information-processed-by-operator/ (accessed 2026-10-20)
- POPIA section 35 (and section 34 prohibition): https://popia.co.za/section-35-general-authorisation-concerning-personal-information-of-children/ (accessed 2026-10-20)
- POPIA section 69: https://popia.co.za/section-69-direct-marketing-by-means-of-unsolicited-electronic-communications/ (accessed 2026-10-20)
- POPIA section 71: https://popia.co.za/section-71-automated-decision-making/ (accessed 2026-10-20)
- POPIA section 72: https://popia.co.za/section-72-transfers-of-personal-information-outside-republic/ (accessed 2026-10-20)
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- Information Regulator eServices security compromise reporting: https://eservices.inforegulator.org.za/compromises/default.aspx (accessed 2026-10-20)
- Moonstone, "Updated POPIA compliance: data breach reporting must be online": https://www.moonstone.co.za/updated-popia-compliance-data-breach-reporting-must-be-online/ (accessed 2026-10-20)
- Michalsons, guideline on notification of security compromises: https://www.michalsons.com/blog/guideline-notification-of-security-compromises-to-the-information-regulator/58993 (accessed 2026-10-20)
- GitHub advisory GHSA-wrjc-x8rr-h8h6 (react-router open redirect): https://github.com/advisories/GHSA-wrjc-x8rr-h8h6 (accessed 2026-10-20, via `docs/review/prep/audit.md`)

4.4 Legal and regulatory advisor (South Africa)

Reviewer file: `docs/review/legal.md`. Findings are listed in full in Appendix A.



Why this date drawer, where the only "not financial advice" line sits.

Verdict

This is not legal advice. It is a simulated professional perspective for a hackathon review, based on public sources accessed on 2026-10-20 and the code at commit d114497; it does not replace an opinion from a South African attorney or a ruling from the FSCA, the NCR or the Information Regulator, and it does not claim knowledge of FNB's internal policy or licences. As a demo with fictional data and nothing sent, Tengo carries little legal risk today, and its "Simulated" labels are honest and well placed on the consent, Privacy and Inbox screens. As a real product the picture changes. Most of what Tengo says ("Enough until", "Safe to spend today", "Spend about R250 less a day") is cash flow guidance that does not relate to a financial product, so on the most likely reading it falls outside the FAIS definition of "advice" and needs no FSP licence. Three things break that comfort. First, "What helps most" can tell a user to pay a bill late, and the engine picks the bill by size alone, so it already tells Thabo to pay his car instalment after its due date (`src/engine/help.ts:62`, `docs/review/screens/thabo-home-light-768.png`). Pushing a credit instalment into arrears is the most harmful thing the app says, and the same logic would pick an insurance or funeral premium, which edges into FAIS advice on the "variation" or "termination" of a financial product. Second, the word "safe" is used as an unqualified promise ("Safe to spend", "Safe to lend up to") while the only "not financial advice" line sits at the bottom of a drawer most people will never open. Third, the daily WhatsApp and SMS messages default to on, carry no opt-out instruction or sender details, and would send a person's financial state through a foreign platform. None of this is fatal, and all of it is fixable in wording and engine rules before a pilot. The cheapest compliant route is to ship inside a licensed bank, under its FAIS licence, conduct standards and POPIA programme, rather than as a standalone fintech.

What works

- **Honest simulation labels.** "Simulated privacy controls. Not a compliance certification." appears on the consent screen and the Privacy overview (`src/pages/Consent.tsx:65`, `src/pages/privacy/Overview.tsx:56`, `docs/review/screens/thabo-consent-light-768.png`). The Inbox says "Simulated. No message is sent." (`src/pages/Inbox.tsx:55`, `docs/review/screens/thabo-inbox-dark-390.png`). A judge or a user cannot reasonably mistake these for compliance claims.
- **Purpose limitation stated up front.** The consent notice says data is used "Not for marketing or credit scoring" (`src/i18n/locales/en.json:144`). This aligns with POPIA section 18 notice duties and, if kept true, keeps Tengo away from POPIA section 71 automated decision rules and the section 69 direct marketing rules.
- **Granular consent with the right default for analytics.** Forecast, messages and analytics are separate toggles; analytics is off by default (`src/pages/Consent.tsx:27-28`). Withdrawal pauses the forecast rather than silently continuing (`src/i18n/locales/en.json:167`, `paused` keys).
- **Explainability exists.** The Why drawer shows the inputs, the method and the rules for Enough, Tight and Short in plain language (`src/i18n/locales/en.json:403-439`, `docs/review/screens/thabo-why-drawer-dark-390.png`). That is the kind of logic disclosure POPIA section 71(3) expects where automated decisions apply, and it supports CPA section 22 plain language.
- **Message content minimisation.** Messages carry "Only your date, cover and safe to spend figure" and "Never your balance or transactions" (`src/i18n/locales/en.json:657-658`), which limits exposure on a lock screen.
- **Loan check is about the user's own money.** It answers "what does lending to family do to my cover", it does not arrange, broker or price credit (`src/engine/forecast.ts:346-367`). Tengo is not a credit provider or intermediary on this design.

What to improve

LEGAL-01: "What helps most" tells users to pay credit instalments and insurance premiums late

- **Severity:** High
- **Evidence:** `src/engine/help.ts:55-71` picks the smallest bill before next money that, moved to after it, lifts the state; candidates are sorted by amount only (line 62) with no check of what the bill is. For Thabo it outputs "Pay Car instalment (R3 450) after Tue 27 Oct instead of before." (`docs/review/prep/persona-snapshot.json`, help.billToMove; `docs/review/screens/thabo-home-light-768.png`). Missing a vehicle finance instalment is a default under a credit agreement; it can lead to a section 129 notice, enforcement, repossession by court order and an adverse credit listing (cars.co.za and Moonstone summaries of NCA s129 and s130). Ayanda's recurring bills include a funeral policy and Thabo's include car insurance (`persona-snapshot.json`). Insurance policies are "financial products" under FAIS, and "advice" includes a recommendation "on the variation of any term or condition applying to a financial product ... or on the termination" of it (FAIS s1(1)(d)). A prompt to pay a premium late, which risks a lapse, is arguably such a recommendation. Tengo also cannot know whether the payee will accept a later date; most of these are debit orders.

- **Recommendation:** (1) Classify bills into credit repayments, insurance and other, using description keywords plus a user confirmation step, and never suggest moving credit or insurance payments. (2) Reword the remaining suggestion to a question that keeps the decision with the user and states the risk: "Rent is due before your next money. If your landlord agrees to a later date, you would reach Tight. Ask first, and do not skip debit orders for loans or insurance." (3) Add a one-line inline note on the card: "Ideas, not advice. Check fees and penalties before changing a payment." (4) Add a unit test that a bill described as instalment, loan, finance, insurance, policy or premium is never returned as `billToMove`.
- **Effort:** M

LEGAL-02: "Safe" is an unqualified promise and the disclaimer is hidden

- **Severity:** High
- **Evidence:** "Safe to spend today", "Safe to lend up to", "Use R320" (`src/i18n/locales/en.json:359,523`; `docs/review/screens/thabo-plan-dark-390.png`). "Safe to lend" for Thabo keeps him only at Tight, meaning a heavy week does not last (`src/i18n/locales/en.json:435,524`). Ayanda is Short yet sees "Safe to spend today R310" (`docs/review/screens/ayanda-home-light-390.png`); the figure assumes income lands by the latest likely date (`src/engine/forecast.ts:295-296`). The only "not financial advice" text is `why.notAdvice`, rendered once at the foot of the Why drawer (`src/components/forecast/WhyDrawer.tsx:145`) and below the fold at 390px (`docs/review/screens/thabo-why-drawer-dark-390.png`). It is absent from Home, Plan, What helps most and every message. Where the CPA applies, a supplier must not use "exaggeration, innuendo or ambiguity" about a material fact (CPA s41(1)(b)) and notices must be understandable by an ordinary consumer with average literacy (s22(2)). A disclaimer also cannot turn advice into non-advice; the FAIS test looks at the substance of the recommendation (FAIS s1(1) and s1(3)(a); FIA commentary). **Unclear law:** whether the CPA covers a free app, or a service supplied by a bank that is regulated under financial sector laws, needs an attorney's view; treat CPA standards as the floor either way.
- **Recommendation:** Rename to estimates: "Spend up to about R295 today" with sub-line "Based on past spending. Assumes your money lands by Tue 27 Oct." Rename "Safe to lend up to" to "You could lend up to" with "and stay at Tight. A heavy week would still run short." When Short, hide the spend figure or show "To last until your next money, keep to about R310 a day" in the Short colour. Put a short persistent line under the Home hero, the Plan cards and the Help card: "An estimate from your past patterns. Not financial advice." Keep the drawer text too.
- **Effort:** S

LEGAL-03: Daily messages are opt-out by default and lack POPIA s69 elements

- **Severity:** High (for any real send), Low (for the demo)
- **Evidence:** The messages toggle starts on (`src/pages/Consent.tsx:27`, `docs/review/screens/thabo-consent-light-768.png`). POPIA defines consent as "any voluntary, specific and informed expression of will" (s1); a pre-set toggle bundled behind "Agree and continue" is weak evidence of that. Templates have no sender contact or opt-out route inside the message (`src/i18n/locales/en.json:664-684`, `src/messages/templates.ts:21-35`); "Turn off any time in Privacy" sits only in the app. The tips "A good week to put a little aside" are service guidance today, but the moment a bank partner adds a product nudge ("open a savings pocket") the message becomes direct marketing by electronic communication, which needs opt-in consent or the customer exception, plus sender identity and a cease address in every message (POPIA s69(1), (3), (4)). The Information Regulator's December 2024 Guidance Note on direct marketing says opting out is not consent. Meta requires opt-in that names the business before business-initiated WhatsApp messages, using approved templates. WASPA's Code has a "Reply STOP" requirement for SMS marketing.
- **Recommendation:** Default messages to off and ask on the Inbox with a separate, named opt-in: "Send me a daily Tengo update on WhatsApp at 07:00. I can reply STOP at any time." Record channel, phone, wording version and timestamp in the consent log. Add "Reply STOP to end. Tengo, [contact]" to every SMS and WhatsApp template (SMS budget: the current SMS fits in one segment, so check the 160 character limit in `templates.test.ts`). Keep service messages free of any product promotion; if a partner wants promotional content, collect a separate marketing consent.
- **Effort:** S

LEGAL-04: No responsible party, terms of use or complaints route for a real launch

- **Severity:** Medium
- **Evidence:** The Information Officer is a placeholder (`src/i18n/locales/en.json:239-244`); there is no operator name, address, terms of use or complaints process anywhere in the locale file (grep for "terms" returns nothing). POPIA s18(1) requires the name and address of the responsible party, the purpose, whether supply is voluntary, cross-border plans and the right to complain. ECTA s43(1) requires suppliers transacting electronically to show full name, legal status, physical address, contact details and privacy policy on the site. **Unclear law:** ECTA s43 is framed around goods or services offered by electronic transaction; a free app may fall outside, but a paid tier or a bank-embedded service likely does not.
- **Recommendation:** Before any pilot, publish: operator legal name, registration number and address; terms of use (estimate only, no guarantee, no liability for decisions taken, how to complain); a POPIA privacy notice; and the Information Officer registered with the Information Regulator. If embedded in a bank, use the bank's existing notices and complaints process instead.
- **Effort:** M

LEGAL-05: Regulatory perimeter not documented; FAIS position depends on product links

- **Severity:** Medium
- **Evidence:** FAIS "advice" means a recommendation, guidance or proposal of a financial nature "in respect of" the purchase of, investment in, transactions on, or variation or termination of a financial product (FAIS s1(1)). Financial products include insurance policies and deposits as defined in the Banks Act (FAIS s1(1), paragraph (f) of "financial product"). Daily spend figures and "spend less" suggestions do not name a product, so the **likely interpretation** is that they are not advice. The **alternative interpretation** is that guidance on drawing down a bank deposit, or on setting money aside ("You could set aside up to R280", `src/i18n/locales/en.json:702`), is guidance in respect of a deposit; FAIS s1(4) limits FAIS for short-term bank deposits to what the code of conduct regulates, which reduces but does not remove this risk. Once a bank links Tengo's "set aside" to a named savings product, it is automated advice and falls under the bank's FAIS licence and the FSCA's fit and proper rules for automated advice (key individual who understands the algorithm, ongoing monitoring; Profile summary of FSCA requirements). The COFI Bill was introduced to the National Assembly on 17 April 2026 (Clyde & Co); I could not source its final advice definitions, so treat activity-based conduct regulation and "fair customer outcomes" as the direction of travel, not settled law.
- **Recommendation:** Write a one-page perimeter memo, reviewed by an attorney, covering FAIS, NCA, CPA and POPIA, with a rule: Tengo never names, compares or links to a financial product unless operated inside a licensed FSP. Keep "set aside" generic. If a bank pilots it, the bank's compliance function should classify it as either factual information (FAIS s1(3)(a)) or automated advice, and test it against the bank's Conduct Standard obligations (FSCA Conduct Standard 3 of 2020 requires banks to prioritise fair treatment of financial customers).
- **Effort:** S (memo), L (bank onboarding)

LEGAL-06: "Bring in R380 more" can steer people towards expensive or informal credit

- **Severity:** Medium
- **Evidence:** `help.extra`: "Bring in {{amount}} more before {{date}}." (`src/i18n/locales/en.json:699`; `src/components/forecast/HelpCard.tsx:45`). For a Short user the fastest way to "bring in" money is often a loan. Tengo would not be giving credit advice in the regulated sense (there is no NCA licence category for general money tips, and debt counselling is the specific section 86 process run by a registered natural person), but the wording invites borrowing without a warning.
- **Recommendation:** Reword to "Earn or receive R380 more before Sun 25 Oct, for example an extra shift or money owed to you. Borrowing to cover a gap usually costs more later." Add a link to the NCR's consumer information and, for Short users, a neutral line pointing to free help such as the NCR's debt counselling

information, without recommending any provider.

- **Effort:** S

LEGAL-07: Cross-border transfer and sensitive inference in WhatsApp messages

- **Severity:** Medium
- **Evidence:** WhatsApp is operated by Meta outside South Africa. A message saying "Your money may run out on Fri 23 Oct ... Cover: Short" (`src/i18n/locales/en.json:668–669`) discloses financial distress to a foreign processor and to anyone who sees a shared or locked phone. POPIA s72(1) permits transfer abroad only with adequate protection, consent, contractual necessity or similar grounds. The Privacy centre's cross-border card is simulated.
- **Recommendation:** Name WhatsApp and Meta in the message opt-in and the privacy notice as a cross-border recipient, rely on explicit consent under s72(1)(b), offer SMS as the alternative, and offer a "discreet" template that omits the state word ("Tengo: your update is ready. Open the app.").
- **Effort:** S

LEGAL-08: State label contradicts itself on Home

- **Severity:** Low
- **Evidence:** Thabo's hero reads "Enough until Tue 27 Oct" next to a "Tight" chip, and the 30 day chart header shows "Enough" (`docs/review/screens/thabo-home-light-768.png`; `src/i18n/locales/en.json:346`). "Enough" is used both as a state and as a date label. Under CPA s22(2) an ordinary consumer should understand the notice "without undue effort"; mixed signals about whether one is covered are a misrepresentation risk under s41(1)(b).
- **Recommendation:** Use "Lasts until Tue 27 Oct" for the hero in all non-Short states, and reserve "Enough" for the state chip only.
- **Effort:** S

LEGAL-09: Future bank use could trigger POPIA s71 and purpose creep

- **Severity:** Low (today), Medium (in a bank)
- **Evidence:** POPIA s71 bars decisions with legal consequences based solely on automated profiling of, among other things, creditworthiness and reliability. Tengo's states are not used for any decision today, and the notice says "Not for marketing or credit scoring" (`src/i18n/locales/en.json:144`). A bank partner may want to reuse Enough, Tight and Short for limits, overdraft offers or collections timing.
- **Recommendation:** Put the purpose limit into any partner contract: Tengo outputs may not be used for credit decisions, pricing, collections or marketing without a fresh, separate consent and a s71 assessment. Keep the "Question this result" route and make it reach a human in production.
- **Effort:** S

Top three risks

1. **Harm from late payment suggestions.** A user follows "Pay Car instalment after Tue 27 Oct" and falls into arrears, fees or repossession steps, or lets a funeral or car insurance policy lapse. This is the most likely route to a real complaint, a reputational incident for a bank partner, and an argument that Tengo gave FAIS advice on an insurance product (LEGAL-01).
2. **"Safe" read as a guarantee.** A Tight or Short user spends or lends the "safe" amount, income arrives late, and the user blames the app. The hidden disclaimer will not help, and CPA s41 standards on ambiguity apply where the CPA applies (LEGAL-02, LEGAL-08).
3. **Messaging consent failure at scale.** Default-on daily messages without in-message opt-out, and any later product nudges, breach POPIA s69 and Meta's opt-in rules, risking Information Regulator enforcement and WhatsApp account suspension (LEGAL-03, LEGAL-07).

Score

5/10: honest demo labels and a sensible privacy posture, but the engine can recommend paying credit and insurance late, "safe" wording overpromises, and the messaging consent design would not survive a real launch.

Sources

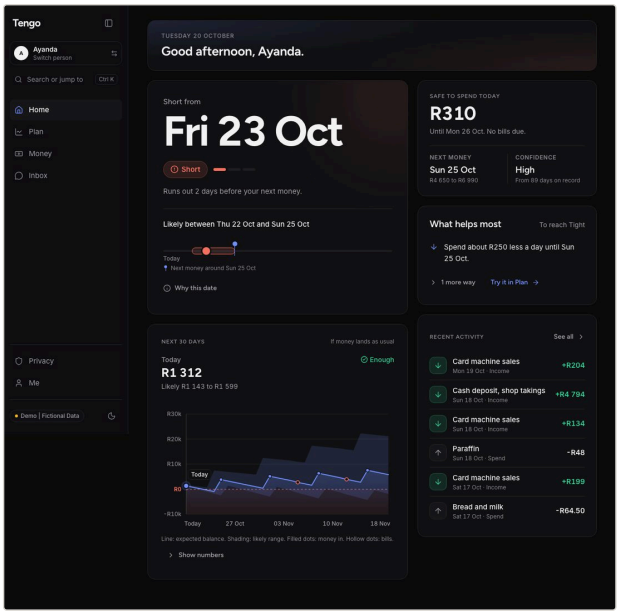
All accessed 2026-10-20.

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- Consumer Protection Act s41: https://www.acts.co.za/consumer-protection/41_false_misleading_or_deceptive_representations
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4.5 Credit and finance risk analyst

Reviewer file: `docs/review/credit-risk.md`. Findings are listed in full in Appendix A.



Ayanda, Home, dark, 1280 px: Short beside "Safe to spend today R310".

Verdict

The engine is small, pure and explainable, and its core choice (count only money already in the account, run it down at a rolling 7 day rate) is sensibly conservative for the headline date. The risk sits elsewhere. The cover state, "Safe to spend" and "Safe to lend up to" all assume the next money will arrive, and nothing checks whether it still does: income that stopped five weeks ago, a three week holiday, or an overdue lump all still produce Enough, high confidence and a lending figure of most of the balance. Every "safe" figure is sized to leave about R0 on payday, and the state looks no further than payday, so a rent payment due the day after shows as Enough with R1 090 "safe to lend". Zero income can show Enough. One-off spends, windfalls and a changed bill amount distort the rates or drop bills entirely. For a product aimed at people whose income is irregular by definition, these are the failure modes that matter most, and the unit tests do not cover any of them. Fixable in days, but not safe for real users as it stands.

What works

- **Conservative headline.** The runway counts only money already in the account and ignores expected income, so the date errs early. Evidence: `src/engine/forecast.ts:196` to `211`, README "How the forecast works" step 4.
- **Rolling 7 day spend instead of single-day medians.** Copes with lumpy stock runs and weekly grocery shops; heavy and light paths use the same basis. Evidence: `src/engine/forecast.ts:59` to `88`, README "Decisions" (Spend paths).
- **Safe to spend uses the latest likely next money date, not the typical one, and nets off bills due before it.** Following Ayanda's R310 every day leaves R17 on 26 Oct rather than running out. Evidence: `src/engine/forecast.ts:291` to `296`; `docs/review/tools/credit-risk/results.md` row P-ayanda.
- **Overdrawn and Short cases fail safe on lending.** A balance of minus R800 gives Short, R0 safe to spend and R0 safe to lend. Evidence: `src/engine/forecast.ts:202`, `358`; results row S03.
- **Overdue interval income never lands in the past.** The overdue rule moves the window to start tomorrow. Evidence: `src/engine/forecast.ts:178` to `182`.
- **Everything is reproducible and inspectable.** One engine powers date, what-if, loan check and help, and the "Why this date" drawer shows the inputs. Engine statement coverage is 97.6 percent. Evidence: `src/engine/help.ts:24` to `26`, `docs/review/prep/coverage.txt`, fact sheet.

What to improve

Stress test: `docs/review/tools/credit-risk/stress.ts` (run with `npx vite-node docs/review/tools/credit-risk/stress.ts`), results in `docs/review/tools/credit-risk/results.md` and `results.csv`. Scenario ids below (S01 to S13, P-persona) refer to those rows. Constructed histories are synthetic: weekly R2 000 pay on Fridays, R60 to R160 daily spend, rent on the 1st unless stated.

RISK-01: Income that has stopped is still treated as coming

Severity: High

Evidence: The weekday rule only needs 60 percent of qualifying deposits on one weekday and has no recency check; the interval rule, when income is overdue, simply restarts the window from tomorrow (`src/engine/forecast.ts:144` to `166`, `178` to `182`). Confidence depends only on history length (`forecast.ts:30` to `34`). Results: - S11, weekly pay stopped 35 days ago (five missed paydays): next money "23 Oct, R2 000", state Enough, confidence high, safe to lend R1 990 of a R2 500 balance. - S09, salon closed for 21 days with no income: Enough, next money Friday, safe to lend R990. - S11b, lumpy deposits, none for 40 days: next money 21 to 23 Oct, Enough, safe to lend R2 110. No unit test covers stopped or paused income (`src/engine/forecast.test.ts` test list).

Recommendation: Count expected paydays that did not happen since the last qualifying deposit. Weekday pattern: if the last qualifying deposit is more than 10 days old, treat the pattern as broken. Interval pattern: if today is past last deposit plus the 80th percentile gap plus one typical gap, treat it as broken. When broken: pattern becomes "paused", the state can be at most Tight, safe to lend is R0, confidence drops to low, and the copy says "Your usual money has not come in since 14 Sep. Tengo is not counting on it until it does." Add unit tests for S09, S11 and S11b.

Effort: S

RISK-02: Every "safe" figure leaves nothing on payday, and the state ignores what is due just after payday

Severity: High

Evidence: Safe to spend spreads the balance to exactly zero at the latest next money (`forecast.ts:294` to `296`): following it leaves R32 (Thabo), R18 (Lerato) and R17 (Ayanda) (rows P-thabo, P-lerato, P-ayanda). Safe to lend is the largest amount that keeps the state (`forecast.ts:357` to `367`); in every Enough scenario the heavy-week path is left at R0 to R10 on the latest payday (column "heavy left after lending", rows S01, S02b, S05, S09, S11, S12b, S13). The state checks only that the money reaches next money (`forecast.ts:217` to `221`), never whether next money covers what follows. S13: rent R3 000 due 25 Oct, the day after the latest next money, weekly pay about R2 000. Engine: Enough, safe to spend R400 a day, safe to lend R1 090. Following either leaves the user unable to pay rent.

Recommendation: (1) Extend the cover check one cycle: Enough also requires balance minus spend to next money, plus the low next-money amount, to cover heavy spend and bills up to the payday after. (2) Keep a buffer in both "safe" figures. **Assumption:** one heavy day of spend or 10 percent of the balance, whichever is larger, is a reasonable starting buffer (plausible range: one heavy day to one heavy week); validate with the backtest. (3) Subtract bills due within 7 days after next money that the low next-money amount cannot cover. (4) Cap safe to lend at the smaller of the state-keeping amount and the balance left after a full heavy week.

Effort: M

RISK-03: No income can show Enough, and Short still talks about "your next money"

Severity: High

Evidence: With no qualifying deposits the engine invents a next money 30 days out (`forecast.ts:107` to `119`, `UNKNOWN_INCOME_DAYS` at `forecast.ts:28`) and then grades cover against it. S02b, 90 days with no income and R9 000 in the account: Enough, safe to lend R2 670. S02, R4 000: Short, and the state card says "Runs out N days before your next money" (`src/i18n/locales/en.json stateCard.short_other`) although there is no next money. S07-5 and S07-7 (5 and 7 days of history, one payday seen) fall into the same 30 day plan. A unit test asserts Enough with no history at all (`src/engine/forecast.test.ts:116` to `122`). A windfall can also trigger this path (RISK-05).

Recommendation: When the pattern is unknown, do not show Enough, Tight or Short. Show "No regular income found" as the state, keep the date as "Money lasts until 2 Dec if nothing comes in", set safe to lend to R0, and let the user enter an expected date and amount. Change the no-history test to expect no state.

Effort: S

RISK-04: "Safe to spend" contradicts the state and is not a safe amount

Severity: High

Evidence: Ayanda is Short from 23 Oct, yet "Safe to spend today R310" (`docs/review/prep/persona-snapshot.json`, walkthrough item 2, `docs/review/screens/ayanda-home-dark-1280.png`). The two numbers use different rules: the state uses the usual spend rate against the typical next money, the figure divides the balance by days to the latest next money (`forecast.ts:217` to `221` versus `291` to `296`). R310 is 50 percent of her usual R623 a day, and about 88 percent of her everyday spend is wholesale stock (R15 137 of the 30 day breakdown), so halving it may cut the income that pays for next week. For Lerato the same label shows R330, 274 percent of her usual R120 (row P-lerato), which reads as permission to spend nearly three times her norm. The SMS repeats it: "money may run out {{date}} (Short). Safe today {{sts}}" (`en.json messages.sms.short`). The "not financial advice" line appears only inside the Why drawer (`src/components/forecast/WhyDrawer.tsx:145`).

Recommendation: Rename and reword by state. Short: "To reach 26 Oct, keep to R310 a day. You usually spend R620." Tight: "Daily limit to reach 27 Oct: R295." Enough: "You can spend up to R330 a day until 25 Oct. You usually spend R120." Never use the word "safe" when the state is Short. Show the usual daily spend beside the figure. Drop "Safe today" from the Short SMS and say "Check Tengo before spending today." Put the estimate line under the figure on Home, not only in the drawer.

Effort: S

RISK-05: One-off spends, windfalls and loans distort rates, income detection and help

Severity: High

Evidence: No outlier handling in spend rates (`forecast.ts:66` to `82`) or income totals (`forecast.ts:128` to `131`). S08, two R9 000 one-offs (car repair, funeral) in 60 days: heavy rate R1 393 a day against R127 normally, state Tight, help says "cut R645 a day". S08b, one R6 000 repair in 20 days of history: band widens to 25 days. S04, a R40 000 stokvel payout lifts "typical monthly income" so the 10 percent threshold (R2 200) exceeds the R2 000 weekly pay: the pattern becomes unknown and a person with steady pay shows Short with "next money" a month away. S04b: a R15 000 loan disbursement counts as income because every positive amount is income (`src/engine/classify.ts:66`).

Recommendation: Winsorise daily spend before the rolling sums (cap each day at, for example, the 95th percentile of daily spend, and list capped items as "one-offs" the user can confirm). Base the significance threshold on median monthly income or exclude the single largest deposit. Tag transfers, reversals and loan payouts (description keywords plus a user toggle) as "not income". Add unit tests for S04, S04b and S08.

Effort: M

RISK-06: Bill detection breaks on a price rise or a skipped month, and arrears are invisible

Severity: Medium

Evidence: All amounts must be within 20 percent of the median and every gap 24 to 36 days (`classify.ts:16` to `22`, `44` to `54`). S10, rent R2 000, R2 000, R2 600: all three become everyday spend, no November rent is projected, the heavy rate triples to R379 and the state is still Enough. S10b, a 15 percent rise: projected rent stays R2 000, the median of the last three (`classify.ts:88`). S05b, a skipped September: rent becomes spend. S05, October rent unpaid: the engine projects the next rent on 1 Nov and treats the missed R2 500 as never owed; state Enough, safe to lend R690. Only monthly bills exist, so weekly stokvel contributions or termly school fees are everyday spend.

Recommendation: Allow one missing cycle (gap up to 2 times the cycle) and judge tolerance against the latest amount, projecting the latest amount when the last two agree. When an expected bill has not appeared 5 days after its usual date, ask "Did you pay Rent this month?" and, if not, add it as due now. Let users mark and edit bills. Support weekly and fortnightly cycles.

Effort: M

RISK-07: Month-end and seasonal spikes are averaged away

Severity: Medium

Evidence: The median of rolling 7 day sums ignores calendar position. S06, R600 a day extra spending from the 25th, today 24 Oct, balance R3 000: usual rate R128, headline "Enough until 16 Nov", safe to spend R425. The known spike costs about R4 260 over 25 to 30 Oct, so the balance is gone around 28 Oct. The README admits no month-end or seasonality modelling (`README.md:319`). Irregular earners in South Africa often face month-end and December pressure (**Assumption**, based on salaried customers and school fees spending around paydays and January; not sourced).

Recommendation: With 60 or more days of history, build a day-of-month uplift (ratio of spend in days 25 to 31 versus the rest) and apply it to the projected days that fall in that window. At minimum, flag "Month end is usually heavier for you" when the ratio exceeds 1.5.

Effort: M

RISK-08: Confidence and the range band do not reflect real uncertainty

Severity: Medium

Evidence: Confidence is history length only (`forecast.ts:30` to `34`). The band is the spread of rolling weekly spend, so with few windows it collapses: S07-5 and S07-7 show a 0 day band at low confidence (one rolling window, so all percentiles are equal and 1.5 times zero is zero); S07-14 shows a 0 day band, an interval pattern inferred from just two paydays, Enough and safe to lend R1 150. S11 shows high confidence after income stopped. The band ignores uncertainty in the income date and amount.

Recommendation: Score confidence on history length, number of qualifying deposits (at least 4), gap regularity, recency of last income and outlier share. Enforce a minimum band (**Assumption:** plus or minus 1 day per week of runway at low confidence, range 0.5 to 2 days). Hide safe to lend at low confidence and say why.

Effort: S

RISK-09: "What helps most" can recommend harmful actions

Severity: Medium

Evidence: The bill to move is any bill before next money whose deferral reaches the target (`src/engine/help.ts:58` to `74`). It names Thabo's R3 450 car instalment (P-thabo, walkthrough item 10) and, with zero income, suggests moving R2 500 rent (S02), where deferral may mean penalties, arrears or eviction risk. The daily cut is the money gap divided by days (`help.ts:56`), so for Ayanda it is R250 a day out of R623, mostly stock, and after a one-off repair it is R645 a day (S08). Copy: "Pay {{name}} ({{amount}}) after {{date}} instead of before." (`en.json help.moveBill`).

Recommendation: Never suggest moving rent, credit instalments, insurance or funeral policies unless the user marks the bill as flexible; when shown, say "Ask {{name}} first if you can pay on {{date}}. Late payments can cost fees or hurt your credit record." Separate business stock from household spend (the spending groups already find "stock") and phrase cuts on household spend only: "Spend about R55 less a day on household costs until 27 Oct." Cap the suggested cut at a share of usual spend (**Assumption:** 25 percent, range 15 to 35) and fall back to "Bring in R1 240 more" beyond that.

Effort: S

RISK-10: Payday morning shows Short

Severity: Medium

Evidence: Next money is "always after today" (`forecast.ts:103`, `106`, `149`), so if the usual payday is today and the payout has not cleared, it is pushed a full week. S12, Tuesday payout due today, R400 in the account: Short, next money 27 Oct, help says "bring in R420". Once paid (S12b) the same person is Enough. This invites alarm and panic borrowing on the morning pay is due.

Recommendation: If today is the expected payday and no qualifying deposit has landed today, set earliest and typical to today and latest to tomorrow, and show "Payday today, not in yet" in place of Short until the latest date passes.

Effort: S

RISK-11: Safe to lend wording implies safety and ignores repayment

Severity: Medium

Evidence: "Safe to lend up to" and "Lending up to this keeps you at {{state}}" (`en.json plan.safeToLend`, `plan.safeToLendHint`, shown at `src/pages/Plan.tsx:204` to `206`), with a one-tap "Use R..." button (`Plan.tsx:236` to `238`). The figure assumes the money never returns but does not say so, and in Tight it keeps only the normal-week path alive, so lending Thabo's R320 leaves his heavy week R695 short (row P-thabo).

Recommendation: Rename to "You could lend up to R320". Hint: "You would still reach payday on a normal week. Only lend what you can manage without getting it back." In Tight add "A heavy week would leave you short." Remove the one-tap fill or show it only in Enough.

Effort: S

RISK-12: Small daily income is ignored in the headline, so states skew low and conflict with the chart

Severity: Low

Evidence: Deposits under 10 percent of monthly income are background income used only in the scrubber (`forecast.ts:135`, `259` to `262`). Ignored per day: Thabo R107, Lerato R170, Ayanda R57 (persona rows). This is part of why Ayanda's headline says Short while the chart for today says Enough (walkthrough item 3).

Recommendation: Add half the background rate to the central path (the heavy path keeps none), or at least show "Not counting about R107 a day in small takings" in the Why drawer. Keep the headline conservative but explain it.

Effort: S

Proposed wording, summary

Where	Now	Proposed
Home figure, Short	Safe to spend today R310	To reach 26 Oct, keep to R310 a day. You usually spend R620.
Home figure, Enough	Safe to spend today R330	You can spend up to R330 a day until 25 Oct. You usually spend R120.
State, no income	Short / Runs out N days before your next money	No regular income found. Money lasts until 2 Dec if nothing comes in.
State, income paused	Enough	Your usual money has not come in since 14 Sep. Tengo is not counting on it.
Payday today	Short	Payday today, not in yet.
Plan	Safe to lend up to R1 090	You could lend up to R1 090. Only lend what you can manage without getting it back.
Help, bill	Pay Car instalment (R3 450) after 27 Oct instead of before.	Ask the lender first if you can pay on 28 Oct. Late payments can cost fees or hurt your credit record.
SMS, Short	Safe today R310.	Check Tengo before spending today.
Under every figure	(drawer only)	An estimate from your past money, not advice.

Top three risks

1. **False Enough when income has stopped or never existed** (RISK-01, RISK-03): a person who lost platform access or closed for the holidays is told they are covered and can lend most of their balance.

2. **Zero-buffer "safe" figures that look only to payday** (RISK-02, RISK-04, RISK-11): following the figures exactly leaves R0 to R30 on payday and can leave rent unpaid the day after, while the word "safe" carries the authority of a bank.
3. **Harmful nudges from distorted inputs** (RISK-05, RISK-09): one-off costs and windfalls swing the state, and the help card can tell someone to defer rent or a car instalment or to halve stock purchases that generate their income.

Score

4/10: clean, explainable and conservative on the date, but the state and "safe" figures fail exactly the irregular-income cases the product is built for, and none of them is tested.

Sources

None. All evidence is from the repository and the stress test in `docs/review/tools/credit-risk/`. Items marked **Assumption** are unsourced judgements with stated ranges.

4.6 Data scientist (forecast validation)

Reviewer file: `docs/review/data-scientist.md`. Findings are listed in full in Appendix A.

> **What this backtest can and cannot show.** All data here is synthetic. The seed generator (`src/data/seed/generate.ts`) and the engine (`src/engine/forecast.ts`) were written by the same author with matching assumptions: income on a fixed weekday or at bounded intervals, bills on a fixed day of the month for a fixed amount, everyday spend drawn from the same distribution every day, no missed payouts, no shocks, no cash the bank cannot see. The engine was in effect designed for this world, so good numbers here are **weak evidence of real-world accuracy**. They show the engine is internally consistent and they locate structural weaknesses (which are real, because they appear even in this friendly world). They do not show that a real irregular earner would see a date within two days of reality. The shipped generator is also thinner than it looks: across 24 cut-off dates it produces only 7 distinct random paths for Thabo, 7 for Lerato and 1 for Ayanda (`results/generator_diversity.csv`). Real-data validation is described at the end of this file.

Verdict

On data from its own generator the engine is reasonably accurate on the headline date (mean absolute error 2.1 days with 90 days of history, bias near zero, beats a naive balance divided by average spend baseline at 3.4 days) and its range band covers the actual date 72 percent of the time overall. That headline hides three structural problems that appear even in this friendly setting: the band is too narrow exactly when the user is close to running out (54 percent coverage when money runs out within 6 days, with 28 percent running out before the "heavy week" edge); the Tight state is followed by running short 24 to 35 percent of the time and Enough is followed by running short 12 percent of the time for the lumpy spaza owner; and with 14 to 28 days of history, which the CSV import accepts, monthly bills are invisible and the engine is worse than the naive baseline and biased optimistic by up to 5.5 days. None of this has been tested on real statements, so every number below is an upper bound on what to expect in the field.

What works

- **Headline date is accurate on its own data.** Seeds, 90 day history, 2 157 observed cases: mean absolute error 2.1 days (95 percent bootstrap interval 1.95 to 2.20), mean signed error 0.0 days, 78 percent within 3 days. Source: `docs/review/backtest/results/summary_headline.csv`, `results/bootstrap_ci.csv`, chart `docs/review/backtest/charts/error_distribution.png`.
- **It beats a naive baseline once it has 60 or more days.** Naive balance divided by 28 day mean outflow: 3.4 days MAE against the engine's 2.1 (90 days) and 2.2 (60 days). The recurring bill detection (`src/engine/classify.ts:16`) is what earns this. Source: `results/summary_by_history.csv`.
- **Overall band coverage is near its implied level and misses are balanced.** The band is the 20th to 80th percentile of weekly spend (`src/engine/forecast.ts:81–82`), which implies roughly 60 percent coverage. Observed: 71.8 percent (interval 69.1 to 74.4), 13.5 percent before the band and 14.7 percent after. Source: `results/summary_headline.csv`, `charts/calibration.png`.
- **Enough is reliable for regular earners.** For Lerato no Enough or Tight case ran short (0 of 595), for Thabo 6 of 450 Enough cases ran short (1.3 percent). Source: `results/cases.csv` via the confusion breakdown in this file.
- **Next money is right for weekday earners.** Thabo and Lerato: next money date exact in 100 percent of cases with 28 or more days of history. Source: `results/summary_by_persona_history_seeds.csv`.
- **The engine is pure, deterministic and fast enough to backtest at scale.** 12 528 forecasts (plus bisection inside `startingBalance`) ran in about 55 seconds with no mocks. This makes continuous accuracy testing cheap. Source: `docs/review/backtest/backtest.ts`, `results/console.txt`.

Method

Data

- **Personas:** thabo (weekly app payout plus daily cash fares), lerato (Saturday takings), ayanda (lumpy cash deposits every 4 to 9 days, irregular stock runs).
- **Cut-offs:** 24 per persona, every 15 days from 2025-10-06 to 2026-09-11, so they cover 12 months, every weekday and month boundaries.
- **Source "shipped":** `generateHistory(persona, cutoff + 30)` returns 90 days ending cutoff + 29. The first 60 days are history, the last 30 the hold-out.
- **Source "seeds":** a copy of the persona rules from `generate.ts` with the seed and length as parameters. The script asserts it reproduces `generateHistory` exactly for the shipped seed on three dates before running. Seeds `tengo:<persona>:backtest<k>:<cutoff>`, k = 1 to 5, 120 days: 90 history, 30 hold-out. This gives 120 independent paths per persona. I added it because the shipped generator repeats itself (see the box above).
- **History length:** truncated to the last 14, 28, 60 (both sources) and 90 days (seeds) before the cut-off, then reclassified with `classifyRows` on the history only, so future bill occurrences cannot leak into classification.
- **Balance at the cut-off (known):** six per case. Three from `startingBalance` with targets enough, tight and short (this uses the engine itself, so it is circular and reported separately). Three engine-independent: 7, 14 and 21 days of the history's mean daily outflow.
- **Case counts:** 9 936 unperturbed cases (1 296 shipped, 8 640 seeds) and 2 592 perturbed cases.

Measures (day index 0 is the cut-off, the engine's "today")

- **Actual runway date (engine definition):** first day on which balance minus cumulative actual outflows (everyday spend and bills, from day 0) is zero or below, ignoring all new income. This matches `zeroDate` (`src/engine/forecast.ts:196`). Censored if not reached within 30 days (0.1 percent of seeds cases).
- **Runway error:** engine central date minus actual runway date, in days. Positive means the forecast was too late, that is too optimistic. MAE is over observed cases.
- **Actual run-out including real income:** first day the real balance (including all real income from day 1) is zero or below. Day 0 income is treated as already in the balance, as the engine does (`src/engine/forecast.ts:124`).
- **Band coverage:** share of decidable cases where the actual runway date is inside [heavy-path date, light-path date] inclusive (`src/engine/forecast.ts:305`). Undecidable when the actual is censored and the band reaches past day 29. Misses are split into "early" (ran out before the heavy-path edge, the dangerous direction) and "late".
- **Next money accuracy:** actual next money is the first day from day 1 whose income is at least the engine's own significance threshold (10 percent of typical monthly income, computed from the history). Reported as MAE of the typical date, percent exact, and percent inside the engine's earliest to latest window.
- **State in hindsight:** let A be the actual next money day and L the balance after outflows on days 0 to A-1 (no income). **Short** if L is zero or below (ran out before next money, the engine's convention that running out on the arrival day counts as covered). **Enough** if L is at least 20 percent of the outflows of that period (it would have survived a heavier stretch). **Tight** otherwise. Sensitivity with 10 and 30 percent is in the CSVs and changes overall accuracy by under 1 point for seeds.
- **Covered accuracy (binary):** engine Enough or Tight against hindsight not Short. **False reassurance:** share of all cases where the engine showed Enough or Tight and the hindsight state was Short.
- **Real shortfall:** the real balance with all real income hit zero before the next money.

- **Baseline:** balance divided by the mean daily outflow (spend and bills) of the last 28 days of history.
- **Uncertainty:** 95 percent bootstrap intervals, resampling the 360 independent seed paths with the six balances of a path kept together (`bootstrap.py`, `results/bootstrap_ci.csv`). Shipped-source numbers have no meaningful interval because they rest on 15 distinct paths.

Backtest summary

Primary row for citing: **seeds, 90 day history, all personas**. "Cov" is band coverage; "early" is the share that ran out before the heavy-week edge. State accuracy is 3-class with the 20 percent buffer rule.

Slice	Cases	MAE days	Bias days	Within 3 days %	Baseline MAE days	Band coverage %	Early miss %	Mean band width days	State accuracy %	Covered accuracy %	False reassurance %	Next money in window %
Seeds, 90 day, all	2 160	**2.1** (1.95 to 2.20)	0.0	78.1	3.4	**71.8** (69.1 to 74.4)	13.5	8.2	**71.9** (70.4 to 73.2)	86.7	7.6 (6.1 to 9.2)	85.8
Seeds, 90 day, Thabo	720	1.3	0.0	92.6	2.3	73.1	15.8	2.5	79.3	90.7	7.2	100.0
Seeds, 90 day, Lerato	720	2.1	1.0	77.1	4.5	83.3	1.4	6.8	73.5	93.1	0.0	100.0
Seeds, 90 day, Ayanda	720	2.9	-0.8	64.4	3.3	59.0	23.3	15.2	62.8	76.2	15.6	57.5
Seeds, 60 day, all	2 160	2.2	0.2	78.2	3.3	71.2	14.8	8.2	71.3	85.7	8.4	83.9
Seeds, 28 day, all	2 160	3.9	1.3	60.6	3.3	71.2	14.1	14.3	64.8	85.6	7.7	81.9
Seeds, 14 day, all	2 160	5.4	1.3	53.9	4.2	59.2	18.2	14.0	64.0	78.8	12.3	63.6
Shipped, 60 day, all	432	1.9	0.3	76.1	3.4	65.7	19.3	14.5	70.6	89.1	7.6	100.0
Shipped, 60 day, Thabo	144	2.2	-1.2	73.6	2.8	61.1	10.4	2.8	80.6	93.8	4.2	100.0
Shipped, 60 day, Lerato	144	2.0	0.4	79.0	4.9	77.6	5.6	7.1	72.2	90.3	2.1	100.0
Shipped, 60 day, Ayanda	144	1.7	1.7	75.7	2.4	58.3	41.7	33.4	59.0	83.3	16.7	100.0
Shipped, 28 day, all	432	4.1	1.5	59.9	3.5	74.0	15.8	30.0	68.5	86.8	8.1	100.0
Shipped, 14 day, all	432	5.1	0.6	62.6	4.7	56.3	25.3	11.0	62.7	80.1	14.1	56.9

Intervals in brackets are 95 percent bootstrap intervals. Lerato at 14 and 28 days: MAE 8.4 and 5.9 days, bias +5.5 and +3.6 days (`results/summary_by_persona_history_seeds.csv`).

Other results

- **By lead time** (seeds, 90 day; `results/summary_by_lead_time.csv`): when money actually ran out on day 0 to 6, band coverage 53.5 percent with 27.5 percent before the heavy edge, state accuracy 49.8 percent, false reassurance 19.0 percent. Day 7 to 13: coverage 80.4 percent. Day 14 to 20: 88.5 percent. Day 21 to 29: 87.1 percent.
- **Shown state against hindsight** (seeds, 90 day; `charts/state_confusion.png`): Enough shown 1 349 times, 93 percent Enough, 2 percent Tight, 4.3 percent Short. Tight shown 434 times, 66 percent Enough, 9 percent Tight, 24.4 percent Short. Short shown 377 times, 67 percent Short, 33 percent covered. By persona, P(Short in hindsight given shown): Thabo Enough 1.3, Tight 32.9; Lerato Enough 0.0, Tight 0.0; Ayanda Enough 11.6, Tight 39.7. Shown Short was right 88.5 percent for Thabo, 60.0 for Lerato, 51.6 for Ayanda.
- **Real money, with real income:** the engine's runway date fell inside the 30 day hold-out in 99.9 percent of seeds cases, but the real balance reached zero within 30 days in only 20.8 percent. When the engine showed Short, the real balance hit zero before the next money in 53.1 percent of cases (background income such as card sales and cash fares carried the rest).
- **At the persona's own demo balance level** (`startingBalance` target enough, seeds 90): Ayanda ran short in hindsight in 29 of 120 Enough cases (24 percent); Thabo 6 of 120; Lerato 0.
- **Perturbations** (shipped, 60 day, 432 cases each; `results/perturbation.csv`, `charts/perturbation.png`): MAE / band coverage / false reassurance went from 1.9 / 65.7 / 7.6 with no change to: noisier daily spend 2.2 / 63.1 / 8.6; new R800 debit order 2.8 / 56.7 / 18.5; prices up 25 percent 2.9 / 52.8 / 19.9; one R2 000 emergency 3.4 / 47.5 / 17.4; income 3 days late 1.9 / 65.7 / 15.0 (next money in window falls from 100 to 0 percent); one payout missed 1.9 / 65.7 / 36.8.

What to improve

DATA-01 Validation is circular and the evidence base is thin

- **Severity:** High
- **Evidence:** The generator encodes the engine's assumptions: monthly bills on a fixed day at a fixed amount (`src/data/seed/generate.ts:18–20`), income on a fixed weekday or a bounded interval (`generate.ts:26, 40, 72–76`), no missed payouts, no shocks. `startingBalance` picks balances with the engine itself (`generate.ts:117–149`). The shipped seed is fixed per persona (`generate.ts:97`) and the random stream only depends on weekday alignment, so 24 cut-offs give 7, 7 and 1 distinct paths (`results/generator_diversity.csv`); Ayanda's demo history is the same on every date. The engine unit tests check rules, not accuracy (`src/engine/forecast.test.ts`). There is no real-data result anywhere in the repository.
- **Recommendation:** Treat every number in this file as an upper bound. Do not quote "2 day accuracy" in the pitch without "on synthetic data". Run the real-data validation below before any pilot claim. Make the generator seed vary with the date so demo data is not one path.
- **Effort:** L

DATA-02 Tight and Enough give false reassurance for lumpy earners

- **Severity:** High
- **Evidence:** Seeds, 90 day: when Tight is shown ("Covered on a normal week, not on a heavy one", `src/i18n/locales/en.json:366`) the user ran out before the next money in 24.4 percent of cases (Ayanda 39.7, Thabo 32.9). Enough ("Covered until your next money, even on a heavy week", `en.json:365`) was followed by running short 11.6 percent of the time for Ayanda and 24 percent at Ayanda's own demo balance level. Overall false reassurance 7.6 percent (6.1 to 9.2). The cause: Enough only stresses spend with a constant 80th percentile weekly rate (`forecast.ts:81`) and income by one day for weekday patterns (`forecast.ts:156`); large irregular outflows such as stock runs are smoothed into the rate.
- **Recommendation:** Set state thresholds from a target error rate, for example "Enough means under 3 percent chance of running short", estimated by simulation over resampled historical weeks and income gaps. Model large irregular outflows as discrete events. Make the Tight copy say plainly that running short is a real possibility. Report P(short) per state as a tracked metric.
- **Effort:** M

DATA-03 The band is too narrow when it matters most and too wide when it does not

- **Severity:** High
- **Evidence:** Seeds, 90 day: when money ran out within 6 days, only 53.5 percent of actual dates fell in the band and 27.5 percent fell before the heavy-week edge; at 14 to 29 days coverage is 87 to 88 percent (`results/summary_by_lead_time.csv`, `charts/calibration.png`). The band applies one constant percentile rate over the whole horizon (`forecast.ts:81–82, 196–213`), which ignores that a single fuel fill or stock run dominates a short horizon while weekly variation averages out over a long one. Ayanda coverage is 59.0 percent with 23.3 percent early misses. The band is labelled "Likely between" (`en.json:353`) with no stated coverage.
- **Recommendation:** Build the band from simulated paths (block bootstrap of historical days or weeks) and take date quantiles, so width grows with horizon correctly and lumpy outflows are represented. Pick and state a nominal coverage (for example 80 percent) and track empirical coverage by lead time.
- **Effort:** M

DATA-04 Short histories hide bills and the engine becomes worse than a naive baseline

- **Severity:** High
- **Evidence:** Import accepts 14 days and only warns under 28 (`src/lib/csv.ts:6–7`). A bill needs 2 occurrences 24 to 36 days apart (`src/engine/classify.ts:16–21`), so no monthly bill can be detected under about 25 days and few under 60. Seeds: MAE 5.4 days at 14 days against a baseline of 4.2, and 3.9 at 28 days against 3.3. Lerato (rent R2 800 on the 1st) is biased optimistic by +5.5 days at 14 days and +3.6 days at 28. The low-confidence widening (`forecast.ts:84–85`) does not fix it: band coverage at 14 days is 59.2 percent.
- **Recommendation:** Treat a single large debit as a provisional monthly bill (or ask the user to confirm "Is this rent? Monthly?"), or fall back to mean total outflow when history is under 60 days. Do not show Enough with under 28 days of history.
- **Effort:** M

DATA-05 Errors grow quickly under shocks the generator never produces

- **Severity:** Medium
- **Evidence:** One payout missed raises false reassurance from 7.6 to 36.8 percent; prices up 25 percent to 19.9; a new R800 debit order to 18.5; one R2 000 emergency to 17.4 with band coverage down to 47.5 percent (`results/perturbation.csv`). These events are ordinary for informal earners. The engine has no notion of income risk beyond a one day delay for weekday patterns (`forecast.ts:156`).
- **Recommendation:** Add an explicit "what if the next payout is missed" check to the state rule for Enough, update rates faster after a regime change (weight recent weeks more), and surface new debit orders as soon as they appear once.
- **Effort:** M

DATA-06 Next money is weak for interval income

- **Severity:** Medium
- **Evidence:** Ayanda (interval pattern, seeds 90): next money inside the earliest to latest window in 57.5 percent of cases, exact 16.7 percent, MAE 2.1 days. Weekday personas score 100 percent only because the generator never misses or shifts a payout; with income 3 days late, window accuracy falls to 0 percent (`results/perturbation.csv`). The weekday window is hard-coded to typical plus one day (`forecast.ts:156`).
- **Recommendation:** Estimate the weekday window from observed lateness in the history, widen the interval window (20th to 80th percentile of 8 to 13 gaps is noisy), and show next money as a range in the copy.
- **Effort:** S

DATA-07 The headline date is a counterfactual that never happens for most users

- **Severity:** Medium
- **Evidence:** The runway ignores all future income (`forecast.ts:196–213`). The date fell within 30 days in 99.9 percent of seeds cases, yet the real balance reached zero in only 20.8 percent. The message copy says "Your money should last until *{{date}}*" and "money lasts until {{date}}" (`en.json:666, 680`), which a user will read as a prediction that their account hits zero then. It also makes real validation harder: the quantity is not observable and must be reconstructed by removing income from statements.
- **Recommendation:** Name it for what it is ("Money you have now lasts until"), and track a second, observable target (real shortfall before next money) as the primary accuracy measure in a pilot.
- **Effort:** S

DATA-08 Short over-warns, especially for Ayanda

- **Severity:** Medium
- **Evidence:** When Short was shown, 33 percent of cases were covered in hindsight and the real balance hit zero before next money in only 53.1 percent (Ayanda: Short right 51.6 percent). Background income (card sales, cash fares) counts toward the scrubber (`forecast.ts:242–245`) but not toward state.
- **Recommendation:** Include a conservative share of background income (for example its 20th percentile) in the state decision, and measure alarm precision as a product metric to avoid alarm fatigue.
- **Effort:** S

DATA-09 No accuracy regression test or field measurement plan in the product

- **Severity:** Low
- **Evidence:** 66 unit tests cover rules (`docs/review/prep/checks.md`); none measures accuracy. Nothing stores past forecasts to compare with outcomes.
- **Recommendation:** Add this harness to CI with thresholds (MAE, coverage by lead time, P(short | Enough)) so engine changes cannot silently regress. In any pilot, store each shown forecast and score it against the statement 30 days later.
- **Effort:** S

DATA-10 Day 0 is treated inconsistently

- **Severity:** Low
- **Evidence:** Income dated today is assumed to be in the balance and next money is searched from tomorrow (`forecast.ts:106, 124`), but a full day of spend is subtracted for today (`forecast.ts:205–209`). If a payout lands today but the balance feed lags, next money skips to next week. In the first backtest run, counting day 0 income as next money moved next money accuracy for weekday personas from about 87 to 100 percent exact.
- **Recommendation:** Use the timestamp of the balance and include same-day credits that are not yet in it; subtract only the rest of today's expected spend.
- **Effort:** S

Real-data validation that is still needed

1. **Data:** consented, de-identified transaction histories of at least 120 days from irregular earners, sourced through a bank partner or user CSV imports with explicit research consent (POPIA review by the security and legal reviewers applies). Stratify by income type (weekly platform payout, weekend trade, lumpy cash deposits, grants, mixed). **Assumption:** a few hundred users per stratum would give intervals of about plus or minus 5 points on rates near 10 percent; the exact number needs a power calculation once the target rates are set.
2. **Rolling-origin evaluation:** many cut-offs per user, as here, with the same metrics, but scored on the observable target (real shortfall before next money, overdraft or declined debit order) as well as the reconstructed runway.
3. **Real-data issues the generator never produces:** pending and reversed transactions, transfers between own accounts, cash withdrawals that hide where money went, cash income never deposited, stokvel contributions and payouts, family transfers, month-end and festive season seasonality, debit orders that move or bounce.
4. **Baselines:** naive mean outflow and the bank's existing balance alert logic. The engine should beat both by a pre-registered margin.
5. **Pre-registered acceptance thresholds** before a pilot, for example P(short | Enough) under 3 percent, band coverage within 10 points of the stated level at every lead time, Short precision over 70 percent. These targets are **Assumptions** for discussion, not sourced standards.
6. **Prospective pilot:** log every forecast shown and score it 30 days later; check whether users who see Tight or Short change behaviour, since that changes the outcome the forecast is scored on.

Top three risks

1. **False reassurance for the most vulnerable pattern.** A lumpy earner shown Enough runs short about 1 time in 9 on the engine's own data (1 in 4 at the demo balance level), and the copy says "even on a heavy week". Real data will be harsher.
2. **Accuracy claims built on circular evidence.** The only accuracy evidence is synthetic data shaped like the engine's assumptions, with one distinct path for Ayanda in the shipped generator. A judge or bank who asks "how accurate is it on real statements" has no answer yet.
3. **Short imports degrade silently.** 14 to 28 days of history, which the import allows, hides rent and other monthly bills and gives an optimistic date by several days, with a low-confidence label that does not convey the direction of the error.

Score

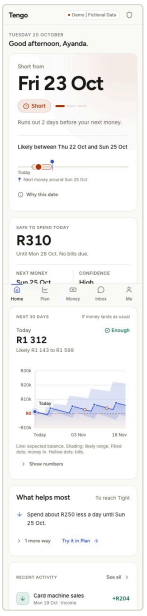
5/10: an accurate and well-built engine on its own synthetic world, with clear structural calibration and state-risk weaknesses and no real-data validation yet.

Sources

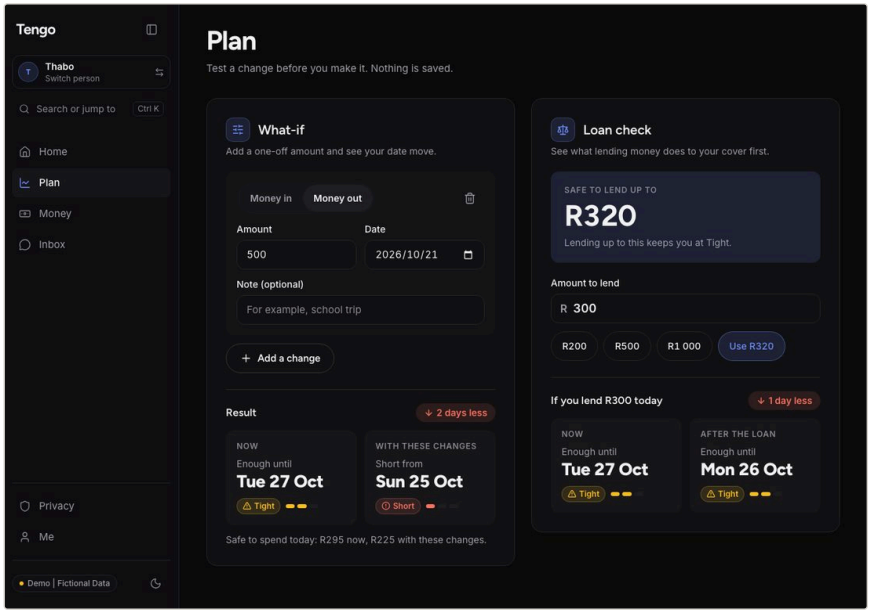
None. All figures come from the backtest in `docs/review/backtest/` run on 2026-10-20 against the code in this repository.

4.7 UI and UX designer

Reviewer file: `docs/review/ux.md`. Findings are listed in full in Appendix A.



Ayanda, Home, light, 390 px.



Plan opens with a pre-filled R500 what-if and R300 loan.

Verdict

Tengo is visually one of the more finished hackathon builds I would expect to see: a calm, consistent card system, a date that genuinely leads the Home screen, state shown by colour, word, icon and bar count together, careful reduced-motion handling and good empty and error states. The weakness is not polish but coherence of the numbers. On a single Home screen a user can meet three different dates (cover date, next money, "safe to spend until") and two verdicts that disagree (the hero says Short, the chart says Enough), and a Short user is told R310 is "safe to spend today" while Plan says R0 is safe to lend. The headline label also says "Enough until" when the state is Tight, which is the first thing a judge sees on the landing page. These are design and copy problems that undercut the one promise the product makes, and they are cheap to fix. In the light theme Tight and Short use almost the same colour. Fix the number story on Home and Plan before adding anything else.

What works

- **Date-first hierarchy is real.** The cover date is the largest element on Home at every width (56px phone, 72 to 88px desktop, `src/components/forecast/DateHero.tsx:22`), with state chip and plain sentence directly beneath. Evidence: `docs/review/screens/ux/ayanda-home-fold-light-390.png`, `thabo-home-light-1280.png`.
- **State is never colour alone.** `StateChip` renders icon plus word plus colour (`src/components/StateChip.tsx:13`), and `CoverBars` adds a 3, 2, 1 bar count. The 30 day chart caption explains line, shading and dot types in words, and "Show numbers" gives a table alternative (`src/components/forecast/TimelineScrubber.tsx:214`).
- **Motion is restrained and respects reduced motion.** Global override at `src/styles/index.css:123`, count-up skips under reduced motion (`src/lib/useCountUp.ts:8`), scroll reveal never hides above-the-fold content (`src/components/Reveal.tsx:15`).

- **Empty and error states are designed, not default.** Forecast paused has a clear cause and two actions (`docs/review/screens/ux/paused-dark-390.png`); a bad CSV shows a specific fix (`docs/review/screens/ux/thabo-import-badfile-light-1280.png`); a negative what-if amount gets inline help with an example (`docs/review/screens/ux/ayanda-plan-negative-light-1280.png`); render crashes have a branded fallback (`src/components/ErrorBoundary.tsx:22`); Home with no transactions routes to import (`src/pages/Home.tsx:97`).
- **Honest demo labelling is consistent.** "Demo | Fictional Data" chip on every app screen, simulated banners on Inbox, Privacy and Consent, a placeholder warning on the Information Officer card. Evidence: `thabo-inbox-dark-1280.png`, `ayanda-privacy-dark-1280.png`.
- **Dark and light parity in layout is complete.** Every page has identical structure across themes; tokens are defined once per theme (`src/styles/index.css:13` to `48`) and 116 of 116 contrast pairs pass (`docs/review/prep/checks.md`, `npm run contrast`). axe passes on all audited pages (`docs/review/prep/playwright.txt`, 221 passed).
- **Desktop layout earns its width.** At 1280 Home becomes a two-column dashboard with hero and chart left, stats, help and activity right (`src/pages/Home.tsx:116`); Plan puts What-if and Loan check side by side. The sidebar, persona switcher and Ctrl K menu are well placed.
- **The Why drawer explains the date in plain words** with the inputs listed (balance, usual daily spend, bills) and a route to question the result (`ayanda-why-drawer-light-390.png`).
- **Microcopy is mostly short and human**: "Covered on a normal week, not on a heavy one.", "Runs out 2 days before your next money.", "Test a change before you make it. Nothing is saved." (`src/i18n/locales/en.json stateCard`).

What to improve

UX-01. Short state sits next to "R310 safe to spend today" and "R0 safe to lend" - **Severity:** High - **Evidence:** Ayanda Home: hero "Short from Fri 23 Oct", "Runs out 2 days before your next money", then card "SAFE TO SPEND TODAY R310. Until Mon 26 Oct. No bills due." (`docs/review/screens/ux/ayanda-home-fold-light-390.png`). Her usual daily spend is R623 (`ayanda-why-drawer-light-390.png`), and What helps most says "Spend about R250 less a day", which implies about R373, not R310. On Plan the same person sees "SAFE TO LEND UP TO R0. You are already short" (`ayanda-plan-light-390.png`). The figures come from different rules (`docs/review/prep/fact-sheet.md`, "Safe to spend today") and the screen does not reconcile them. A Short user can read R310 as permission. - **Recommendation:** When state is Short, relabel the card ("Daily limit to reach your next money" or "To get through to Sun 25 Oct, keep to R310 a day"), show it beside usual spend ("You usually spend R623"), and use the same base in What helps most. Make Plan's lend wording consistent with that figure. - **Effort:** S

UX-02. The chart says Enough on the same screen where the hero says Short or Tight - **Severity:** High - **Evidence:** Ayanda hero "Short" while the Next 30 days readout shows a green "Enough" for Today (`ayanda-home-light-390.png`, `ayanda-home-dark-1280.png`); Thabo hero "Tight", chart "Enough" (`thabo-home-dark-768.png`). The readout uses `day.state` from a projection that includes expected income (`src/components/forecast/TimelineScrubber.tsx:110`). The qualifier "If money lands as usual" is 12px subtle text in the card corner (`TimelineScrubber.tsx:95`). - **Recommendation:** Do not show a state word on the chart readout for today. Show balance only, or label it "If your money lands as usual: Enough" in the readout itself. Better, show both paths on the chart (with and without the next income) so the hero's logic is visible. - **Effort:** S

UX-03. Headline label says "Enough until" when the state is Tight - **Severity:** High - **Evidence:** `src/components/forecast/DateHero.tsx:9` uses `hero.enoughUntil` for every non-Short state. Thabo reads "Enough until Tue 27 Oct" beside an amber "Tight" chip (`thabo-home-light-1280.png`), and this is the product mock on the landing page hero (`landing-light-1280.png`, `landing-dark-390.png`). Plan repeats it in the Now and After cards (`thabo-plan-dark-1280.png`). The Inbox message already words it better: "should last until Tue 27 Oct on a normal week. Cover: Tight." (`thabo-inbox-dark-1280.png`). - **Recommendation:** Use a state-specific label: Enough "Covered until", Tight "Lasts until (on a normal week)", Short "Short from". Keep the Inbox phrasing as the model. - **Effort:** S

UX-04. Three different dates for "next money" on one screen - **Severity:** Medium - **Evidence:** Ayanda: hero "Short from Fri 23 Oct", band "Next money around Sun 25 Oct", safe to spend "Until Mon 26 Oct" (`ux/ayanda-home-fold-light-390.png`). Lerato: next money Sat 24 Oct, safe to spend "Until Sun 25 Oct" (`lerato-home-dark-390.png`). Cause: Stats uses `nextMoney.latest` for the safe-to-spend date and `nextMoney.typical` for the Next money figure (`src/pages/Home.tsx:56`, `74`). - **Recommendation:** Pick one date the user sees (typical) and explain the buffer in words ("We plan to Mon 26 Oct in case your money is a day late"), or show next money as a range everywhere. - **Effort:** S

UX-05. Tight and Short are nearly the same colour in the light theme - **Severity:** Medium - **Evidence:** Light tokens `--tight: 155 71 8` and `--short: 169 57 10` (`src/styles/index.css:47`, `48`): contrast ratio between them 1.01 (computed with the WCAG luminance formula), both read as rust brown. Compare picker chips (`picker-light-390.png`) and Me (`lerato-me-light-390.png`). In dark they are clearly amber versus coral. Icon, word and bar count still differ, so this is not a WCAG failure, but the colour signal for the most important distinction is lost in light. - **Recommendation:** Move light Tight towards a true amber or ochre (for example a darker yellow-brown with higher luminance on the soft fill) and keep Short as a red. Re-run the contrast script for text on soft fills. - **Effort:** S

UX-06. Plan opens with a pre-filled R500 what-if and a R300 loan - **Severity:** Medium - **Evidence:** `src/pages/Plan.tsx:59` (`amount: '500'`) and `:187` (`useState('300')`). The page loads already showing "2 days less" and "Short from Sun 25 Oct" for Thabo (`thabo-plan-dark-1280.png`) before he has done anything. For Ayanda the What-if says "1 day less" on arrival (`ayanda-plan-light-390.png`). A first-time user may think something has happened to their money. On Thabo's Loan check "Use R320" is styled as selected while the field holds 300. - **Recommendation:** Start empty with a neutral result ("Add an amount to see your date move"), and keep the quick chips. If a demo needs a filled state, use the placeholder or a "Try R500" chip. - **Effort:** S

UX-07. The range band has no visible legend for its main marker, and the headline date can sit at the edge of "likely" - **Severity:** Medium - **Evidence:** `src/components/forecast/RangeBand.tsx:29` to `48` draws band, a state-coloured dot (the headline date) and an accent pin (next money); only the pin has a legend line (`:49`). The whole graphic is `aria-hidden`. Lerato's date 1 Nov sits on the far right edge of "Likely between Mon 26 Oct and Sun 01 Nov" (`lerato-home-dark-390.png`), so "likely" and the headline look inconsistent to a lay reader. - **Recommendation:** Label the dot ("Your date") and the band ("Likely range") directly on the track, and explain in the Why drawer why the headline can sit at one end of the range. - **Effort:** S

UX-08. The 30 day chart scale is driven by the optimistic band, so the balance line hugs zero - **Severity:** Medium - **Evidence:** `TimelineScrubber.tsx:59`, `60`: y-axis max is the highest optimistic value. Ayanda's axis runs R30k to minus R10k while her balance is about R1.3k, so the expected line is squashed into the bottom fifth of the plot and the shading dips well below R0 (`ayanda-home-light-390.png`). Below-zero shading on a "money lasts" chart looks like debt. - **Recommendation:** Scale to the central line plus a capped band (for example 80th percentile), clip shading at R0 or tint it differently, and mark the cover date on the chart so it ties back to the hero. - **Effort:** M

UX-09. Daily messages is switched on by default at consent - **Severity:** Medium - **Evidence:** `src/pages/Consent.tsx:27 useState(true)`; the toggle shows "On" before the user chooses (`thabo-consent-dark-390.png`). Analytics is off by default, so the pattern is inconsistent, and a pre-ticked optional choice reads as a nudge. - **Recommendation:** Default Daily messages to off and offer it as a positive choice after the first Home view ("Want this date each morning on WhatsApp?"). - **Effort:** S

UX-10. Import step 2 is dense and its summary tiles are ambiguous - **Severity:** Low - **Evidence:** After "Use sample statement" the phone shows six column dropdowns, a preview table without an amount column, then tiles "DAYS 45, INCOME 14, SPEND 37, BILL 2" (`thabo-import-sample-light-390.png`). "INCOME 14" reads as R14, not 14 transactions. The column mapping is shown even when every guess is right. - **Recommendation:** Collapse the mapping behind "Columns look wrong?" when detection succeeds, add the amount column to the preview, and relabel tiles ("14 money-in rows"). - **Effort:** S

UX-11. Microcopy and format inconsistencies - **Severity:** Low - **Evidence:** "Pay Car instalment (R3 450) after Tue 27 Oct instead of before." capitalises the bill mid-sentence (`thabo-home-light-1280.png`, `en.json:697` inserts the raw description). "To reach Tight" frames Tight as a goal (`ayanda-home-dark-1280.png`, `en.json:704`). Lerato sees "safe to spend R330" and "You could set aside up to R280 today" on the same screen (`lerato-home-dark-390.png`). Dates show "Sun 01 Nov" and "Sept" from the en-ZA locale, while the native date input shows "2026/10/21" (`thabo-plan-dark-1280.png`). - **Recommendation:** Lower-case bill names in

sentences, reword the help goal ("To avoid running short" or "To be covered on a heavy week"), explain the R330 versus R280 difference or show one figure, and format days without a leading zero. - **Effort:** S

UX-12. Phone first fold and tablet layout - **Severity:** Low - **Evidence:** At 390×844 the hero fills the fold and the safe-to-spend card is cut by the bottom bar, with Next money and Confidence hidden (`ux/ayanda-home-fold-light-390.png`). At 768 the layout is the phone stack stretched wide, with a greeting banner taking prime space above the hero (`thabo-home-dark-768.png`). Privacy is only an unlabelled shield icon in the top bar on phones (`src/layouts/nav.ts`, any `*-390.png`). - **Recommendation:** Tighten hero padding on phones so the safe-to-spend figure is visible without scrolling; fold the greeting into a single line on tablet; use the two-column layout from 768; give the shield a visible label or put Privacy under Me. - **Effort:** M

UX-13. Persona lists truncate key context and show a fixed starting state - **Severity:** Low - **Evidence:** Me truncates "E-hailing driver, Tsh…" and "Spaza shop owner, …" (`lerato-me-light-390.png`). Picker and Me chips show the designed `startState`, not the current one (`src/pages/Picker.tsx`, `docs/review/prep/walkthrough.md` item 6). - **Recommendation:** Allow two lines for the role, and compute the chip from the current forecast or drop it. - **Effort:** S

Nielsen heuristic evaluation

Scale: 1 (serious problems) to 5 (excellent).

#	Heuristic	Score	Evidence
1	Visibility of system status	4	Demo chip on every screen, "Last changed" on consent, Plan result badges ("2 days less", "No change"), import row counts and skipped-row warning (`thabo-import-sample-light-390.png`). Count-up and re-keyed date animate on change (`DateHero.tsx:19`).
2	Match with the real world	3	Plain language ("next money", "a heavy week") fits the personas. Undercut by "Enough until" for Tight (UX-03), "To reach Tight" (UX-11) and "Income 14" counts (UX-10).
3	User control and freedom	4	Remove what-if rows, Not now on consent, Resume forecast, Cancel on import, reset demo data, persona switch everywhere. Pre-filled Plan values add changes the user did not ask for (UX-06).
4	Consistency and standards	2	Same card system and chips across pages, but the numbers are inconsistent: three next-money dates (UX-04), chart state versus hero state (UX-02), R310 versus R0 versus R373 (UX-01), R330 versus R280 (UX-11), Tight and Short colours in light (UX-05).
5	Error prevention	3	Import refuses under 14 days and asks for column mapping; delete needs a checkbox. Consent pre-enables messages (UX-09); a Short user is shown a spendable figure without a caveat (UX-01).
6	Recognition rather than recall	4	Labelled bottom nav and sidebar, chart caption explains marks, Why drawer lists inputs. Range band marker unlabelled (UX-07); Privacy is an unlabelled icon on phones (UX-12).
7	Flexibility and efficiency	4	Ctrl or Cmd K menu, quick amount chips, filters and search on Money, chart scrubbing plus table view.
8	Aesthetic and minimalist design	4	Clean hierarchy and generous spacing. Import step 2 and the five-card consent screen are long on phones (`thabo-consent-dark-390.png`).
9	Help users recognise, diagnose and recover from errors	4	Specific inline messages with examples (`ux/ayanda-plan-negative-light-1280.png`, `ux/thabo-import-badfile-light-1280.png`), paused state with actions, crash fallback.
10	Help and documentation	4	Why this date drawer, "How the forecast works" in Me, "What the file needs" on import. No help for what Tight means beyond one sentence, and nothing explains why safe to spend differs from usual spend.

Average 3.6 of 5. The low point is consistency of the numbers, which is the heuristic that matters most for a money app.

Cognitive walkthrough: first two minutes

Thabo (e-hailing driver, Tight). - 0 to 20 s: Landing shows "Money you can plan around" and a mock with his own figures. He must understand that the date is when money runs out. Risk: the mock says "Enough until Tue 27 Oct" next to "Tight" (UX-03); he may read Enough as the verdict. - 20 to 50 s: Picker, then a consent page several screens tall on a phone (`thabo-consent-dark-390.png`). He must understand he can agree. Risk: Daily messages already on (UX-09); he may scroll past without reading. - 50 to 90 s: Home. He sees Tue 27 Oct, Tight, "Covered on a normal week, not on a heavy one." This works. Next he sees R295 safe to spend "Until Tue 27 Oct" but Next money Mon 26 Oct (UX-04), then a chart saying Enough (UX-02). He must work out which one to believe. - 90 to 120 s: What helps most says "Pay Car instalment after Tue 27 Oct". He may not know if this is allowed, and the bill shows as due Sun 25 Oct on Money (`thabo-money-light-1280.png`). Tapping Try it in Plan lands on a page already showing "2 days less" and Short (UX-06); he may think his position just got worse.

Lerato (hairdresser, Enough). - 0 to 40 s: Picker shows her as Enough with three green bars; consent as above. - 40 to 90 s: Home shows Enough until Sun 01 Nov, green chip, "Covered until your next money, even on a heavy week." Clear and reassuring. Risk: next money is Sat 24 Oct, so she may wonder why the date is 1 Nov and why the dot sits at the very end of "Likely between Mon 26 Oct and Sun 01 Nov" (UX-07). - 90 to 120 s: R330 safe to spend "Until Sun 25 Oct", then "You could set aside up to R280 today" (UX-11). She must decide if she can spend R330 and save R280, or only one of them. Nothing on the screen answers that.

Ayanda (spaza owner, Short). - 0 to 40 s: Picker shows Short with one bar. Consent as above. - 40 to 70 s: Home shows Short from Fri 23 Oct and "Runs out 2 days before your next money." This is the clearest moment in the product: the warning is unmissable. - 70 to 100 s: Directly below: "SAFE TO SPEND TODAY R310. No bills due." (UX-01). She must understand this is a ceiling to stretch money to Mon 26 Oct, not a free amount, while she usually spends R623. Likely misreading: "I am short, but I can still spend R310 today." - 100 to 120 s: Scrolling, the chart shows a green "Enough" for today (UX-02) with an axis up to R30k (UX-08). She may conclude the warning was a false alarm. If she opens Plan, the loan card says R0 is safe to lend and the what-if already shows "1 day less" from a R500 she did not enter (UX-06).

Top three risks

1. **False reassurance for the Short user.** R310 safe to spend plus a green Enough on the chart can override the Short headline for exactly the person who most needs the warning (UX-01, UX-02).
2. **Loss of trust from contradictory numbers.** Three next-money dates and two state verdicts on one screen make the forecast look unreliable to judges and users even where the engine is correct (UX-03, UX-04).
3. **First impression on the landing page.** The hero mock shows "Enough until" beside "Tight", so the core concept is muddled in the first five seconds of the demo (UX-03).

Score

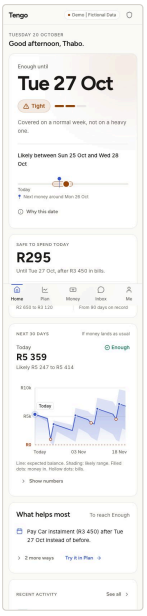
6 out of 10: polished, accessible and consistent visual design, held back by Home and Plan figures that contradict each other in ways a stressed user could act on.

Sources

None.

4.8 Accessibility and inclusive design specialist

Reviewer file: `docs/review/accessibility.md`. Findings are listed in full in Appendix A.



Thabo, Home, light, 390 px: fixed tab bar at the bottom.

Verdict

Tengo has a stronger accessibility base than most hackathon builds. axe-core 4.14 with the WCAG 2.2 AA tags finds one violation across 15 pages and 4 overlays in both themes at 390 and 1280 px. Nothing scrolls sideways at 320 CSS px, every target is at least 24 px, reduced motion is honoured in CSS and in JavaScript, and the drawer and command menu follow the expected keyboard patterns. The tools miss the problems that matter for this audience. When a user picks Tshivenda, Sepedi or any other language, the page declares that language while every word is still English (WCAG 3.1.1). The chart slider tells a screen reader "Enough" for a person whose headline says "Short", and it hides the bills and income that explain each jump. Keyboard focus goes behind the fixed bottom tab bar on phones (2.4.11). The app shows a blank page offline after a first visit, which is when loadshedding is most likely to catch a new user. Beyond WCAG, the Home screen asks people who may be anxious about numbers to reconcile about 20 figures that sometimes contradict each other. Nothing protects private money data on a shared phone. This is a careful build with a few high-impact gaps, all fixable in days rather than weeks.

What works

- **Automated scan is nearly clean.** My own axe run (tags wcag2a, wcag2aa, wcag21a, wcag21aa, wcag22aa) over 15 routes plus the Why drawer, chart table, command menu and a Tshivenda session, in dark and light at 390 and 1280 px, found 1 violation: the chart table scroll region, A11Y-08. Evidence: `docs/review/tools/accessibility/results/axe-{dark,light}-{390,1280}.json`, script `docs/review/tools/accessibility/axe.a11y.ts`. The "incomplete" contrast items are SVG axis text and labels over translucent bars. The shipped token pairs pass (`docs/review/prep/checks.md`: 116 of 116 pairs).
- **Reflow at 320 CSS px (1.4.10).** On 12 routes at 320 × 640, `scrollWidth` equals the viewport width and no text runs off screen. Evidence: `results/manual.json` key `reflow320`. 320 px is the 400 percent equivalent of a 1280 px window. Nothing blocks zoom: no `maximum-scale` or `user-scalable` in `index.html`, and the manifest sets no orientation lock (1.3.4).
- **Target size (2.5.8).** Every control measured at 320 px is at least 24 × 24, and most are 44 px tall. The smallest are the 28 px privacy switch labels, which also have a 44 px input. Evidence: `results/manual.json` `reflow320.*.small`.
- **Dragging alternative (2.5.7).** The chart is a native `input type=range` laid over the plot (`src/components/forecast/TimelineScrubber.tsx:190`). A single tap sets the day (my tap at 60 percent set value 16, `results/manual.json` `chart.tapSetsValue`). Arrow keys, Home and End work, and "Show numbers" gives a full table. `touch-action: pan-y` (`src/styles/index.css:198`) keeps vertical page scroll working on phones.
- **Overlays behave.** The Why drawer moves focus to Close, keeps Tab inside the dialog, closes on Escape and returns focus to "Why this date" (`results/manual.json` `drawer`, `src/components/Drawer.tsx:24`). The command menu follows the combobox pattern. Typing "plan" then Enter navigates and moves focus to `main`. A theme command is announced in the live region ("Light theme on"). Evidence: `results/command-menu.json`.
- **Screen reader basics are thoughtful.** The date hero reads "Short from Friday 23 October" as an h1, with the short visual date hidden (`src/components/forecast/DateHero.tsx:16`). Amounts read "money in, R204". The count-up figure has a static sr-only copy (`src/pages/Home.tsx:61`). State is shown with colour, word and icon (`src/components/StateChip.tsx:12`, 1.4.1). Route changes move focus to `#main` and announce the page title (`src/components/A11y.tsx:44`). There is a skip link.
- **Reduced motion.** A global CSS override (`src/styles/index.css:123`), and `useCountUp` and `Reveal` both check `prefers-reduced-motion` (`src/lib/useCountUp.ts:8`, `src/components/Reveal.tsx:14`).
- **Language plumbing is ready.** Each language option carries its own `lang` (`src/components/LanguageSelector.tsx:46`, 3.1.2). Inbox messages carry `lang` (`src/pages/Inbox.tsx:117`). Headings switch to Inter for Tshivenda because Figtree lacks ḓ ḽ ṅ ṋ ṱ (`src/styles/index.css:84`).
- **Plain English.** A rough Flesch-Kincaid estimate over all 565 English strings gives about grade 3. Plan and Help sections score under grade 2 and the Why explanation about grade 2. This is a heuristic on short UI strings and tends to underestimate difficulty. Form errors give an example, for instance "Enter an amount above zero, for example 250." (`src/i18n/locales/en.json:532`, 3.3.3).
- **Data cost is small.** A cold Home load transfers 149 KB (`docs/review/prep/perf-home.json`). At ICASA's reported average of about R71 for a 1 GB prepaid bundle, that is roughly R0.01 per cold load.

What to improve

A11Y-01 Page language says Tshivenda or Sepedi while the text is English

- **Severity:** High
- **Evidence:** `src/components/Preferences.tsx:12` sets `document.documentElement.lang = language` for any choice, but `src/i18n/status.json` marks all 10 non-English languages "pending" and they render English. Measured: with `ve`, `nso` or `zu` selected, `html lang` is `ve`, `nso` or `zu` and the h1 is "Short from Friday 23 October" (`results/manual.json` key `lang`). This fails WCAG 3.1.1 Language of Page because the declared language is wrong. A screen reader with a matching voice installed, or a translate prompt, will treat English as Tshivenda. axe cannot detect this, and the Tshivenda scan passes.
- **Recommendation:** Set `lang` to the language actually rendered. Use `en` while a locale is pending, and when a locale is partly translated, put `lang="en"` on any string that falls back to English. Add a Playwright check that the `html lang` value matches the status.
- **Effort:** S

A11Y-02 Blank page offline after a first visit (loadshedding)

- **Severity:** High
- **Evidence:** `public/sw.js` precaches only `/`, the manifest and two icons. The hashed JS and CSS are cached only when fetched while the worker is in control, and on a first visit they load before it takes control. Test: one visit, worker "activated", cache holds `/`, manifest, icons and two fonts but no JS or CSS. After going offline and reloading, the page has no h1 and an empty body (`results/offline-single-visit.json`, script `offline.a11y.ts`). The repo's own test passes only because it reloads once while online before going offline (`tests/flows.spec.ts:131`). There is no offline message.
- **Recommendation:** Precache the build output at install (Workbox or vite-plugin-pwa injecting the asset manifest). Add a static offline fallback that says in plain words that the app needs one connection to finish setting up. Change the test to go offline straight after the first visit.
- **Effort:** S

A11Y-03 The chart slider tells screen reader users something different from the headline and hides what moves the balance

- **Severity:** High
- **Evidence:** For Ayanda, the h1 reads "Short from Friday 23 October", while the slider's first value reads "Tuesday 20 October, about R1 312, Enough" (`results/manual.json chart.ayanda`). The "If money lands as usual" caveat sits only in the section description, not in the value text (`src/i18n/locales/en.json:389`). Income and bill notes for the chosen day are `aria-hidden` (`TimelineScrubber.tsx:118`) and absent from `valueText` (`TimelineScrubber.tsx:79`). So Thabo's balance falls to R971 on Sunday and rises to R3 643 on Monday with no spoken reason. Sighted users see "Expected money +R2 860". The likely range is also visual only. Information shown visually is not exposed programmatically (1.3.1). For a blind user the unexplained "Enough" is a false reassurance risk.
- **Recommendation:** Put events and assumptions into `aria-valuetext`, for example ", about , Enough if money lands as usual. due, minus . Likely to ." Give the slider a name that includes "if money lands as usual". Consider showing the day state as "Covered if paid" so the chart and the headline do not use the same word for different rules.
- **Effort:** S

A11Y-04 Keyboard focus hidden behind the fixed bottom tab bar

- **Severity:** Medium
- **Evidence:** At 390 × 844, tabbing to "Try it in Plan" on Home and to "Edit Paraffin" on Money leaves the focused control under the tab bar. The bar's top edge is at 783 px, the controls sit at 789 to 844 px, and `elementFromPoint` returns the tab bar (`results/focus-obscured-app-390.png`, `results/focus-obscured-app-money-390.png`). Seven transaction edit buttons on Money were fully hidden (`results/manual.json obscured-/app/money`). This fails WCAG 2.4.11 Focus Not Obscured (Minimum). The bar is defined in `src/layouts/AppLayout.tsx:166` and the codebase has no `scroll-padding`.
- **Recommendation:** Add `html { scroll-padding-top: 64px; scroll-padding-bottom: calc(76px + env(safe-area-inset-bottom)); }` below the lg breakpoint. Add a Playwright check that the focused element is not covered.
- **Effort:** S

A11Y-05 Too many numbers, some of them contradicting each other, for people anxious about money

- **Severity:** Medium
- **Evidence:** Ayanda's first phone screen shows "Short", "Runs out 2 days before your next money", "Safe to spend today R310" (her usual daily spend is R623), and a chart readout of "R1 312 … Enough". The chart axis runs from minus R10k to R30k, and the range on the last chart day is "Likely -R1 882 to R21 173" (`docs/review/screens/ayanda-home-light-390.png`, `results/manual.json chart.ayanda.main`). Above "What helps most" I count about 20 distinct figures. Not a WCAG failure. For readers with low numeracy, a red "Short" next to a confident "R310 safe" invites them to keep the number they would rather believe.
- **Recommendation:** Lead with one sentence and one action, for example "Your money runs out on Friday, 2 days before Sunday's money. Spend about R250 less a day to close the gap." Move the range, confidence and 30 day chart behind "See the detail". Reconcile safe to spend with Short, or label it "only if you also cut R250 a day". Test with 5 users scoring low on a numeracy screener.
- **Effort:** M

A11Y-06 No privacy on a shared phone

- **Severity:** Medium
- **Evidence:** Full transaction history, balance, consent and audit log sit in plain localStorage, 16.5 KB for one persona (`results/manual.json storage`, `src/data/localRepository.ts:15`). There is no PIN, no "hide amounts" mode and no sign-out, and the session reopens straight to Home. The planned messages contain the date and cover state (`src/i18n/locales/en.json:666`), which would appear on a lock screen. **Assumption:** phone sharing within households is common among lower-income South African users. I did not find a current sourced figure, so the range is wide: 10 to 40 percent of the target segment.
- **Recommendation:** Before any pilot, add an optional 4 digit app lock or device biometric, a one tap "hide amounts" toggle, and a discreet message mode with no amounts or state on the lock screen ("Tengo has an update"). Make sure the lock uses no cognitive test and allows paste and password managers (3.3.8 Accessible Authentication).
- **Effort:** M

A11Y-07 Form fields and chart shading below 3:1 non-text contrast

- **Severity:** Medium
- **Evidence:** `.field` uses `border-line` on `bg-surface`. The border token gives 1.24:1 against surface in dark and 1.26:1 in light, and the field fill is the same colour as the card (`src/styles/index.css:16,40`, my calculation from the tokens). The border is the only cue that identifies the input, so it needs 3:1 under 1.4.11. The chart's likely-range band is 1.25:1 to 1.27:1 against the card. The band is supplementary because the table and readout carry the range, so it matters less. The repo's contrast check covers text pairs only (`docs/review/prep/checks.md`).
- **Recommendation:** Give inputs a `border-strong` that reaches 3:1, for example around rgb(110 110 118) in dark and rgb(140 136 128) in light, and verify with the contrast script. Add non-text pairs (field border, switch track, focus ring) to `scripts/contrast-check.mjs`.
- **Effort:** S

A11Y-08 Chart data table cannot be scrolled by keyboard

- **Severity:** Medium
- **Evidence:** axe `scrollable-region-focusable` (serious, WCAG 2.1.1) on the `max-h-[360px] overflow-auto` table container in all four runs (`results/axe-*-*.json`, overlay `chart-table`; `TimelineScrubber.tsx:224`). The table has no focusable content, so keyboard users cannot reach days below the fold. The table is also wider than a 390 px phone (`min-w-[460px]`).
- **Recommendation:** Add `tabIndex={0}`, `role="region"` and an `aria-label` to the container, as the drawer body already does (`Drawer.tsx:79`). On phones, show 2 columns (day, expected) and put the range on a second line.
- **Effort:** S

A11Y-09 The multilingual, low-literacy audience gets English only

- **Severity:** Medium
- **Evidence:** 10 of 11 locales are English copies (`src/i18n/status.json`). PIRLS 2021 found 81 percent of South African Grade 4 learners could not read for meaning in any language (Daily Maverick, source below), so reading load matters for adults in the target segment too. Most strings are short, but some privacy and Me strings estimate at grade 9 to 11 (`me.sim.translations`, `privacy.breach.contact.body`, `privacy.complaints`). Nothing is available as audio or voice.
- **Recommendation:** Before claiming multilingual reach, ship 2 or 3 languages that match the personas' areas (for example isiZulu, Sepedi, Setswana), each reviewed by a first-language speaker. Add a "read this to me" button on the date hero using the device's speech synthesis where a voice exists. Rewrite the privacy strings to grade 5 or lower.
- **Effort:** L

A11Y-10 Import errors are not explained or announced

- **Severity:** Low
- **Evidence:** "This file could not be read as a CSV." (`en.json:582`) gives no next step (3.3.3), and the `Notice` that shows it has no live role (`src/components/Feedback.tsx:54`, `src/pages/Import.tsx:131`), so screen reader users are not told (4.1.3). An invalid balance sets `aria-invalid` with no error text and only disables the Save button (`src/pages/Import.tsx:248`, 3.3.1).
- **Recommendation:** Say what to do ("Save the statement from your banking app as CSV, or try the sample"). Render errors with `role="alert"`. Add a visible error under the balance field and keep the button enabled, so pressing it explains the problem.
- **Effort:** S

A11Y-11 Negative amounts may be read without "minus"

- **Severity:** Low
- **Evidence:** Spoken text uses a hyphen-minus, for example "Friday 23 October, about -R384, Short" (`results/manual.json` `chart.ayanda.slider`, `src/lib/format.ts:13`). **Assumption:** at default punctuation levels some screen readers and TalkBack voices skip or misread a leading hyphen. I did not test with a real screen reader, so this needs checking on NVDA and TalkBack.
- **Recommendation:** In `aria-valuetext` and sr-only copies, use words: "R384 below zero" or "minus R384".
- **Effort:** S

A11Y-12 Truncated text loses meaning at small widths and high zoom

- **Severity:** Low
- **Evidence:** Transaction names and persona roles are cut with an ellipsis: "Cash deposit, shop takin..." at 390 px, "E-hailing ..." and "Spaza s..." at 320 px (`docs/review/screens/ayanda-home-light-390.png`, `results/home-320.png` from the Me page). Screen readers get the full text, so this is not a WCAG failure, but sighted low-literacy users lose the words they would recognise.
- **Recommendation:** Allow 2 lines (`line-clamp-2`) instead of `truncate` for descriptions and roles.
- **Effort:** S

A11Y-13 Command menu is keyboard-only and its groups are not announced

- **Severity:** Low
- **Evidence:** Below the lg breakpoint there is no visible control to open it (`results/manual.json` `commandMenu.mobileOpenButtonVisible: 0`, opener only in the sidebar, `AppLayout.tsx:89`). Group headings are `<p role="presentation">` inside `role="presentation"` items (`CommandPalette.tsx:197`), so "Pages", "People" and "Language" are not announced. Ctrl+K also opens it from inside a text field (`results/command-menu.json` `opensFromTextField: 1`), which overrides a common editing shortcut in some browsers.
- **Recommendation:** Use `role="group"` with `aria-labelledby` for each section. Ignore the shortcut when focus is in an editable field, unless the field is empty. On phones this is optional, because everything is reachable in Me.
- **Effort:** S

A11Y-14 Low-end Android parses translations that are not there yet

- **Severity:** Low
- **Evidence:** One 696 KB JS file (142 KB gzip), of which 337 KB is locale JSON for 11 languages, 10 of them English copies (`docs/review/prep/audit.md`). Cold hero at 4x CPU slowdown is 1.5 s (`perf-home.json`), which is acceptable. **Assumption:** entry-level Android phones with 1 to 2 GB RAM and older CPUs are 2 to 4 times slower than the 4x throttle, which would put the cold hero at 2 to 4 s.
- **Recommendation:** Load only the active locale on demand, and cache it in the service worker. Test once on a real entry-level Android Go device.
- **Effort:** S

A11Y-15 Before a pilot: authentication, help and accessibility testing with people

- **Severity:** Low
- **Evidence:** There is no login yet, so 3.3.8 Accessible Authentication does not apply today. The privacy pages have a "Question this result" help route, but nothing appears in the same place on every page (3.2.6 Consistent Help applies once a real help channel exists). No test with screen reader users or low-literacy users is recorded (`docs/review/prep/walkthrough.md` is a developer walkthrough).
- **Recommendation:** Design the bank-grade login to allow biometrics, paste and password managers, with no puzzles. Put one "Help" link in the same place on every page. Run a session with 3 TalkBack users and 5 low-literacy users before a pilot.
- **Effort:** M

Top three risks

1. **Misleading reassurance for screen reader users.** The slider says "Enough" while the headline says "Short", and the bills that explain the drops are hidden from assistive tech (A11Y-03). A blind user could act on the wrong signal.

2. **The app fails exactly when power and network drop.** After one visit it shows a blank page offline (A11Y-02), and the first visit is the moment a new user is most likely to lose signal during loadshedding.
3. **Multilingual claim versus reality.** Picking a language changes `lang` but not the words (A11Y-01, A11Y-09). A judge or user who picks isiZulu sees English, and assistive tech gets the wrong language, which undercuts the inclusion pitch.

Score

6/10: a carefully built accessible base with near-clean automated results, held back by a language-of-page failure, offline failure on first visit, contradictory screen reader output on the chart, focus hidden under the tab bar, and a Home screen too dense in numbers for its audience.

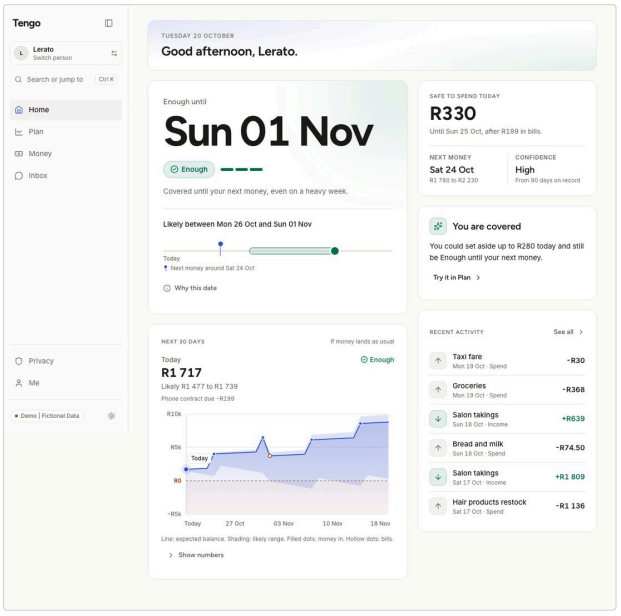
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Tools and raw outputs: `docs/review/tools/accessibility/` (`axe.a11y.ts`, `manual.a11y.ts`, `cmd.a11y.ts`, `obscured.a11y.ts`, `offline.a11y.ts`, `a11y.config.ts`) and `docs/review/tools/accessibility/results/`. Run with `npx playwright test -c docs/review/tools/accessibility/a11y.config.ts`. The first command menu reading in `manual.json` was a timing artefact. `command-menu.json` is the corrected run.

4.9 Behavioural economist and financial inclusion researcher

Reviewer file: `docs/review/behavioural.md`. Findings are listed in full in Appendix A.



Lerato, Home, light, 1280 px.

Verdict

Tengo asks the right question for people paid in lumps, and its date-first answer, daily figure and loan check are a credible attempt to turn a balance into a decision. The behavioural case is weaker than the pitch suggests. No published study shows that a runway date changes behaviour more than a balance does, so that claim is a hypothesis to test, not a finding. Three frames also carry real risk of harm. "Safe to spend today" can license spending well above a person's normal rate and drop them a level (Lerato: R330 a day against a usual R120, with rent of R2 800 due a week later and not mentioned). "Safe to lend up to" turns a ceiling into a default with a one-tap button and ignores the money already promised to safe to spend and to "set aside". The model only sees one bank account, while the people it targets live largely in cash, stokvels, family transfers and several wallets, none of which appear in the seed data or the engine. The product is promising as a prompt for reflection and not yet safe as a guide to action. It needs the frames fixed and a small field test with real earners in Tshwane, Mamelodi and Tembisa before anyone claims it builds money confidence.

What works

- **The question matches lived reality.** Irregular, lumpy income is the defining feature of low-income household finance in South Africa (Collins, Morduch, Rutherford and Ruthven, Portfolios of the Poor, which included South African financial diaries). The tagline "is my money enough until the next money" names that problem in plain words. Evidence: `README.md` lines 1 to 9, `src/i18n/locales/en.json:346–368`.
- **Daily framing is well chosen for adoption.** Framing a recurring amount per day rather than per month quadrupled enrolment in a savings programme and removed the income gap in take-up (Hershfield, Shu and Benartzi 2020). "Safe to spend today" uses the same unit as the day-to-day decisions of a driver, hairdresser or shop owner. Evidence: `src/engine/forecast.ts` (`safeToSpendToday`), `docs/review/screens/thabo-home-light-390.png`.
- **Salience of a specific upcoming cost.** Reminders work best when they name a specific future lumpy expense (Karlan, McConnell, Mullainathan and Zinman 2016), and salience of overdraft fees reduced overdrafts, most among less financially literate customers (Stango and Zinman 2014). Tengo's date, "Bills before next money" and the Why this date drawer make future costs salient in the same way. Evidence: `docs/review/screens/ayanda-why-drawer-light-390.png`.
- **Transparent, conservative headline.** The headline counts only money already in the account and says so ("Money you have not received yet is not counted"), and Enough requires the heavy week path to reach the latest likely next money. This avoids the most common reassurance error of income forecasts. Evidence: `README.md:302`, `src/engine/forecast.ts:217`, `ayanda-why-drawer-light-390.png`.
- **Short is paired with an action, not only a warning.** Short states show "What helps most" with a concrete daily cut and a "Try it in Plan" link, and the daily message ends with "Check Plan in Tengo to see what helps most". Action-first messages reduce the chance that bad news leads to avoidance (see BEH-06). Evidence: `ayanda-home-light-390.png`, `src/i18n/locales/en.json:674–676`.
- **The loan check exists at all.** Lending and giving to family is a core behaviour that most budgeting tools ignore. Showing "days less cover" before lending gives people a concrete, shareable reason for a smaller amount. Evidence: `docs/review/screens/thabo-plan-dark-1280.png`.
- **Privacy-aware message content.** The daily message promises "Never your balance or transactions", which matters on shared phones. Evidence: `docs/review/screens/ayanda-inbox-light-390.png`.

What to improve

BEH-01 "Safe to spend today" can license overspending and hides the next big bill

Severity: High **Evidence:** Safe to spend is balance minus bills before the latest next money, divided by days to it (`src/engine/forecast.ts`, `perDay`). It is not tied to the person's spending rate or to the cover state. For Lerato it is R330 while her usual daily spend is R120 and her heavy week rate is R276 (`docs/review/prep/persona-snapshot.json`). My probe with the real engine (`docs/review/tools/behavioural/frames.ts`) shows that if Lerato spends the R330 every day until next money, she drops from Enough to Tight and her date moves from 1 Nov to 25 Oct. Her rent of R2 800 on 1 Nov is the reason her headline date is 1 Nov, yet neither the safe to spend card nor "You are covered" mentions it (`lerato-home-dark-1280.png`). Daily framing makes amounts feel small and acceptable (Atlas and Bartels, periodic pricing; Hershfield, Shu and Benartzi), so a daily figure framed as "safe" will be read as an allowance. A preliminary conference poster also suggests budget feedback can speed up spending towards the limit (Huang and Liang, UBC poster, treat as preliminary). **Recommendation:** Make safe to spend the largest daily amount that keeps the current state (the same search already used for safe to lend), never above the heavy week rate when Enough. Rename it "Spend up to R... a day to stay Enough". Always show the next large bill after next money ("Rent R2 800 on Sun 1 Nov"). Test the wording in the user study (questions C4 and C5). **Effort:** S

BEH-02 The model sees one bank account; the target users live in cash, wallets and several accounts

Severity: High **Evidence:** There is no cash ledger, no manual income or spend entry, no detection of transfers between own accounts, and no handling of cash withdrawals (grep of `src/engine` and `src/lib` finds no `ATM`, `transfer` or `withdraw` rule except the CSV debit column name at `src/lib/csv.ts:50`). A withdrawal to cash is counted as everyday spend, a transfer to an e-wallet or savings pocket as spend, and money moved in from another own account as income that can become "next money". FinScope SA Consumer 2023 reports that 94 percent withdraw cash monthly, and the SARB Payments Study found cash is 56 percent of payments by volume. The seed data understates this: Thabo's "Cash fares" are deposited into the bank most days (`src/data/seed/generate.ts`), which few drivers would do in practice (Assumption: from the FinScope cash withdrawal pattern, most daily cash is spent from the pocket). The persona copy says "Cash fares most days" and "Cash takings come in lumps" (`src/i18n/locales/en.json:122,130`), but the engine only sees cash that reaches the bank. **Recommendation:** Add a "Cash in hand" figure the user updates in one tap, a quick "money in" and "money out" entry, a transfer rule (same amount out and in within two days, or a user tag "my other account or wallet"), and a prompt on large deposits: "Is this your income, a stokvel payout, or your own money moved?". Show "Based on your bank account only" under the date until cash is added. **Effort:** M

BEH-03 "Safe to lend up to" anchors a default and double counts the same money

Severity: High **Evidence:** The loan check shows a large "Safe to lend up to R320" and a one-tap "Use R320" chip (`src/pages/Plan.tsx:204–238`, `thabo-plan-dark-1280.png`). A displayed maximum with a button to apply it acts as an anchor and a default. Safe to lend is computed independently of safe to spend and of "set aside": Lerato is told she can set aside R280 (Home) and lend R280 (Plan), and spend R330 today, all from the same buffer (`src/engine/help.ts:49` reuses `safeToLend` for `canSetAside`). My probe shows that lending R280 and spending R330 on the same day drops Lerato to Tight, and Thabo lending R320 plus spending R295 moves his date a day earlier (`frames.ts` output). The loan is modelled as gone, with no option for an expected repayment date or partial repayment, and the Short message "no amount is safe to lend" is a moral verdict that people facing family obligations will override (Mangoma and Wilson-Prangley 2019 find transfers to family are substantial and usually monthly; Magubane 2017 finds black tax can crowd out savings). Positively, a number from a neutral tool can be a socially acceptable reason to give less, which is worth keeping. **Recommendation:** Reframe as "Give or lend without dropping a level", show one shared budget ("Today you can spend R… or lend R…, not both"), remove the "Use" chip or relabel it "Show me the most", add "Expect it back on" with a date that feeds the what-if, and in Short show the cost in days ("Lending R200 brings your date forward 1 day") instead of "no amount is safe". Add a short, optional script the user can share ("I can do R… this week, more after the 25th"). **Effort:** M

BEH-04 Conflicting signals undermine comprehension and trust

Severity: Medium **Evidence:** (1) Thabo's hero reads "Enough until Tue 27 Oct" with a "Tight" chip (`src/components/forecast/DateHero.tsx:9` uses "Enough until" for every non-Short state; `thabo-home-light-390.png`). (2) Ayanda's hero says Short while the 30 day chart says Enough for today (`ayanda-home-light-390.png`, `walkthrough.md` item 3). (3) Ayanda gets two daily targets: safe to spend R310 and "Spend about R250 less a day", which implies about R373 against her R623 usual rate, with different end dates (26 Oct and 25 Oct) (`frames.ts` output, `src/engine/help.ts`). (4) Lerato's headline date equals the end of her range (band 26 Oct to 1 Nov, headline 1 Nov), so the "likely" date sits at the edge rather than the middle (`persona-snapshot.json`). People under financial strain have less attention to reconcile mixed messages (Mani, Mullainathan, Shafir and Zhao 2013; note Carvalho, Meier and Wang 2016 found no cognitive effect around payday, so the size of this effect is contested). **Recommendation:** Use a state-neutral label ("Money lasts until"). Give one daily number per state and make "What helps most" express its cut as that number. Reconcile the chart and hero with a visible line ("If your money lands as usual on Sun 25 Oct, you are back to Enough"). Test comprehension of the range with question C3. **Effort:** S

BEH-05 Business inputs are treated as household spending to cut

Severity: Medium **Evidence:** Ayanda's top spend is stock (R15 137 over 90 days) and "What helps most" tells her to spend R250 less a day (`persona-snapshot.json`, `ayanda-home-light-390.png`). For a spaza owner, cutting stock cuts future takings. Thabo's fuel is the input to his income. The bill-move suggestion for Thabo names his car instalment (`walkthrough.md` item 10), which for an e-hailing driver is the asset that earns the income. A Short state that appears every restock cycle risks alert fatigue for a shop owner whose business normally runs that way (Assumption: wholesale runs are periodic, so Short may recur every cycle). **Recommendation:** Add a "business stock and fuel" group, keep it out of the daily cut, and phrase help for business users as timing ("Restock after Sun 25 Oct if you can"). Exclude vehicle finance and funeral policies from bill-move suggestions, or warn that moving them may cost fees or cover. **Effort:** M

BEH-06 Daily bad news risks avoidance, anxiety and exposure on shared phones

Severity: Medium **Evidence:** The Inbox design sends a message every morning at 07:00 that reads "Your money may run out on Fri 23 Oct… Cover: Short" (`ayanda-inbox-light-390.png`, `src/i18n/locales/en.json:666–682`). Investors look at accounts less after bad news (Karlsson, Loewenstein and Seppi 2009, the ostrich effect), so a daily Short message may lead to muting rather than action. Nudges can also have side effects: payment reminders cut late fees but raised overdraft fees for some users (Medina 2021). The message preview on a lock screen shows state and safe to spend, which on a shared or family phone can invite requests (if not Short) or shame (if Short). The privacy promise covers balance and transactions only. **Recommendation:** Default to messages on change of state or two days before a Short date, not daily. Offer a neutral lock-screen preview ("Your Tengo update is ready"). Lead with the next action and a specific upcoming cost, which Karlan et al. found more effective. Track opt-out rates by state in any pilot. **Effort:** S

BEH-07 Stokvels, burial societies, family transfers and informal credit are absent

Severity: Medium **Evidence:** The seed descriptions contain no stokvel, burial society, family transfer, mashonisa or airtime advance entries (`src/data/seed/generate.ts`). FinScope SA Consumer 2023 reports informal savings at 30 percent of adults (10 percent in 2003), burial societies at 29 percent and borrowing to buy food at 40 percent. NASASA estimates are reported as over 11 million stokvel members (Cape Argus 2024). A rotating stokvel payout is a large, infrequent deposit that can pass the 10 percent "next money" threshold and distort the income pattern (`src/engine/forecast.ts:24`, Assumption: payout sizes of one to several months of contributions). Contributions are only treated as bills if the amount and gap fit the monthly rule (`src/engine/classify.ts:16`), so weekly contributions become everyday spend. **Recommendation:** Add a stokvel contribution and payout, a family transfer and an informal loan repayment to at least one persona. Let users mark a deposit "one-off or stokvel payout" so it is excluded from the next money pattern, and mark a regular contribution as a commitment rather than spend. **Effort:** M

BEH-08 The date-first behavioural claim is unproven

Severity: Low **Evidence:** The pitch line "A balance tells you what you have. Tengo tells you until when" (`README.md:5`) implies a date changes behaviour more than a balance. I found no published experiment comparing a runway date with a plain balance. The nearest evidence is adjacent (daily framing, salience of costs, reminders). Monzo shows a "left to spend" estimate, which suggests the idea is in use but not that it works better (Monzo help page). **Recommendation:** Present the date-first claim as a hypothesis in the pitch and run the comparison in the user study (task T2). **Effort:** S

BEH-09 Account-level view can be gamed or used strategically

Severity: Low **Evidence:** The state depends only on the balance of the imported account (`src/engine/forecast.ts`, `forecast`). A user can move money to another account or wallet to make the state look Short, for example to show a relative, or keep income elsewhere so the app looks better than reality. Assumption: strategic display is plausible where requests from family are frequent; there is no data on how often it would occur. **Recommendation:** Treat this as a design input, not a flaw to police. Ask in testing how people would show or hide the screen (question F3), and keep the shared message free of amounts if users prefer. **Effort:** S

User-testing plan (5 to 8 participants, hackathon budget)

Aim. Check comprehension of the date, range, states and safe to spend; whether a date prompts different decisions than a balance; and how the loan check fits family obligations. This is qualitative, not a statistical test.

Participants. Six, with two optional extras if time allows. - Two e-hailing or minibus taxi drivers (Tshwane), two hairdressers or salon workers (Mamelodi), two spaza or informal traders (Tembisa). - Screener: aged 18 or over; income varies week to week; uses a smartphone with WhatsApp; has a bank account or wallet; at least three of six include someone in a stokvel and someone who regularly sends money to family. Mix of gender and age. Exclude anyone who works in banking or fintech. - Recruitment: through a community organisation, a salon or taxi rank contact, or a stokvel chairperson, with a short spoken explanation in the person's preferred language. Avoid friends of the team.

Setting. At the participant's place of work or a nearby quiet spot, outside peak trading hours, 40 to 45 minutes. Use a mid-range Android phone provided by the team with the demo installed and offline-capable, so no participant data is used and no data cost falls on them. One facilitator, one note taker who speaks the participant's language. Use the fictional personas only. Do not ask for real balances.

Consent. Plain-language verbal and written consent in the participant's language: what the study is, that the app is a demo with fictional data, that nothing they say is linked to their bank, that they can stop at any time and still receive the incentive, whether audio is recorded (ask separately), and that notes are anonymised and deleted after the hackathon. No photos of faces.

Incentive. Cash or a data or airtime voucher of about R200 to R300 per person plus transport if they travel (Assumption: a common range for short community interviews in Gauteng; confirm locally). Total for six: about R1 500 to R2 000, plus about R500 for refreshments and printing (Assumption).

Tasks and exact questions.

Warm-up (5 min) - W1. "How do you know if your money will last until the next money comes in?" - W2. "Which accounts, wallets, cash or groups does your money go through in a normal week?" - W3. "When family asks to borrow, how do you decide how much?"

T1. First look at Home (Ayanda's or Lerato's screen, 2 minutes, no help) - C1. "What does this date mean?" Success: says the money already there runs out around that day, before new money is counted. - C2. "What does Short (or Enough) mean here?" Success: links the word to next money, not to a general judgement. - C3. "The screen says likely between Thu 22 and Sun 25 Oct. What does that mean for you?" Success: understands it is uncertain, could be earlier. - C4. "What does Safe to spend today R330 mean? If you spent R330 every day this week, what would happen?" Success: says it might not be safe every day or that rent comes later. Failure: treats it as a daily allowance. - C5. "Is anything big coming after your next money?" (Lerato) Success: spots rent without opening drawers. Expected failure today, by design of BEH-01.

T2. Date versus balance (paper cards, order alternated between participants) - Card A shows only "Balance R1 877". Card B shows "Short from Fri 23 Oct, safe to spend R310". Same situation. - Q: "A friend asks you for R300 today. With this card, what would you do?" then "Would the other card change your answer? Why?" Record the amount each card leads to. Success for the hypothesis: Card B leads to a smaller or delayed amount for at least four of six, with reasons tied to the date.

T3. Loan check (Thabo's Plan screen) - Q: "Your sister needs R500 for school shoes. Use the app to decide." Observe whether they tap "Use R320". - Q: "What does Safe to lend up to R320 mean to you?" "Would you show this screen to your sister?" "If it said no amount is safe, what would you do?" - Success: understands it as a limit, not a recommendation; finds a way to give less without shame.

T4. What helps most (Ayanda) - Q: "It says spend about R250 less a day. What would you cut?" Success: does not cut stock or fuel; if they do, record as BEH-05 evidence.

T5. Message (Inbox) - Q: "If this came every morning at 7, how would you feel? Would you keep it on?" "Who else sees your phone?"

Wrap-up (5 min) - F1. "Would you trust this date? What would make you trust it more?" - F2. "What does it miss about how your money really works?" (probe cash, stokvel, family, airtime) - F3. "Would you ever want this to look better or worse than it is? When?" - F4. "On a scale of 1 to 5, how sure do you feel about your money until next week after using this?" Ask the same item before T1 to see direction only.

What counts as success or failure. - Pass to proceed: at least five of six answer C1 and C2 correctly, at least four of six answer C3 correctly, and no one describes Safe to spend as a daily allowance in C4. - Red flags that stop a pilot until fixed: two or more treat safe to spend or safe to lend as permission; two or more cut stock or fuel in T4; two or more say they would turn messages off in T5 because they feel judged. - Record cash, stokvel, family and wallet flows in W2 for each participant to size BEH-02 and BEH-07.

Running it on a hackathon budget. Two team members, one afternoon per area across two days, six sessions. Paper cards and the offline demo, no recording equipment beyond a phone if consent is given. Write up a one-page table of answers per question and three quotes per finding within 24 hours. Total cost about R2 000 to R2 500 plus transport (Assumption).

Top three risks

1. **Harmful action on a "safe" figure.** People treat safe to spend or safe to lend as permission and drop a level or miss rent, while the app said "safe" (BEH-01, BEH-03).
2. **False precision from a partial view.** Cash, wallets, stokvels and family transfers make the single-account date wrong in ways the user cannot see, which damages trust after the first miss (BEH-02, BEH-07).
3. **Disengagement of the people who need it most.** Daily Short messages and advice that cuts business inputs push the most stretched users to mute or abandon the app (BEH-05, BEH-06).

Score

5 out of 10. The question, daily framing and transparency are strong, but the "safe" frames can mislead, the single-account view misses how the personas actually handle money, and the core behavioural claim is untested.

Sources

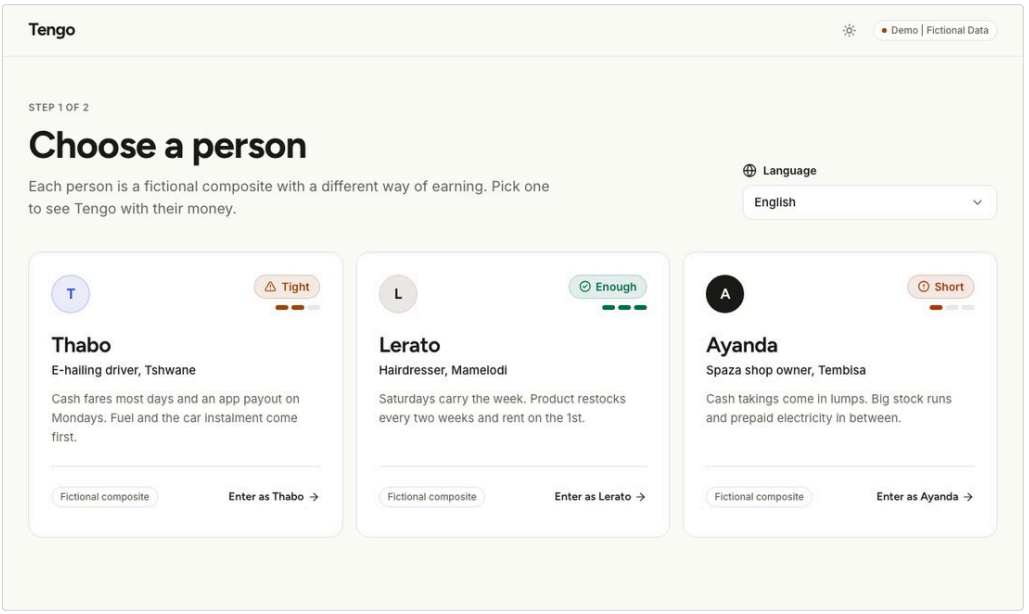
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4.10 Localisation and language reviewer

Reviewer file: `docs/review/localisation.md`. Findings are listed in full in Appendix A.



Picker with the language selector.

Verdict

Tengo has good plumbing for 11 languages and no translations. All 10 non-English locale files are byte-identical copies of `en.json`, and `status.json` marks every one `pending`. The pipeline that is meant to fill them has a blocking defect: Google Cloud Translation's supported-language list does not include Tshivenda. Because the script processes `ve` ninth and only writes `status.json` at the end, a full run would fail partway and leave state that the sync script then overwrites with English. The parts that exist are well thought out. Keys stay in step, the picker labels each language's translation status, `<html lang>` follows the chosen language, Tshivenda headings switch to a font that has the letters, placeholders are protected, and dates are kept in en-ZA instead of thin CLDR data. What is missing matters more for a money product: no glossary for the three state words, no per-string review status, hard-coded word order in several places, an SMS length check that only works for English, and no first-language review process. As it stands, a judge who picks isiZulu sees English with a notice saying so. If the script is run as written and shipped, people would get machine-translated money advice of uneven quality, with nothing in the message itself saying so.

What works

- **Key parity is enforced.** `npm run i18n:check` passes with "All 11 locales have the same 565 keys" (run on 2026-10-20). `scripts/i18n-sync.mjs:33` refreshes pending languages from English and makes changed strings in machine or reviewed languages fall back to English. Note: the fact sheet says 577 keys. The check reports 565.
- **The status is honest and visible where language is chosen.** `src/components/LanguageSelector.tsx:53–57` shows "Translation on the way. This language shows English for now." for any language that is not English. The note is linked to the select by `aria-describedby` (line 42). Each option carries its own `lang` attribute (line 46).
- **`<html lang>` follows the selection** (`src/components/Preferences.tsx:12`, set before first paint in `index.html:18`). The Inbox message bubble has its own `lang={message.language}` (`src/pages/Inbox.tsx:117`), so the message language can differ from the app language.
- **Tshivenda headings are handled correctly at the CSS level.** `src/styles/index.css:84–86` switches `--font-heading` to Inter for `html[lang='ve']`. My glyph check (`docs/review/tools/localisation/glyphs.mjs`) confirms Figtree lacks ḓ ḽ ṅ ṋ ṱ and their capitals, and Inter's bundled latin-ext subset has all ten. The pseudo-localisation run confirms the computed `h1` font is `"Inter Variable"` when `lang` is `ve` (`docs/review/tools/localisation/expansion-flat-home-390.json`).
- **The translation script has sensible guards** (`scripts/translate.mjs:51–66,111–116`). `{{placeholders}}` and brand words are wrapped in `translate="no"`, `*bold*` becomes `<b>`, house-rule characters are stripped, and any string whose placeholders change is kept in English. The script runs at build time only, so the app never calls Google.
- **Plural keys use i18next `_one` and `_other`** (18 pairs in `en.json`), which i18next resolves through `Intl.PluralRules`. In Node 26 (ICU 78.3, CLDR 48), all 11 codes are supported and return `one/other` (`docs/review/tools/localisation/plurals.mjs`).
- **Keeping dates in en-ZA was a reasonable call.** CLDR output for these locales is thin or odd. Node prints `zu` as "Bil, Okt 27" and `xh` as "Okt 27, Lwesb", and `nr`, `ss`, `ve` and `ts` fall back to en-US ("Tue, Oct 27") (`plurals.mjs` output). Using en-ZA for every language (`src/i18n/languages.ts:39`) avoids US-ordered dates.
- **Rand formatting is consistent.** `formatZAR` (`src/lib/format.ts:6–15`) produces "R1 250" with a no-break space and a decimal point. Intl en-ZA would give "R 1 250,50", so the hand-rolled formatter avoids a comma decimal that many readers would not expect.
- **The unconfirmed name origin is stated in the UI rather than hidden** (`en.json:460`: "still to be confirmed by a Tshivenda speaker").

What to improve

L10N-01 Ten of the 11 languages are English, so the 11-language claim is not yet true

Severity: High **Evidence:** `cmp` shows every file in `src/i18n/locales/*.json` is identical to `en.json`. `src/i18n/status.json` lists all 10 as `pending`, and `src/i18n/translated-from.json` is `{}`. The fact sheet's "Languages" section agrees. A user who picks isiZulu, Sepedi or Tshivenda gets English plus a notice. **Recommendation:** Do not present "11 languages" as a feature in the pitch until at least the pilot languages are translated and checked. Say instead "built for 11, with N live". For the hackathon, pick two or three languages that fit the personas' area (Gauteng home languages, Census 2022: isiZulu 23.1 percent, Sesotho 13.1, Sepedi 12.6, Setswana 10.4; Stats SA P0301.4 Table 2.9). Have a first-language speaker write or post-edit at least the `messages.*`, `state.*`, `hero.*`, `stateCard.*`, `sts.*` and `nav.*` keys, and mark them reviewed. **Effort:** M

L10N-02 Tshivenda is not in Google Cloud Translation, and a full run would fail and leave the files in an inconsistent state

Severity: High **Evidence:** The Cloud Translation supported-language page (accessed 2026-10-20) lists `af` and `zu` under Translation LLM, and `xh`, `st`, `nso`, `tn`, `ts`, `ss` and `nr` under NMT. It does not mention Venda, Tshivenda or `ve`. Consumer Google Translate added Venda in June 2024, but the API page used by the script does not list it. `scripts/translate.mjs:16` includes `ve` in `ALL`, and line 76 throws on any non-OK response. The loop writes each locale file as it finishes (line 123), but `status.json` and `translated-from.json` are written only after the loop (lines 127-128). A run that fails at `ve` (ninth of ten) therefore leaves `af` to `ss` translated on disk but still `pending` in `status.json`. The next `npm run i18n:sync` then replaces them with English (`scripts/i18n-sync.mjs:33`), wasting about 137 000 characters of quota. Not run, because there is no API key. The exact error code is an expectation based on the documentation. Separately, the README's estimate of "about 155,000 characters" for 10 languages is low: `en.json` has 17 160 characters of strings, or about 171 600 for 10. **Recommendation:** Remove `ve` from the machine translation list and have a human translate Tshivenda. Make the script write `status.json` and `translated-from.json` after each language, or catch per-language errors and continue. Add a dry-run that checks each target against the API's `languages` endpoint first. Correct the README figure. **Effort:** S

L10N-03 Machine-translated money messages would ship without a glossary or context, and with no caveat in the message itself

Severity: High **Evidence:** `scripts/translate.mjs:74` sends bare strings with `source: 'en'`. It sends no key path or context, and uses no glossary: v2 has no glossary support, and the script uses v2. 156 of 565 strings are 12 characters or fewer, including `state.enough`, `state.tight` and `state.short` ("Enough", "Tight", "Short"). Out of context, "Short" can be read as a length and "Tight" as a fit (**Assumption**: likely for at least some languages, which only a speaker can confirm). The state word is interpolated into every message: `messages.whatsapp.*` "Cover: {{state}}" and `messages.sms.*` "({{state}})" (`src/messages/templates.ts:17`). The "Machine translated with Google Translate" notice appears only beside the language select (`LanguageSelector.tsx:53-57`, used in `Picker.tsx:65`, `Me.tsx:82` and `Inbox.tsx:135`). It does not appear on Home or in the WhatsApp or SMS text that would reach the user. Public evidence on quality is weak. Google said of the 2022 zero-shot languages, including Sepedi and Tsonga, that the method "isn't perfect". It added Venda and Swati in 2024 with PaLM 2 and noted that it is "impossible to pick a 'right' variety". A Stellenbosch study of Afrikaans to English Google output found omission, mistranslation, non-translation and grammar to be the main error types, and quality stopped improving after 2012. News24 (July 2024) reported Tshivenda and siSwati output that diverged from the intended meaning (headline and search summary only, because the page returned 403). **Recommendation:** Never show machine output for the `messages`, `state`, `hero`, `stateCard`, `sts`, `plan` loan-check and `help` keys until a first-language reviewer has signed them off. Fall back to English for those keys until then. Build a short term base (Enough, Tight, Short, Safe to spend, Cover, Next money, Plan) per language, written by speakers, and enforce it. Move to Cloud Translation v3 with a glossary if MT stays in the loop. Send each string with its key and an English context note. Keep the machine-translation notice in both the language and English, and show it in the app shell when the status is `machine`. **Effort:** M

L10N-04 Review status is per language, so "checked by a first-language speaker" can become untrue

Severity: Medium **Evidence:** `scripts/translate.mjs:124` leaves a `reviewed` language as `reviewed` even after it machine-translates new or changed strings into it, and `--force` re-translates every string while keeping the status. `LanguageSelector.tsx:56` then shows `language.reviewed`: "Translated and checked by a first-language speaker." `TranslationStatus` in `src/i18n/languages.ts:29` has no per-key notion. **Recommendation:** Track status per key, for example `reviewed-from.json` that maps each key to the English text and the reviewer. Show "reviewed" only when every user-facing key in the language is reviewed. Refuse `--force` on reviewed languages unless a flag names the keys. **Effort:** M

L10N-05 The page declares a language its text is not in

Severity: Medium **Evidence:** Choosing Tshivenda sets `<html lang="ve">` (`Preferences.tsx:12`), but the content is English because the status is `pending`. The pseudo-localisation run records `lang: "ve"` on English pages. The Inbox bubble is marked `lang="ve"` over English text (`Inbox.tsx:117`). After a machine run, any string that falls back to English (`i18n-sync.mjs:19-20`, `translate.mjs:113`) also sits under the wrong `lang`. WCAG 3.1.1 and 3.1.2 require the declared language to match the content, and a screen reader may read English with a Tshivenda or default voice. **Recommendation:** Set `<html lang>` to the language actually rendered: `en` while the status is `pending`. Wrap English fallback strings in `lang="en"`, for example by having a small `T` component compare the resolved string with English. **Effort:** S

L10N-06 Word order and agreement are fixed in code and by interpolated fragments

Severity: Medium **Evidence:** Code joins a fragment and a date in English order: `WhyDrawer.tsx:105` and `Import.tsx:260` (`{t('hero.enoughUntil')} {date}`), and `privacy/Review.tsx:22` (`` `${label} ${date} (${state})` ``). `DateHero.tsx:9` shows "Enough until" above the date even for Tight, which conflicts with the "Tight" chip below it (screenshot `docs/review/tools/localisation/shots/pseudo-ve-flat-home-360.png`). Translated sentences receive a translated state word (`{{state}}`) and an English date ("Fri 23 Oct", from `formatDayShort` with en-ZA, `src/lib/format.ts:31`). In conjunctively written languages (isiZulu, isiXhosa, siSwati, isiNdebele), the word around a date or number usually takes a prefix or concord. **Assumption**: a translator will want forms such as a locative prefix attached to the date. That cannot be done when the date is a fixed English token, and it can break WhatsApp `*bold*` if the asterisk lands inside a word. `messages.whatsapp.tip.short` names a UI label ("Check Plan in Tengo") that MT translates separately from `nav.plan`. **Recommendation:** Replace every concatenation with a single key that contains `{{date}}`, for example `hero.enoughUntilDate`. Give each state word its own full sentence per state instead of interpolating `{{state}}`. Add month and weekday names to the locale files so translators control them and the format, while keeping en-ZA as the fallback. Write the "Plan" reference as `$t(nav.plan)` nesting so it always matches. Ask reviewers to check bold markers on a real WhatsApp client. **Effort:** M

L10N-07 The SMS length check only works for English and GSM characters

Severity: Medium **Evidence:** `SMS_LIMIT = 160` (`src/messages/MessageWriter.ts:28`). The Inbox counter uses `message.text.length` (`Inbox.tsx:129`), and the only test checks English (`src/messages/templates.test.ts:17-21`). The GSM 7-bit default alphabet has no ê, ô, ë, š, ḓ, ṱ or ṅ. A message containing any of them is sent as UCS-2, with at most 70 characters per SMS (Wikipedia, GSM 03.38). These letters are needed for Afrikaans (ê, ë, ô), Sepedi (š) and Tshivenda. Today's Short SMS for Ayanda is 101 characters. At 40 percent expansion it would be about 141 characters, which is more than two UCS-2 segments (**Assumption**: 30 to 50 percent expansion for these languages). **Recommendation:** Count segments by encoding (GSM-7 or UCS-2), not by characters. Add a test that renders every SMS template in every language with long names and dates and checks segment count. Allow a shorter per-language SMS variant. Bring per-segment cost into the financial model. **Effort:** S

L10N-08 The layout clips expanded text in the bottom nav, persona rows and segmented controls

Severity: Medium **Evidence:** I ran a pseudo-localisation test (`docs/review/tools/localisation/expansion.l10n.ts`) at 360 and 390px with `lang="ve"`, using Tshivenda letters as padding. With a flat 40 percent expansion, the main reading paths wrap without page overflow. With the W3C/IBM size bands (short strings at least double), the bottom tab bar truncates "Money" and "Inbox" on every page (`grid-cols-5`, `text-[12px]`, `truncate`, `src/layouts/AppLayout.tsx:171,185`; screenshot `shots/pseudo-ve-ibm-inbox-360.png`). On Me, `whitespace-nowrap` state chips (`src/components/StateChip.tsx:18`) squeeze persona descriptions to "E-hailing d..." and "Spaza shop o..." (`shots/pseudo-ve-flat-me-360.png`). The WhatsApp or SMS channel control loses its shape and stacks (`shots/pseudo-ve-ibm-inbox-360.png`). In the header, the demo chip pushes the privacy icon to the edge, although that chip is kept in English, so real translation would affect it less. The other clipped items in the JSON are screen-reader-only spans and user-entered descriptions. **Recommendation:** Allow nav labels to wrap to two lines or use shorter per-language labels approved by reviewers. Let state chips shrink or move below the name on narrow screens. Make the segmented control wrap cleanly. Add the pseudo-locale as a permanent Playwright project. **Effort:** S

L10N-09 The quality gates do not check translation quality

Severity: Medium **Evidence:** `scripts/i18n-check.mjs` checks only that keys match and that no string contains an exclamation mark. It does not check placeholder parity across locales (only the translate script does, at the moment of translation), `*bold*` parity, em or en dashes, or how many strings are identical to English. `scripts/fonts-check.mjs` passes, but only because every locale file is English. It has never seen a Tshivenda letter in a locale file. **Recommendation:** Extend `i18n:check` to compare `{{var}}` sets and bold-marker counts with English for every key, reject U+2013 and U+2014, and report per language the share of strings still identical to English, so the status cannot say `machine` or `reviewed` while 100 percent is English. Add a fixture with real Tshivenda text to the font check. **Effort:** S

L10N-10 Plural rules need briefing and testing

Severity: Low **Evidence:** `Intl.PluralRules` for `zu` and `nso` returns `one` for 0 as well as 1. i18next therefore renders `_one` for a count of 0 (`plurals.mjs`: "zu: count=0 → '0 day'"). Counts can be 0 in `privacy.deleteTx`, `privacy.deleteAudit`, `money.count` and `import.skipped`. Node 26 supports all 11 codes. **Assumption**: older Android WebViews may ship trimmed ICU data without `nr`, `ss`, `ve` or `ts`, in which case the rules resolve to the default locale. This is untested in the app. There are no plural tests for any language other than English. **Recommendation:** Tell translators that the `_one` form covers 0 in isiZulu and Sepedi. Add a unit test that renders every plural key in every language for counts 0, 1 and 2. Avoid count-0 sentences where an empty state reads better. **Effort:** S

L10N-11 Decomposed Tshivenda characters would render in a fallback font

Severity: Low **Evidence:** The bundled Inter has the precomposed letters but not the combining marks U+032D (circumflex below) or U+0307 (dot above) (`glyphs.mjs` output: "MISSING" for both fonts). Tshivenda text typed or pasted as base letter plus combining mark, which some keyboards and sources produce, would draw the mark from a system font. `translate.mjs` does not normalise to NFC. `fonts-check.mjs` would catch this in locale files, but not in imported CSV descriptions or names. **Recommendation:** Apply `.normalize('NFC')` to translated strings, imported descriptions and names. Add NFC as a rule in `i18n:check`. **Effort:** S

L10N-12 The name origin is still unconfirmed, and "value" may really be "price"

Severity: Low **Evidence:** `README.md:106–110` and `BRIEF.md:11` record "mutengo", Tshivenda for value, as unconfirmed, and the UI says so (`en.json:460`). My search found no Tshivenda dictionary entry. The only hit lists "mutengo: price" on a Shona vocabulary page (search result summary only, because the page returned 403, so this is unverified). "Price" and "value" carry different meanings for a money brand. **Recommendation:** Confirm with two Tshivenda speakers, and ideally a lexicographer from a university African languages department, before the pitch. Record the exact gloss and use it consistently in the pitch and the About copy. Have the `me.nameBody` string in Tshivenda written by a person, not MT. **Effort:** S

L10N-13 The demo safety label is English in every language

Severity: Low **Evidence:** `translate.mjs:18` keeps `common.demoChip` as "Demo | Fictional Data" in every language. This is the label that tells users the figures are not theirs. **Recommendation:** Show it in the chosen language and in English, for example "Demo | Fictional Data" plus a translated line. **Effort:** S

L10N-14 Every locale is bundled into the main script

Severity: Low **Evidence:** `src/i18n/index.ts:5` imports all locale files eagerly. Per the fact sheet, the JSON accounts for 337 KB of the 696 KB bundle. Translated text with non-ASCII letters will grow slightly. A user downloads ten languages they will not read, on prepaid data. **Recommendation:** Bundle English, and load the selected language with a dynamic import that the service worker caches for offline use. **Effort:** S

First-language review process (step by step)

1. **Recruit two reviewers per language**, one translator and one checker, both first-language speakers. At least one should live in or near the target users' area (for these personas, Gauteng). Pay them, and record their names and dates against each key.
2. **Brief them before they see any text.** Explain who the users are (irregular earners, low data, shared phones), what the product does and does not promise, and that it is not financial advice. Give the house style (plain words, no exclamation marks or dashes) and the tone: calm, direct and never shaming, especially for Short.
3. **Agree a term base first:** Enough, Tight, Short, Cover, Safe to spend, Next money, What helps most, Plan, Money, Inbox, Privacy, statement, bill, income. For each term, record the chosen word, the rejected alternatives and why. Test the three state words with three or more lay speakers for the intended meaning (enough money, nearly not enough, not enough).
4. **Translate or post-edit the money paths first:** `messages.*`, `state.*`, `hero.*`, `stateCard.*`, `sts.*`, `next.*`, `why.*`, `plan.*` (the loan check especially) and `help.*`. Then consent and privacy. Then everything else.
5. **Check meaning against the numbers.** For each persona's real output (Thabo Tight, Lerato Enough, Ayanda Short), have the reviewer read the Home screen and the WhatsApp and SMS messages, then say back what they would do. A wrong answer is a defect, whatever the grammar.
6. **Check grammar around variables:** prefixes or concords on `{{date}}`, `{{count}}` and `{{state}}`, the plural forms (0, 1, 2, 10), and `*bold*` boundaries on a real WhatsApp client.
7. **Check register:** formal or informal address, respect forms, and whether "Hi {{name}}" fits. Check that no line sounds like an instruction to borrow, lend or spend, so that "safe to spend" does not become "you may spend".
8. **Check on screen at 360px** in both themes, for truncation, wrapping and the Tshivenda glyphs in headings.
9. **Check accessibility by ear.** Listen to the main screens with a screen reader set to the language, if a voice exists, or confirm that `lang` is set correctly.
10. **Sign off per key** in a reviewed file. Re-review any key whose English changes. Spot-check 10 percent of keys after each release.
11. **Test with five users per pilot language** before any real data is used, using the same comprehension questions as step 5.

Risk ranking by language

Speaker shares are Census 2022 "language spoken most often in the household" (Stats SA P0301.4, Table 2.8). Counts are the share applied to the 62 027 503 total population. They are approximate, because the table excludes infants under 1.

Rank	Language	Census 2022 share (approx. speakers)	Why the risk
1	Tshivenda (ve)	2.5% (about 1.6m)	Not offered by the Cloud Translation API the script uses (L10N-02). Added to consumer Translate only in 2024. Special letters (L10N-11). The brand name rests on an unconfirmed Tshivenda word (L10N-12). Few reviewers.
2	isiNdebele (nr)	1.7% (about 1.1m)	Smallest written language. Added to Google in 2024. Conjunctive writing makes placeholders and agreement hard (L10N-06). Least public evidence of MT quality.
3	siSwati (ss)	2.8% (about 1.7m)	Added in 2024. Conjunctive. News24 reported poor 2024 output.
4	Sepedi (nso)	10.0% (about 6.2m)	2022 zero-shot model ("isn't perfect"). š breaks GSM SMS (L10N-07). 12.6% of Gauteng, so many target users.
5	Xitsonga (ts)	4.7% (about 2.9m)	2022 zero-shot model. 7.0% of Gauteng.
6	isiZulu (zu)	24.4% (about 15.1m)	Better MT support (listed under Translation LLM), but errors reach the most people, including 23.1% of Gauteng. Conjunctive. 0 takes the singular plural form (L10N-10).
7	Sesotho (st), Setswana (tn), isiXhosa (xh)	7.8%, 8.3%, 16.3%	Longer-supported NMT languages. Still need human review of the money terms.
8	Afrikaans (af)	10.6% (about 6.6m)	Strongest MT support and many reviewers. ê, ë and ô still force UCS-2 SMS.

Top three risks

1. **A machine-translated state word or loan check gives the wrong meaning in a money decision.** For example, Short read as fine, or "safe to lend" read as an instruction, sent by WhatsApp or SMS with no caveat (L10N-03, L10N-06).
2. **The 11-language claim is challenged at judging.** Every language shows English, and the chosen MT tool cannot produce one of the 11 languages, which is also the language of the brand's origin (L10N-01, L10N-02, L10N-12).
3. **Trust labels drift from reality.** "Checked by a first-language speaker" stays on after unchecked machine strings arrive, and `lang` declares a language the text is not in (L10N-04, L10N-05).

Score

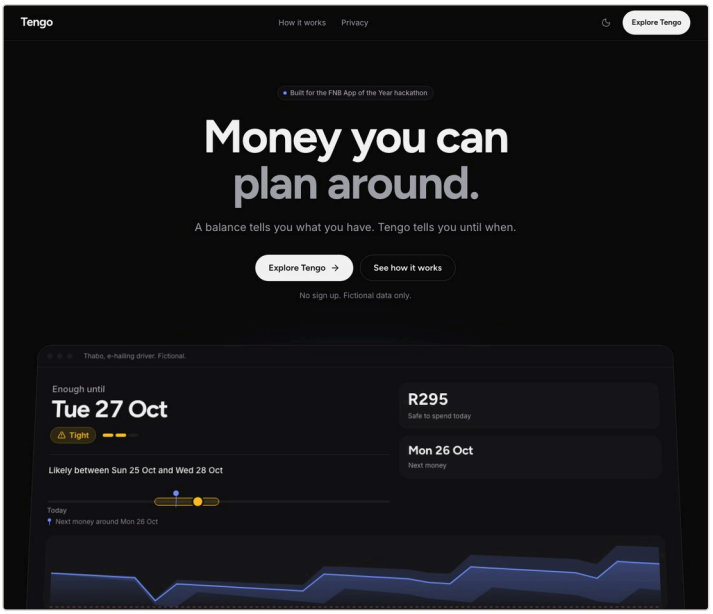
4/10. The infrastructure is above average for a hackathon build, but no language is translated, the pipeline cannot produce Tshivenda, and there is no review process for money messages.

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- Statistics South Africa, Census 2022 Statistical Release P0301.4, Tables 2.8 and 2.9, https://census.statssa.gov.za/assets/documents/2022/P03014_Census_2022_Statistical_Release.pdf (accessed 2026-10-20)
- Wikipedia, "GSM 03.38", https://en.wikipedia.org/wiki/GSM_03.38 (accessed 2026-10-20)
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4.11 Product strategist and commercial lead

Reviewer file: `docs/review/commercial.md`. Findings are listed in full in Appendix A.



Landing page, dark, 1280 px.

Verdict

Tengo asks a sharp question that South African bank apps do not answer today: not "what is my balance" but "until when does it last, given how I actually get paid". I found no verified South African bank or app feature that shows a run-out date against an irregular income cadence. The nearest are FNB nav» Money's "available funds after known upcoming payments" and Standard Bank's "Future Payments" 14 to 30 day estimate. That gap is real, but it is narrow and cheap for a bank to close. The engine is a 368 line file of transparent percentile maths (`src/engine/forecast.ts`), it has no proprietary data, and every bank already holds the transaction history it needs. The only business model that fits is a B2B2C licence embedded in a bank app, paid as a pilot fee and then a per-active-user fee with a platform floor, with sponsor funding as a secondary track for users outside the partner bank. On my sourced inputs (`docs/review/model-inputs.md`), one bank at base adoption gives about 44 000 monthly active users. At R5 per user that does not cover a lean team and compliance. Daily paid WhatsApp or SMS messages would cost more per user than the bank is likely to pay. The consumer subscription model does not fit this segment. The pitch is strong as a feature concept for FNB and weak as a standalone company today.

What works

- **A question no verified competitor answers.** Home leads with a date, a range and a state (Enough, Tight, Short), with the next money date estimated from weekday or interval patterns (`src/engine/forecast.ts`, fact sheet "Forecast engine"). Of the competitor features I could verify, FNB nav» Money shows funds left after scheduled payments, with no income projection (BusinessTech, 2018). Standard Bank's Future Payments estimates upcoming payments and spending for 14 to 30 days (Standard Bank). Nedbank MoneyTracker and Discovery's Financial Analyser are budget and category tools with no stated balance forecast (Nedbank, Discovery). The date-first framing for irregular earners is a real point of difference.
- **The segment is large and under-served on resilience.** Informal sector employment is 3.6 million and private household employment 1.1 million in Q2 2026 (Stats SA QLFS Q2 2026 presentation, slide 18). Only 30.8 percent of South African adults say it is not difficult to raise extra money within 30 days, and 18.3 percent in the poorest 40 percent (World Bank Little Data Book on Financial Inclusion 2025, South Africa). Account ownership is 81.1 percent, so most of the segment is reachable through a bank.
- **Built to be embedded.** All data access goes through `TengoRepository` (`src/data/repository.ts:75`), and messages go through a `MessageWriter` interface (`src/messages/templates.ts`). That is the right shape for a bank licence, where the bank supplies data and channel.
- **Cheap to run, apart from messaging.** The forecast is local, deterministic statistics with no model inference. Messages are fixed templates today, and even an LLM rewrite would cost about R0.002 per message on Claude Haiku 5.5 at USD 0.10 and 0.50 per million tokens (Claude pricing) with my assumed 600 input and 120 output tokens.
- **Culturally specific features.** The loan check ("safe to lend up to") answers a common real question about lending to family. The 11 language plumbing exists (`npm run i18n:check`, fact sheet "Languages"). No verified bank feature offers either.
- **A privacy posture that fits bank procurement.** Consent, export, deletion with receipt, an audit log and data minimisation all work against the local store and are clearly labelled as simulated (fact sheet "What is simulated"). This shortens the conversation with a bank's privacy team, even though none of it is production grade.

What to improve

COMM-01. No defensible moat against the bank building it - **Severity:** High - **Evidence:** The engine is transparent percentile maths over date, amount and description (`src/engine/forecast.ts`, `src/engine/classify.ts:16`). It uses no proprietary data. Banks already ship adjacent features: FNB nav» Money "My available funds" since 2018 (BusinessTech), Standard Bank Future Payments (Standard Bank), Discovery's AI budget tool (Discovery). FNB reports 8.16 million digitally active customers and its own large digital build (BusinessDay, 2026-09-16). - **Recommendation:** Present the defensible assets honestly: (1) backtested forecast accuracy on irregular income, which no competitor publishes; (2) a tested, plain-language, multilingual content set for money states; (3) a pilot playbook with outcome measurement. Offer the engine under licence with the bank owning the deployment, so "build versus buy" becomes "buy and adapt now". Do not claim a technology moat. - **Effort:** M

COMM-02. Single-bank unit economics do not break even - **Severity:** High - **Evidence:** Base funnel in `model-inputs.md` section 2: 8.16m digitally active x 15 percent irregular x 60 percent shown x 15 percent activate x 40 percent stay active = about 44 000 MAU (range 6 000 to 220 000). At R5 per MAU (Assumption, R2 to R12) revenue is about R220 000 a month. Variable cost with in-app push is about R1.23 per MAU (hosting R0.15, compute R0.02, LLM R0.06, support R1.00), so

contribution is about R166 000 a month against assumed fixed costs of R310 000 (team R250 000, compliance R60 000). Break-even needs about 82 000 MAU at R5. No public per-user prices exist for bank money-insight vendors to anchor a higher fee, and Capitec's entire main account costs R7.50 a month (Capitec 2026 fees). - **Recommendation:** Price as pilot fee (R150 000 to R1m) plus per-MAU fee plus an annual platform floor (R0.6m to R3m), and plan for a second bank or a white-label channel within 18 months. Model break-even per scenario in `model.xlsx` with the floor switched on and off. - **Effort:** M

COMM-03. Daily paid messaging costs more than the licence fee - **Severity:** High - **Evidence:** The brief and Inbox describe a daily WhatsApp or SMS message (`src/messages/templates.ts`, fact sheet "Screens"). A WhatsApp utility template in South Africa is USD 0.0095 per delivered message from 1 October 2026 (ChatMaxima, quoting Meta list rate; Meta confirms a South African utility increase on that date at Meta pricing docs). 30 messages x USD 0.0095 x R16.13 (valutafx) = R4.60 per user per month before BSP fees. If Meta classed the nudges as marketing (USD 0.0379), it would be R18.34. Daily SMS at R0.24 (Panacea Mobile) is R7.20. Each exceeds the base licence fee of R5. - **Recommendation:** Make in-app push inside the bank app the default channel (no per-message cost). Send WhatsApp only on state changes and at most weekly, and design replies that open the free 24 hour service window. Keep templates rather than an LLM for routine messages. Put message frequency in the model as a top tornado driver. - **Effort:** S

COMM-04. The segment's cash income is invisible to the product - **Severity:** High - **Evidence:** Two of three personas earn cash: Thabo "Cash fares most days", Ayanda "Cash takings come in lumps" (`src/i18n/locales/en.json`, `personas`). The repository has no way to add a transaction, only replace history or edit one (`src/data/repository.ts:75` to `96`). A what-if is temporary. The seed treats cash as bank deposits (`src/data/seed/generate.ts:52`), which real cash earners often do not make. 19.6 percent of adults save through clubs or people outside the family (Little Data Book 2025), which Tengo also does not model. A bank-data-only forecast will read cash earners as Short when they are not, which damages trust and the commercial case. - **Recommendation:** Add a quick "cash in" and "cash out" entry and a "cash on hand" figure that feeds the forecast, plus a stokvel contribution as a recurring bill type. Pitch this as the reason Tengo is different from bank budgeting tools. - **Effort:** M

COMM-05. No evidence of outcomes for a bank buyer - **Severity:** Medium - **Evidence:** All data is synthetic and fictional (fact sheet "Data"). There are no users, no test sessions and no measured behaviour change. Every bank value input in `model-inputs.md` section 5 beyond published fees is an Assumption (unpaid reduction 0 to 30 percent, deposit uplift R0 to R1 000). - **Recommendation:** Before pitching a pilot, run 5 to 8 user sessions with real irregular earners, and publish the backtest. Offer a pilot design with a control group and four pre-agreed KPIs: activation, 30 day retention, unpaid debit orders per user, and a confidence survey item. - **Effort:** M

COMM-06. A consumer subscription does not fit the segment - **Severity:** Medium - **Evidence:** The target pays R7.50 a month for a whole Capitec account (Capitec 2026 fees). Free budgeting tools exist inside FNB, Standard Bank, Nedbank and Discovery apps (sources above), and Vault22, the platform behind 22seven, offers a free tier (Vault22). Card collection costs 3.2 percent plus R2 (PayFast fees, via search index), about 10 percent of a R29 subscription. Public finance app day 30 retention figures are 4.2 to 15 percent (Panto, weak source). At 2 percent conversion, revenue is about R0.44 per MAU against R1.23 cost and an assumed R120 acquisition cost. - **Recommendation:** Keep Tengo free to the user. Drop the consumer subscription from the pitch except as a later premium for small business owners such as spaza shops (stock planning), tested separately. - **Effort:** S

COMM-07. The distribution story conflicts with the product as shown - **Severity:** Medium - **Evidence:** The landing page counts "Bank logins needed" as a selling point (`src/i18n/locales/en.json` `landing.stats.logins`) and onboarding depends on a CSV statement export (`src/pages/Import.tsx`). Few informal earners will export a CSV, so this does not scale. Open banking in South Africa is still being designed: trackers give expected dates, but I found no SARB or IFWG document confirming a 2026 start (Open Banking Tracker, weak). The realistic channel is inside the bank app, using the bank's own data. - **Recommendation:** Tell one distribution story: "inside your bank app, using data the bank already has, with consent". Keep CSV as the demo path and say so. Treat open finance as later upside, not the plan. - **Effort:** S

COMM-08. The pitch overclaims languages - **Severity:** Medium - **Evidence:** The landing page shows an "Official languages" stat (`src/i18n/locales/en.json` `landing.stats.languages`). All 10 non-English locales contain English text (fact sheet "Languages", `src/i18n/status.json`). A judge who picks isiZulu sees English (`docs/review/prep/walkthrough.md` item 5). - **Recommendation:** Before submission, show one or two human-reviewed languages and change the stat to "11 languages ready for translation", or remove it. - **Effort:** S

COMM-09. One buyer, one channel - **Severity:** Low - **Evidence:** The product is built for an FNB competition (README line 9). There is no second channel in the code or README. Comparable digital bases exist at Capitec (15.3m active app clients, Capitec FY2026 summary), Absa (5.4m active digital, BusinessDay) and Nedbank (3.2m Money app active, TechFinancials). - **Recommendation:** Keep FNB as the lead, and prepare a white-label variant and a platform-partner track (for example e-hailing driver programmes; Bolt reports about 500 000 drivers onboarded since 2016, ITWeb). - **Effort:** M

COMM-10. Brand assets are not commercially cleared - **Severity:** Low - **Evidence:** The README notes that the Public Stunt font is free for personal use only and needs a commercial licence, and the name origin is still to be confirmed by a Venda speaker (`BRIEF.md` sections 1 and 5). The fact sheet lists the logo and icons as placeholders. - **Recommendation:** Clear the font licence, confirm the name and run a trademark search before any pilot or public launch. - **Effort:** S

COMM-11. No written business case in the repository - **Severity:** Low - **Evidence:** The README and BRIEF cover product and build only, with no section on the buyer, pricing or metrics (`README.md`, `BRIEF.md`). - **Recommendation:** Add a one-page business case: buyer, model (a) with pricing, pilot KPIs, unit cost per MAU and break-even, linked to `model.xlsx`. - **Effort:** S

Business model comparison

Model	Fit	Why
(a) B2B2C licence, per active user, embedded in a bank app	**Best**	The bank has the data, the trust, the channel (free push) and the cost of the problem (unpaid items, arrears, contact centre). No CAC for Tengo. Weakness: small per-user fee, long procurement, build versus buy (COMM-01, COMM-02).
(b) Revenue share on bank outcomes	Later add-on	Aligns incentives, but attribution needs a control group and agreed baselines. Unpaid fees are small per user (R6 at Capitec, R25 pay as you transact and R130 in a bundle after 3 free at Standard Bank, Standard Bank Access 2026), so the share alone will not fund the company. Use as a bonus on top of (a).
(c) Freemium consumer subscription	Poor	Price sensitive segment, free alternatives, payment fees about 10 percent, weak retention, high CAC (COMM-06).
(d) Sponsored or partner funded for inclusion	Secondary	Fits users outside the partner bank (cash earners, platform workers). Grants found are small (USD 5 000 per startup in the 2023 SAIB fintech challenge, Engineering News). Useful for research and impact evidence, not as the core revenue.

Pricing

- Bank: pilot fee R150 000 to R1m for 6 months; then R2 to R12 per MAU per month (base R5) with an annual platform floor of R0.6m to R3m. All Assumption, reasoned in `model-inputs.md` section 4.
- Outcome bonus: 10 to 30 percent of measured, control-group-verified benefit (Assumption).
- Consumer: free. No paid tier until a small business variant is tested.
- Sponsor: R20 to R150 per reached user per year to cover messaging and support (Assumption).

Channel strategy and partners

1. Bank app embed (lead): in-app card on the home screen for customers whose deposits look irregular, push notifications, the bank's own consent flow. FNB is the natural first buyer because of the competition; FNB also reports R5.7bn in stokvel deposits, which shows it already serves informal money practices (BusinessDay).
2. WhatsApp as a light secondary channel, on state change only (COMM-03).
3. Platform partners: e-hailing and delivery platforms (payout data is regular and digital), and card acceptance providers for small traders such as Yoco, whose settlement data shows trader takings (Yoco fees). No partnership exists; these are targets.
4. Earned wage access providers such as Paymenow, which serves nearly 500 000 users per BusinessDay 2025 (from search snippet), as a possible partner: a "until when" forecast could reduce over-drawing.
5. NGOs and inclusion funders for cash earners outside banks (model d).

Competitive landscape (South Africa)

All accessed 2026-10-20. "Not verified" means I could not confirm the feature from a primary or reliable source.

Competitor	Type	Relevant features	Forecasts run-out date for irregular income?	Source
FNB app (nav» Money)	Bank	Smart budget with auto-sorted spend and alerts at halfway, near limit and at limit (2020); "My available funds" after known upcoming payments and debit orders (2018). Current 2026 menu not verified.	No (scheduled payments only, no income projection)	Sunday World 2020, BusinessTech 2018
Capitec	Bank	Automatic spend categorisation, money in and out notifications, up to 10 savings accounts. Official page blocked by a bot check; seen only through search snippets. Budget limits not verified.	Not verified	Capitec blog (snippet only)
Absa	Bank	Savings Coach: goals, automatic "auto savers", tips, up to 7.75 percent interest. No budgeting or forecasting tool found for the South African app.	No evidence found	Absa Savings Coach
Standard Bank	Bank	Budget Manager (pre-populated budget from past spend, goals, category limits); Future Payments estimates upcoming payments and expected spend for the next 14 to 30 days; Money Movements; Money Reels recap (Dec 2025).	Partly: near-term spend estimate, no stated run-out date or income cadence	Standard Bank app guide, BusinessTech Money Reels
Nedbank	Bank	MoneyTracker: free, category budgets, up to 8 dashboards incl. side hustle, invoicing.	No (page does not mention forecasting)	Nedbank MoneyTracker
TymeBank (GoTyme Bank)	Digital bank	GoalSave up to 10 goals, up to 10 percent interest with conditions; about 12 million customers (date unclear); old app retired 31 Aug 2026 (search snippet).	No evidence found	GoTyme GoalSave, Daily Investor
Discovery Bank	Bank	Vitality Money Financial Analyser: 160+ categories, automated or manual budgets, smart alerts, points for budget goals (2023); AI spending insights on WhatsApp (2025). About 1.6m clients.	No (article does not mention forecasting)	Discovery, News24 2025
Vault22 (22seven)	Budgeting app	22seven.com now redirects to Vault22: account linking, budgeting, free, Basic and Premium tiers (prices not shown); shareholders include Old Mutual and SC Ventures.	No evidence found	Vault22
Paymenow	Earned wage access (formal employees)	Salary advances; nearly 500 000 users (2025 report). Not built for informal earners.	No	BusinessDay 2025 (snippet), tech.africa
Yoco	Card acceptance for small traders	Card machines and online payments; in-person debit rates from 2.30 percent up to R50k turnover (Core). Money-management features for traders not verified.	Not verified	Yoco fees
MTN MoMo	Wallet	Relaunched in SA in 2020; card top-ups added with Mastercard (Oct 2025). Current SA user count not verified.	No evidence found	ITWeb 2020, Engineering News 2025
Vodacom VodaPay	Wallet and super app	Wallet with Bidvest Bank; 1.6m registered users in May 2022. Current users not verified.	No evidence found	Telecom Review Africa, Connecting Africa
Personetics, Meniga (B2B vendors)	Bank insight engines	Entered SA through local partners (2014, 2016). Current SA bank clients not verified. These are the real "buy" alternatives a bank will compare Tengo with.	Not verified for SA	Personetics 2014, Meniga 2016

Other informal-income tools (for example Kazang, spaza ordering apps) were not checked and are not claimed.

Differentiation and defensibility

- Differentiation: a date, not a balance; irregular income cadence (weekday or interval); three plain states; safe to spend and safe to lend; multilingual messaging. Verified competitors offer budgets, categories and near-term payment views instead.
- Defensibility: low. The method is published in the README and is easy to copy. What can become defensible is accuracy evidence from real pilots, tested copy in 11 languages, and a cash-entry habit loop (COMM-04) that bank tools ignore.

Go-to-market

1. Now to submission: fix COMM-07 and COMM-08 copy, add cash entry if time allows, and pitch model (a).
2. Next 3 months: 5 to 8 user tests with drivers, hairdressers and spaza owners; publish backtest; one-page business case (COMM-11).
3. Months 3 to 9: one bank pilot with 5 000 to 20 000 invited customers, push only, control group, four KPIs.
4. Months 9 to 24: commercial licence with platform floor; second bank or white-label; sponsor-funded track for cash earners.

Strongest and weakest parts of the pitch

- Strongest: "A balance tells you what you have. Tengo tells you until when." It names a gap no verified competitor fills, for a large, low-resilience segment, in a form that fits inside a bank app.
- Weakest: there is no reason a bank cannot build it, the per-user economics at one bank are thin, the product ignores cash, and there is no user evidence yet.

Top three risks

1. **Build instead of buy.** A bank with FNB's digital scale can reproduce the engine quickly. Tengo is then a hackathon idea, not a supplier (COMM-01).
2. **Thin economics.** A low per-user fee, a long procurement cycle and messaging costs can leave the company loss-making even after a successful pilot (COMM-02, COMM-03).
3. **Trust failure with cash earners.** Forecasts that ignore cash will say Short to people who are fine, and they will stop using it. Activation and retention will then miss the pilot KPIs (COMM-04, COMM-05).

Score

4 out of 10: a clear, differentiated feature concept with the right embed architecture, but no moat, no user evidence, a blind spot for cash income, and unit economics that only work with a platform floor or more than one bank.

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- Further sources for model inputs only are listed in `docs/review/model-inputs.md`.

4.12 Investor and sceptic

Reviewer file: `docs/review/investor.md`. Findings are listed in full in Appendix A.

Verdict

Tengo is a well built demo of a sensible idea, but nothing in it would yet convince me to fund it. The product has no users, no real data, no data access path beyond a manual CSV, no buyer, and an engine of about 370 lines of percentile arithmetic (`src/engine/forecast.ts`) that any bank with transaction data could rebuild in weeks. The one thing it claims as its edge, a date instead of a balance, is a framing, not an asset: FNB has shown "what's left after scheduled payments and debit orders" since 2018, and several consumer apps already sell "safe to spend until payday". Worse, the demo contradicts itself on the screen that matters most (Ayanda is "Short" while being told R310 is "safe to spend today" and the chart below says "Enough"), and the landing page advertises 11 languages that are all English. The honest pitch today is "a credible prototype of a feature a bank could adopt", not "a company". That is fine for a hackathon. It is a no for an investment until there is evidence that real irregular earners understand, trust and act on the date.

What works

- The question is sharp and easy to say. "A balance tells you what you have. Tengo tells you until when." Evidence: `docs/review/screens/landing-light-1280.png`, `BRIEF.md` section 1.
- The engine is transparent and explainable, which is a real asset for a bank pitch. Every number has a simple rule a compliance team can read: median and percentiles of rolling 7 day spend (`src/engine/forecast.ts:59-88`), recurring bill rule (`src/engine/classify.ts:16-22`), state rule (`src/engine/forecast.ts:217-221`). There is no black box to defend.
- Engine quality is high for a prototype: 97.6 percent statement coverage in `src/engine`, 66 unit tests passing. Evidence: `docs/review/prep/fact-sheet.md` (Quality signals).
- Help is computed with the same engine as the date, so suggestions are internally consistent with the headline. Evidence: `src/engine/help.ts:45-76`.
- Privacy posture is a selling point for a bank: no third party requests, data stays in the browser, and simulated controls are labelled as such. Evidence: `docs/review/prep/fact-sheet.md` (What is simulated), `BRIEF.md` section 2.
- Front end performance and accessibility scores are strong: Lighthouse mobile landing 94 performance and 100 accessibility. Evidence: `docs/review/prep/fact-sheet.md`, `docs/review/prep/lighthouse/`.

What to improve

INV-01. Zero evidence of demand or behaviour change - **Severity:** High - **Evidence:** No users, no interviews, no test sessions recorded anywhere in the repo. The personas are fictional composites (`README.md:11`, `src/data/seed/personas.ts`). The whole thesis, that a date changes behaviour more than a balance, is untested. - **Recommendation:** Run 5 to 8 sessions with real e-hailing drivers, hairdressers or spaza owners before submission. Show the Home screen, ask "what does this tell you", "what would you do today", and record whether they read the date correctly. Quote them in the pitch. - **Effort:** M

INV-02. The core insight is easy to copy and partly exists already - **Severity:** High - **Evidence:** The forecast is 368 lines (`src/engine/forecast.ts`) using median and 20th and 80th percentiles. FNB's nav Money "My available funds" has shown what remains after scheduled payments and debit orders since 2018 (BusinessTech, 11 June 2018). Standalone apps already market "safe to spend before your next paycheck" and day by day balance to payday (Product Hunt listing for Pocket Runway; 11:FS payday article). - **Recommendation:** Stop pitching the algorithm as the moat. Pitch the irregular income specific parts (income rhythm detection without a payday, range band, Short and Tight states, loan to family check) and the design and language work as the asset, and position Tengo as a feature a bank licenses or acquires, not a standalone company. - **Effort:** S

INV-03. Headline, safe to spend and chart contradict each other - **Severity:** High - **Evidence:** Ayanda: hero "Short from Fri 23 Oct", stats card "Safe to spend today R310 until Mon 26 Oct", chart readout for today "Enough" (`docs/review/screens/ayanda-home-dark-390.png`; `persona-snapshot.json`). Her usual daily spend is R623. Safe to spend divides the balance over days to the latest next money (`src/engine/forecast.ts:291-296`) while the date uses the spend rate (`forecast.ts:282-288`) and the chart adds expected income (`forecast.ts:234-267`). Also, the label "Enough until" is shown for the Tight state (`src/components/forecast/DateHero.tsx:9`), visible on the landing hero: "Enough until Tue 27 Oct" next to a "Tight" chip (`landing-light-1280.png`). - **Recommendation:** One source of truth per screen. When Short, replace safe to spend with "Spend at most R X a day to reach Sun 25 Oct" derived from the same path, label Tight as "Tight until", and make the chart's today state match the hero or remove it. - **Effort:** S

INV-04. No path to real data - **Severity:** High - **Evidence:** No bank connection, no open banking, CSV upload only, PDF import "coming next" and not built (`docs/review/prep/fact-sheet.md`, Data). South Africa's open finance framework is still being worked on by the IFWG and FSCA, with formal adoption not yet confirmed (Moonstone, 2026; Stitch, 10 July 2026). - **Recommendation:** Name the data route explicitly: embedded in a bank app using the bank's own transaction feed is the only route that works this year. Show a mock integration contract (inputs Tengo needs: balance, 60 days of transactions, scheduled debit orders). - **Effort:** M

INV-05. Overclaiming languages - **Severity:** High (credibility) - **Evidence:** Landing stat "11 Official languages" (`landing-light-1280.png`). All 10 non-English locales contain English text (`src/i18n/status.json`, `npm run i18n:check` in `docs/review/prep/checks.md`). The About copy says translations are "machine made or on the way" (`src/i18n/locales/en.json:475`) when none has been machine translated. - **Recommendation:** Change the stat to "English now, 10 languages planned" or ship and get first-language review for at least isiZulu and Sepedi or Setswana before submission. A judge who taps isiZulu and sees English will discount everything else. - **Effort:** S (copy) to M (two reviewed languages)

INV-06. Accuracy is unproven and the demo is tuned to look right - **Severity:** High - **Evidence:** The seed balance is chosen by the engine so each persona starts in Tight, Enough and Short (`docs/review/prep/fact-sheet.md`, Data; `src/data/seed/generate.ts`). The synthetic history was generated by the same team that wrote the detector, so a correct forecast on it proves internal consistency, not accuracy. - **Recommendation:** Backtest on at least a handful of real, consented, anonymised statements from irregular earners and publish error in days and band coverage. Until then, say "accuracy not yet measured on real data" in the pitch. - **Effort:** M

INV-07. Who pays is undefined - **Severity:** Medium - **Evidence:** No pricing, no business model in `README.md` or `BRIEF.md`. The intended users are, by definition, short of money, which rules out a meaningful consumer subscription. **Assumption:** the only realistic payer is a bank or a funder, at a per active user fee in the low single rands per month (plausible range R1 to R10), because banks already give budgeting tools away free (nav Money is described as free, Engineering News, 28 March 2022). - **Recommendation:** Pick one model (B2B2C licence to a bank) and state the bank's benefit as a hypothesis to test: fewer failed debit orders and unpaid fees among irregular earners. - **Effort:** S

INV-08. The headline date ignores income, so it means different things to different people - **Severity:** Medium - **Evidence:** Runway runs the balance down with no future income (`src/engine/forecast.ts:195-211`). Lerato is "Enough until 1 Nov" because rent of R2 800 lands on 1 Nov (`persona-snapshot.json`), even though she is paid on 24 Oct and again before 1 Nov. For a weekly earner like Thabo the date is almost always about a week out, so it adds little over "next payout". - **Recommendation:** Test with users which question they think the date answers. Consider phrasing as "Your money now lasts until" and making the state, not the date, the primary signal when next money comes first. - **Effort:** S

INV-09. Advice-like outputs raise regulatory and harm exposure - **Severity:** Medium - **Evidence:** "Safe to spend", "safe to lend" (`src/engine/forecast.ts:357–368`) and a suggestion to move Thabo's car instalment (`persona-snapshot.json`, help.billToMove; `src/engine/help.ts:58–73`). Moving a vehicle finance payment can trigger penalties or arrears. **Assumption:** a bank's legal team would require disclaimers and possibly reframing before a pilot; see the legal reviewer for the regulatory analysis. - **Recommendation:** Exclude credit agreement instalments from "bill to move", rename "safe to lend" to "you could lend up to and stay Tight", and add a one line "estimate, not advice" under every figure. - **Effort:** S

INV-10. Cash and informal money flows are invisible - **Severity:** Medium - **Evidence:** The engine sees only bank transactions (`src/engine/forecast.ts:105–191`). Ayanda's income is "Cash deposit, shop takings" and card machine sales (`ayanda-home-dark-390.png`). **Assumption:** a spaza owner pays suppliers and family partly in cash before banking, so the bank balance understates both income and spend; the effect size is unknown and could be large. - **Recommendation:** Add a manual "cash in hand" and "money I owe or am owed" entry, and test with a cash heavy user. - **Effort:** M

INV-11. Placeholder brand and unconfirmed name - **Severity:** Low - **Evidence:** Logo and icons are placeholders, and the name's Tshivenda meaning is "to be confirmed by a Venda speaker" (`BRIEF.md` section 1; `docs/review/prep/fact-sheet.md`). - **Recommendation:** Confirm the name with a first-language speaker and make a simple wordmark before submission. - **Effort:** S

INV-12. Dependency vulnerabilities and one large bundle - **Severity:** Low - **Evidence:** npm audit shows 2 critical and 5 high (mostly dev), 2 moderate in production (react-router); one 696 KB JavaScript file including all locales (`docs/review/prep/fact-sheet.md`, `docs/review/prep/audit.md`). - **Recommendation:** Update react-router and split locale files so a judge's due diligence checklist comes back clean. - **Effort:** S

Top three risks

1. **The bank builds it.** The feature is a few hundred lines on top of data the bank already owns, and FNB has had a forward looking "available funds" view since 2018. If the idea is good, the cheapest path for a bank is to copy the framing, not to buy Tengo (INV-02).
2. **Users do not trust or understand the date.** No one has used it. On the demo's own Short persona, three numbers on one screen disagree. A wrong "safe to spend" for someone already short is the exact harm the product claims to prevent (INV-01, INV-03, INV-06).
3. **No data, no product.** Without a bank feed the product depends on users exporting CSV statements, which this segment is unlikely to do repeatedly. **Assumption:** repeat manual import rates below 10 percent of users per month (INV-04).

Ten hardest questions

1. **Question:** How many real irregular earners have used this, and what did they do differently? **Best honest answer today:** None. It is a demo with fictional data. The design is based on three composite personas and published research on income volatility, not on sessions with users. **What would strengthen it:** Five to eight recorded sessions with drivers, hairdressers or spaza owners, with quotes and a simple measure such as "read the date correctly" and "said they would change a spend today".
2. **Question:** How accurate is the date? **Best honest answer today:** We do not know on real data. On synthetic data the engine is consistent with its own rules, and it shows a range band and a confidence level instead of pretending to be exact. **What would strengthen it:** A backtest on real, consented statements reporting mean error in days, how often the actual run-out date falls inside the band, and how often the state was right in hindsight.
3. **Question:** FNB already shows money left after scheduled debits, and budgeting apps show "safe to spend until payday". What is new? **Best honest answer today:** The framing and the segment. Those tools assume a payday; Tengo infers next money from irregular deposits (weekday rhythm or gap percentiles), shows a range, and turns it into Enough, Tight or Short. The arithmetic is not new. **What would strengthen it:** Evidence that irregular earners misread existing tools and read Tengo correctly, from a side by side test.
4. **Question:** Why would a bank buy this instead of building it in a sprint? **Best honest answer today:** It probably would build it. Tengo's value is a tested design, copy and explanation layer that is ready to drop in, plus the irregular income detection rules. That is a licence or acqui-hire case, not a moat. **What would strengthen it:** A named bank sponsor, a pilot letter of intent, or a design that has been through user testing in several languages that the bank has not done.
5. **Question:** Where does the data come from? **Best honest answer today:** Today, a CSV upload parsed in the browser. In production, from the host bank's own transaction feed. Open finance in South Africa is not yet a reliable third party route. **What would strengthen it:** A one page integration spec and a working import of a real bank's CSV and PDF statement formats.
6. **Question:** Who pays, and how much? **Best honest answer today:** Not decided. The users cannot be the payers. The likely model is a per active user licence to a bank or a funder for financial inclusion. Any price is an assumption. **What would strengthen it:** A bank value hypothesis with numbers from public sources, for example fee and failed debit order costs for this segment, and a pilot to measure it.
7. **Question:** Your Short persona is told R310 is safe to spend today and the chart says Enough. Which number should she believe? **Best honest answer today:** That is a real inconsistency in the demo. The figures come from three rules: run-out without income, balance spread to the latest next money, and a 30 day path that includes income. It should be fixed so one rule drives the screen. **What would strengthen it:** Fix it before submission and show the fixed screen in the demo.
8. **Question:** Is "safe to spend" or "safe to lend" financial advice, and who is liable if it is wrong? **Best honest answer today:** It is presented as an estimate, not advice, but the wording is close to advice and has not had legal review. Inside a bank it would sit under the bank's own disclosure and compliance process. **What would strengthen it:** A legal opinion on the wording, and renamed outputs that describe consequences ("you would stay Tight") rather than permissions ("safe").
9. **Question:** Your users run on cash, stokvels and family transfers. How does a bank statement capture that? **Best honest answer today:** It does not. The engine only sees bank transactions, so cash income and cash spend are invisible. **What would strengthen it:** A manual cash entry, and data from user sessions on what share of money moves outside the account.
10. **Question:** You say 11 official languages. Can I see isiZulu? **Best honest answer today:** No. The structure for 11 languages exists and every key is in place, but all 10 non-English files are still English. Translation has not been run or reviewed. **What would strengthen it:** Two languages translated and checked by first-language speakers, and an honest stat on the landing page until the rest are done.

Deal breakers

- Evidence after testing that users read the date as a payday or a deadline and make worse decisions because of it, or that a Short user acts on "safe to spend".
- Backtest on real statements showing the actual run-out date falls inside the band much less often than the band implies, with no fix in sight.
- No bank or funder willing to host it, since there is no viable consumer revenue in this segment and no third party data route.
- A bank shipping the same view inside its own app before a pilot, which removes the only distribution path.
- Continued public claims that are not true (11 languages, "machine made") after being flagged.

What would change my mind

- Recorded sessions with 5 to 8 real irregular earners where most read the date correctly and describe a specific decision they would change.
- A backtest on real, consented data with error in days and band coverage that is reported honestly, even if modest.

- A letter of intent or named sponsor at a bank for a small pilot, with an agreed success measure (for example fewer failed debit orders or unpaid fees in the pilot group).
- The three contradictions on Home fixed and at least two languages properly translated.

Is the core insight defensible

As a technology, no. The date is balance divided by a spend rate, minus known bills, which any bank with transaction history can compute, and FNB already computes the after-debits figure (BusinessTech, 11 June 2018). As a product decision for irregular earners, it is partly defensible: detecting next money without a payday, showing a range, and saying Enough, Tight or Short in plain language in many languages is real design work that a bank may not prioritise for this segment. Defensibility would come from user evidence, tested copy in 11 languages, and a bank relationship, not from code. The likely outcome if the idea proves out is that a bank copies or licenses it. The founder should pitch for that outcome openly.

Score

4 out of 10. A careful, honest prototype of an easily copied feature, with no users, no data route, no buyer and visible contradictions on its key screen.

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4.13 Hackathon judge panel

Reviewer file: `docs/review/judges.md`. Findings are listed in full in Appendix A.

Simulated judging panel. We are not FNB judges, have no access to FNB's internal rubric or weighting, and score only from the code, the shared preparation in `docs/review/prep/`, the screenshots, and the twelve other reviewer files, which this panel was told to read.

Criteria used. The brief names seven criteria: Product, Creativity, Usability, Sustainability, Impact, Presentation and Intelligence. The public FNB App of the Year FAQ (https://appoftheyear.co.za/faqs/, accessed 2026-10-20) defines the same seven for the App of the Year awards and says round 1 judges "play around" with apps loaded on devices. I did not find a hackathon-specific scoring sheet for 2026, or anything public about the "Financial inclusion and money confidence" track, so I applied the FAQ wording to this entry as follows:

Criterion	Public FAQ wording (summarised)	How this panel applied it
Product	Complete, polished, works as a user would expect, solves the stated need, market need, traction	Does the build work end to end and answer "is my money enough until the next money" correctly and consistently? Traction counts as zero today.
Creativity	New application or service, creative approach, new approach to an old problem	Is the date-first, irregular-income framing new compared with what South African bank apps publicly offer?
Usability	Easy to use, clear, simple, intuitive, obvious what to do next	First two minutes on a phone for each persona, plus accessibility and offline evidence.
Sustainability	Potential of app and team, next level, path to revenue	Business model, data route, defensibility, readiness for a real backend.
Impact	Size of problem, indispensability, importance	Size and fit of the segment, minus the risk of harm and the parts of the segment the product cannot see.
Presentation	How well the team communicates the idea and its relevance to society	What a judge sees without the founder: landing page, in-app story, README, and honesty of claims. No pitch deck or demo script was in the repository, so this is scored on the product's own story.
Intelligence	Can the app forecast and predict likely future behaviour by analysing patterns in data	Quality of the forecast engine, its validation, and how it behaves when the patterns break.

Verdict

Tengo is a well-made, honest prototype with one clear and good idea: tell an irregular earner the date their money runs out and whether it reaches the next money, not just a balance. It works end to end, it is fast and accessible by tool measures, and its engine is pure, tested and explainable, which maps directly onto the Intelligence criterion. It would lose points in judging for three reasons that are visible within a minute of use: the Home screen contradicts itself (Short beside "Safe to spend R310" and a green "Enough" on the chart, "Enough until" beside a Tight chip, the latter also on the landing hero); the landing page claims 11 official languages when all 10 non-English files are English; and some outputs could cause harm (it tells Thabo to pay his car instalment late). Most of these are copy and rule fixes measured in hours. Sustainability and Impact are capped by things a hackathon cannot fully fix (no users, no data route, a thin single-bank revenue case), but an honest business page and a handful of user sessions would move both. On the evidence today we would shortlist it in round 1 and expect it to struggle in a final unless the Home contradictions and the language claim are fixed first.

What works

- **A sharp, explainable core answer.** The date is the largest element on Home at every width, with state shown by word, icon, colour and bar count (`ayanda-home-dark-1280.png`, `thabo-home-light-390.png`; UX "What works"; `src/components/StateChip.tsx`). Product lead and commercial reviewers each found no verified South African bank feature that projects a run-out date against irregular income cadence (PROD verdict, COMM "What works").
- **Intelligence that is real, not decorative.** The engine detects weekday and interval income rhythms, recurring bills, and rolling spend percentiles (`src/engine/forecast.ts`, `classify.ts:16`). On the backtest, with 90 days of history, mean absolute error is 2.1 days against a naive baseline of 3.4, with near-zero bias and 71.8 percent band coverage (`docs/review/backtest/results/summary_headline.csv`, data scientist "What works"). The engine has 97.6 percent statement coverage and 66 passing unit tests (`docs/review/prep/coverage.txt`, `checks.md`).
- **Polish and accessibility above hackathon norm.** Lighthouse mobile landing 94 performance and 100 accessibility; 221 Playwright tests pass; axe finds 1 violation over 15 routes in both themes (fact sheet; `docs/review/prep/playwright.txt`; A11Y "What works"). Reduced motion, reflow at 320 px and target sizes all pass (A11Y "What works").
- **Honest demo labelling.** "Demo | Fictional Data" appears on every app screen and simulated privacy and messaging are labelled (`landing-dark-1280.png` footer; LEGAL "What works"; UX "What works").
- **Features that fit the segment.** The family loan check and daily safe-to-spend unit match how informal earners decide (BEH "What works", citing Hershfield, Shu and Benartzi 2020). The privacy centre demonstrates consent, export, deletion and human review in working screens (SEC "What works").
- **Built to be embedded.** `TengoRepository` and `MessageWriter` give a bank the seams to supply data and channel (PROD, COMM, TECH "What works").

Criterion scores

Criterion	Score (0 to 10)	Justification and evidence	Biggest gap	Single change that would raise it most
Product	6	Complete and working across 10 screens, both themes and three widths (fact sheet "Screens", 221 Playwright passes). Answers its one question for all three personas. Loses points because the main screen gives conflicting answers (UX-01, UX-02, UX-03, PROD-03, INV-03), data never ages after import so the state drifts to Enough (TECH-01), and there is no traction or real data (INV-01, DATA-01).	The headline, safe to spend figure and chart disagree on the screen judges see first.	Make one rule drive Home: state-specific hero label, a daily limit tied to the state, no state word on the chart readout (JUDGE-01).

Criterion	Score (0 to 10)	Justification and evidence	Biggest gap	Single change that would raise it most
Creativity	7	Date-first framing for people without a payday, an inferred "next money" window, three plain states and a loan-to-family check are not publicly offered by the SA banks reviewed (PROD verdict, COMM competitor table). Not a 9 because the arithmetic is simple and "safe to spend until payday" exists in other apps (INV-02, COMM-01), and the date-beats-balance claim is untested (BEH-08).	The idea's novelty rests on the framing; the cash and stokvel reality that would make it uniquely South African is missing (BEH-02, BEH-07).	A one-tap "cash in hand" entry that feeds the forecast (JUDGE-08).
Usability	6	Strong base: clean hierarchy, designed empty and error states, keyboard and drawer patterns, Nielsen average 3.6 of 5 (UX heuristic table). Pulled down by the lowest heuristic, consistency of numbers (2 of 5), about 20 figures on Ayanda's first screen (A11Y-05), three different next-money dates (UX-04), a pre-filled Plan that shows a loss before the user acts (UX-06), and focus hidden under the phone tab bar (A11Y-04). Offline: the app's own test passes an offline reload (`tests/flows.spec.ts:131`) and the accessibility reviewer found a blank page offline after one visit (A11Y-02). Both are true: offline works only from the second load (TECH-05).	A stressed user cannot tell which number to believe.	Same as Product: reconcile Home (JUDGE-01), then start Plan empty (JUDGE-10).
Sustainability	4	Right shape for a bank licence (COMM "What works", PROD "What works") and cheap to run if messages go in-app (COMM-03). But no business model in the repo (COMM-11, INV-07), no data route beyond CSV (PROD-06, INV-04), no moat against the bank building it (COMM-01, INV-02), and single-bank economics short of break-even at the base fee (COMM-02: about 44 000 MAU against about 82 000 needed at R5, all marked Assumption in `model-inputs.md`). The repository contract is not ready for a real backend (TECH-03).	No stated buyer, price or data route.	A one-page business case: bank-embedded licence, pilot KPIs, data contract, honest moat (JUDGE-06).
Impact	5	The problem is large: 3.6 million in the informal sector and only 18.3 percent of the poorest 40 percent can easily raise emergency money (COMM "What works", Stats SA and World Bank sources). But the product sees one bank account while the segment lives in cash and stokvels (BEH-02, COMM-04, INV-10), the engine shows Enough when income has stopped (RISK-01, RISK-03), lumpy earners shown Enough ran short 11.6 percent of the time on synthetic data (DATA-02), some advice could harm (LEGAL-01, RISK-09), and the multilingual reach is not real yet (L10N-01). No user has tested it (INV-01).	Possible harm and no evidence of benefit.	Remove harmful suggestions (JUDGE-03), then run 5 to 8 user sessions and quote them (JUDGE-07).
Presentation	5	The tagline "A balance tells you what you have. Tengo tells you until when" is the strongest line in the entry (COMM, INV "What works"), and the landing page explains three steps clearly (`landing-dark-1280.png`). But the landing hero itself shows "Enough until Tue 27 Oct" beside "Tight" (UX-03), the stats row claims "11 Official languages" (COMM-08, INV-05, L10N-01) and "0 Bank logins needed", which conflicts with the bank-embed story (COMM-07). Privacy screens show green ticks beside controls that do not exist (SEC-02). No pitch deck, demo script or user quotes are in the repository.	The first screen a judge sees contains a contradiction and an untrue claim.	Fix the landing hero label and replace the language stat with an honest one (JUDGE-02), then rehearse a demo built on the fixed screens (JUDGE-12).
Intelligence	6	This is the criterion the engine was built for: it analyses patterns to forecast income timing and run-out, and the backtest shows it beats a naive baseline with 60 or more days (DATA "What works"). Held back because validation is circular (DATA-01: Ayanda's shipped history is one path), the band is too narrow near run-out (DATA-03: 53.5 percent coverage within 6 days), short imports are worse than the baseline (DATA-04), and pattern breaks are not detected: stopped income, payday morning, windfalls (RISK-01, RISK-03, RISK-05, RISK-10, DATA-05).	The engine assumes the pattern holds when the point of the product is that it often does not.	Add pattern-break guards (missed paydays, no income, payday today, stale data) with tests, and show the backtest honestly (JUDGE-04, JUDGE-05).

Total: 39 of 70. Average: 5.6 of 10.

The spread is deliberate. Creativity is highest because the framing is distinct. Sustainability is lowest because nothing in the repository answers who pays or where real data comes from. Reviewer scores ranged from 4 (credit risk, localisation, commercial, investor) to 6 (technology, UX, accessibility), which matches this panel's split between a strong build and a weak case.

What to improve

Ranked by expected judging score gain per hour of work before submission. Effort is for the change itself, not for every underlying finding.

JUDGE-01 Make one rule drive the Home screen

- **Severity:** High
- **Evidence:** Ayanda: hero "Short from Fri 23 Oct", card "Safe to spend today R310", chart readout "Enough" (`ayanda-home-dark-1280.png`, seen by this panel). Thabo: "Enough until" above a "Tight" chip (`thabo-home-light-390.png`, `landing-dark-1280.png`). See UX-01, UX-02, UX-03, UX-04, PROD-03, INV-03, BEH-04, LEGAL-08, RISK-04, A11Y-03. Seven reviewers raised it independently. The credit risk reviewer notes R310 is in fact a conservative ceiling (following it leaves R17 on 26 Oct, RISK "What works"), so the fix is mainly the label and placement, not the maths.
- **Recommendation:** State-specific hero label ("Lasts until (on a normal week)" for Tight). For Short, show "To reach Mon 26 Oct, keep to R310 a day. You usually spend R620." Drop the state word from today's chart readout or say "Enough if money lands as usual" in the readout and in `aria-valuetext`. Show one next-money date. Re-capture the landing mock.
- **Effort:** S (about half a day). Raises Product, Usability and Presentation.

JUDGE-02 Replace the 11-languages claim and fix the page language

- **Severity:** High
- **Evidence:** Landing stat "11 Official languages" (`landing-dark-1280.png`); all 10 non-English files are byte-identical to English (L10N-01); `html lang` says Tshivenda over English text (A11Y-01, L10N-05); the translation script cannot produce Tshivenda (L10N-02). See COMM-08, INV-05. A judge who picks isiZulu sees English and discounts the rest.
- **Recommendation:** Change the stat to "Built for 11 languages" or "English now, more coming". Set `lang="en"` while a locale is pending. If time allows, have a first-language speaker write the money keys (state, hero, safe to spend, messages) for one Gauteng language and mark it reviewed.
- **Effort:** S for copy and `lang` (under an hour); M for one reviewed language. Raises Presentation and Impact.

JUDGE-03 Remove the outputs that could cause harm

- **Severity:** High
- **Evidence:** "Pay Car instalment (R3 450) after Tue 27 Oct instead of before" (`thabo-home-light-390.png`; `src/engine/help.ts:62` sorts by amount only). "Safe to lend up to" with a one-tap "Use" button; Lerato told she can spend R330, set aside R280 and lend R280 from the same buffer. The only "not advice" line is at the

foot of a drawer. See LEGAL-01, LEGAL-02, RISK-09, RISK-11, BEH-01, BEH-03, BEH-05, PROD-04, INV-09.

- **Recommendation:** Never suggest moving credit, insurance, funeral or rent payments; add a unit test. Rename "Safe to lend up to" to "You could lend up to" with "and stay Tight". Exclude stock and fuel from the daily cut. Put "An estimate from your past money, not advice" under the hero, Plan and help card.
- **Effort:** S to M (about a day). Raises Impact and Product, and removes the question a judge is most likely to end on.

JUDGE-04 Teach the engine to notice when the pattern breaks

- **Severity:** High
- **Evidence:** Income stopped 35 days ago still gives Enough with high confidence and R1 990 safe to lend (RISK-01, S11). No income at all can show Enough (RISK-03; `forecast.test.ts:116` asserts it). Payday morning shows Short (RISK-10). Imported data never ages, so Ayanda drifts from Short to Enough after 30 days (TECH-01). A missed payout raises false reassurance from 7.6 to 36.8 percent (DATA-05).
- **Recommendation:** Add four guarded rules with tests: missed paydays mark the pattern "paused" (state at most Tight, lend R0); no pattern shows "No regular income found"; "Payday today, not in yet" instead of Short; store an `asOf` date and show "Your data is N days old". Show one of these in the demo; a judge scoring Intelligence will value an engine that knows when not to trust itself.
- **Effort:** S each, about a day together. Raises Intelligence and Impact.

JUDGE-05 Put the backtest in the pitch, with its limits

- **Severity:** Medium
- **Evidence:** MAE 2.1 days against a naive 3.4, band coverage 71.8 percent, Enough reliable for regular earners (DATA "What works", `docs/review/backtest/results/summary_headline.csv`, `charts/calibration.png`). But all of it is synthetic and partly circular (DATA-01), and Tight was followed by running short 24.4 percent of the time (DATA-02). No competitor publishes accuracy for irregular income (COMM-01).
- **Recommendation:** One slide or About section: "On synthetic data: within about 2 days, beats a simple baseline, worst for lumpy earners. Not yet tested on real statements." Rename "Likely between" to state its coverage. Optionally tighten Tight copy to say running short is possible.
- **Effort:** S (two hours). Raises Intelligence and Presentation.

JUDGE-06 Write the one-page business case and pick one distribution story

- **Severity:** Medium
- **Evidence:** No buyer, pricing or metrics in README or BRIEF (COMM-11, INV-07). Landing claims "0 Bank logins needed" while the only realistic route is inside a bank app (COMM-07, PROD-06, INV-04). Overlap with nav» Money and Standard Bank Future Payments is unaddressed (PROD-02). Single-bank economics need a platform floor (COMM-02); daily paid WhatsApp costs more than the likely fee (COMM-03).
- **Recommendation:** Add a page: B2B2C licence embedded in a bank app, pilot fee plus per-active-user fee plus floor, in-app push by default, a "bank already has versus Tengo adds" table, a data contract (balance, facility limit, debit orders, 60 days of transactions; PROD-01), and four pilot KPIs. State plainly that the moat is tested copy, accuracy evidence and segment design, not code.
- **Effort:** S (half a day). Raises Sustainability by the most of any single change.

JUDGE-07 Run five to eight user sessions and quote them

- **Severity:** Medium
- **Evidence:** No users, interviews or sessions anywhere in the repo (INV-01, COMM-05). The date-first claim is a hypothesis (BEH-08). The behavioural reviewer wrote a ready plan with tasks, pass criteria and an assumed budget of about R2 000 to R2 500 (BEH user-testing plan).
- **Recommendation:** Run the BEH plan's T1 and T2 with six drivers, hairdressers or traders after JUDGE-01 and JUDGE-03 land. Report how many read the date correctly and quote three people.
- **Effort:** M (two days). Lower score per hour than the fixes above, but the only change that moves Impact and Product traction with evidence.

JUDGE-08 Add cash in hand

- **Severity:** Medium
- **Evidence:** The engine sees one bank account; FinScope 2023 reports 94 percent withdraw cash monthly (BEH-02); two of three personas earn cash (COMM-04, INV-10); stokvels and family transfers are absent (BEH-07). The product lead would pilot gig earners paid into the account first (PROD-07).
- **Recommendation:** A "Cash in hand" figure updated in one tap and a quick money in or out entry that feeds the forecast, with "Based on your bank account only" under the date until it is used.
- **Effort:** M (one to two days). Raises Creativity and Impact; it is the clearest answer to "why not just a bank feature".

JUDGE-09 Make offline work on the first visit

- **Severity:** Medium
- **Evidence:** The service worker precaches only `/`, the manifest and icons, and registers on `load`, so first-visit JS and CSS are not cached (A11Y-02, TECH-05). The repo test reloads online before going offline (`tests/flows.spec.ts:131–140`), so it passes while a first-time user offline gets a blank page. Both reviewers are right; they tested different sequences.
- **Recommendation:** Precache the build output at install (vite-plugin-pwa or a generated manifest), add a plain offline fallback, and change the test to go offline straight after the first visit. Then the loadshedding claim holds in a live demo.
- **Effort:** S (two to three hours). Raises Usability and Impact.

JUDGE-10 Small first-impression fixes on Plan, consent and privacy

- **Severity:** Low
- **Evidence:** Plan opens with a R500 what-if and R300 loan already applied (UX-06). Daily messages default on at consent (UX-09, LEGAL-03, SEC-06). Privacy shows green ticks beside encryption and access claims and a retention rule that never runs (SEC-01, SEC-02). Tight and Short are almost the same colour in light theme (UX-05).
- **Recommendation:** Start Plan empty, default messages off, swap ticks for a neutral "planned" icon and write future controls as "would", shift light Tight to amber.
- **Effort:** S (two hours together). Raises Usability and Presentation, and avoids a judge reading simulated controls as real claims.

JUDGE-11 Confirm the name and clear the brand basics

- **Severity:** Low
- **Evidence:** The Tshivenda meaning of the name is unconfirmed and may mean "price" rather than "value" (L10N-12, INV-11); the heading font needs a commercial licence and the logo is a placeholder (COMM-10).
- **Recommendation:** Confirm the name with two Tshivenda speakers before telling the name story on stage.
- **Effort:** S. Raises Presentation.

JUDGE-12 Script the demo around the fixed screens

- **Severity:** Low
- **Evidence:** No demo script, deck or video is in the repository. The strongest moments are Ayanda's Short warning paired with an action, the Why drawer and the loan check (UX walkthrough, BEH "What works"); the weakest are the language picker and the help card (walkthrough items 5 and 10).
- **Recommendation:** Three minutes: tagline, Ayanda Short with the reconciled daily limit, Why this date, loan check on Thabo, one pattern-break case (JUDGE-04), backtest slide with its caveat, business page. Avoid the language picker unless JUDGE-02 shipped a reviewed language.
- **Effort:** S. Raises Presentation.

Top three risks

1. **A judge finds the contradiction first.** The landing hero and Ayanda's Home disagree with themselves; a judge who notices will doubt the forecast even where it is right (JUDGE-01; UX-01 to UX-03, INV-03).
2. **A false claim costs credibility across all criteria.** "11 Official languages" and green privacy ticks are checkable in seconds and are not true (JUDGE-02, JUDGE-10; L10N-01, SEC-02).
3. **The judge ends on harm.** "Pay your car instalment late" or "Safe to lend" to someone already Tight turns an inclusion pitch into a liability question (JUDGE-03; LEGAL-01, RISK-09, RISK-11).

Score

6 out of 10 (criterion average 5.6). A distinct, well-built and honest prototype whose main screen contradicts itself and whose case for revenue, reach and real-world accuracy is not yet made; the top five fixes are about three working days and would move it to roughly 7.

Sources

- FNB App of the Year, FAQs (judging rounds and the seven criteria): https://appoftheyear.co.za/faqs/ (accessed 2026-10-20)
- FNB App of the Year, 2025 Hackathon programme (found in search; listed as the closest hackathon document, covers schedule rather than scoring, not used for criteria): https://appoftheyear.co.za/wp-content/uploads/2025/10/2025-Hackathon-programme.pdf (accessed 2026-10-20)
- All other external facts are taken from the other reviewers' files, where their sources and access dates are listed.

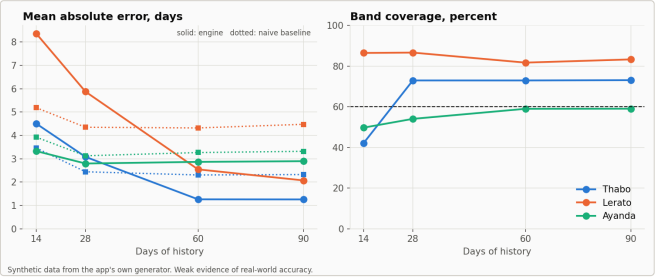
5 Forecast backtest

The data scientist backtested the engine on held-out days generated by the app's own seed generator (method and code in `docs/review/backtest/`). The credit risk analyst separately stress-tested edge cases the generator does not produce. The synthetic data and the engine share assumptions, so these results show internal consistency and relative behaviour, not real-world accuracy (DATA-01).

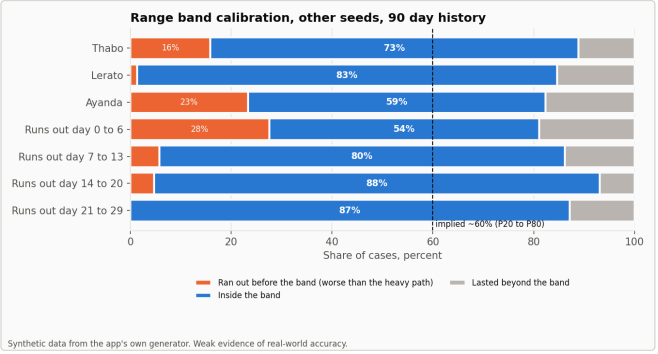
5.1 Headline results

Slice	Cases	MAE days	Naive MAE	Within 3 days	Band coverage	State accuracy	False reassurance
shipped generator, 60 day history, all balances	432	1.94	3.36	76.1%	65.7%	70.6%	7.6%
other seeds, 90 day history, all balances	2160	2.07	3.37	78.1%	71.8%	71.9%	7.6%
other seeds, 60 day history, all balances	2160	2.22	3.29	78.2%	71.2%	71.3%	8.4%
shipped, 60 day, engine-independent balances only	216	2.03	3.84	75.8%	70.2%	91.7%	0.5%
shipped, 60 day, startingBalance targets only	216	1.86	2.88	76.4%	61.1%	49.5%	14.8%

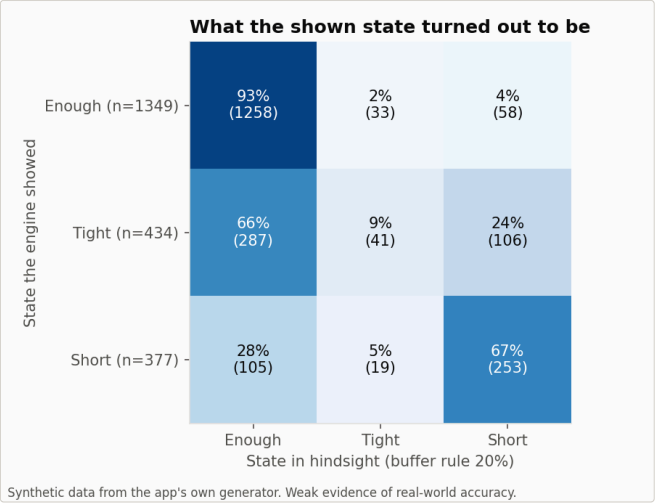
Source: `docs/review/backtest/results/summary_headline.csv`. MAE is mean absolute error in days between the forecast runway date and the actual date computed the same way on held-out data. Band coverage is how often the actual date fell inside the stated range. State accuracy compares the state shown with hindsight using the data scientist's 20 percent buffer rule.



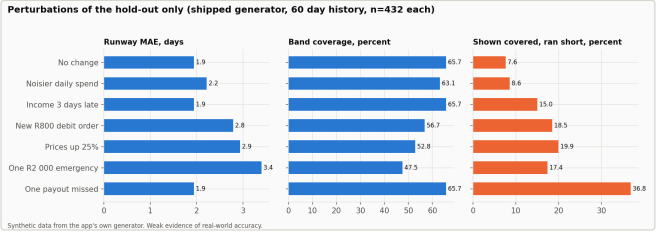
Error grows quickly below 28 days of history.



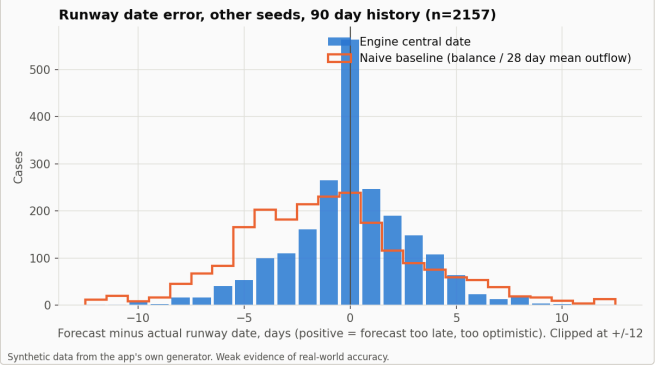
Calibration of the range band.



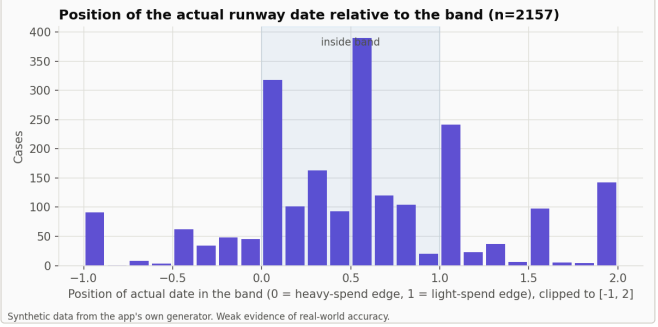
State shown against hindsight.



Shocks the generator does not produce.



Distribution of errors in days.



Where the actual date fell relative to the band.

5.2 How to rerun

Backtest of `src/engine/forecast.ts` against synthetic histories from `src/data/seed/generate.ts`. Findings and definitions are in `docs/review/data-scientist.md`.

Read this first. The generator and the engine were written by the same author with matching assumptions (weekly or interval income, monthly bills on a fixed day, stationary everyday spend). Good numbers here show the engine is internally consistent with the world it was designed for. They are weak evidence of accuracy on real bank statements.

How to rerun

From the repository root:

```sh
# 1. Backtest (about 1 minute). Writes CSV files to docs/review/backtest/results/.
npx vite-node docs/review/backtest/backtest.ts

# 2. Bootstrap intervals for the headline metrics (standard library only).
python3 -I docs/review/backtest/bootstrap.py

# 3. Charts. Needs matplotlib in any Python 3 environment.
python3 -m venv /tmp/tengo-bt && /tmp/tengo-bt/bin/python -m pip install matplotlib
/tmp/tengo-bt/bin/python -I docs/review/backtest/charts.py
```

Everything is deterministic: same code, same results. The script imports the real engine and generator and changes nothing in `src/`.

Design in one paragraph

3 personas x 24 cut-off dates (every 15 days from 2025-10-06, so they cover 12 months and rotate through all weekdays). Two data sources: **shipped** is the app's own `generateHistory(persona, cutoff + 30)` split at the cut-off (60 days history, 30 days hold-out); **seeds** is a verified line-for-line copy of the generator's persona rules run with 5 other seeds per cut-off (120 independent paths per persona, 90 days history, 30 days hold-out). The script checks that the copy reproduces `generateHistory` exactly before running. History is truncated to 14, 28, 60 and 90 days and reclassified on the history alone. Six balances per case: three from the app's `startingBalance` (enough, tight, short targets) and three engine-independent ones (7, 14, 21 days of mean daily outflow). Perturbations (shipped, 60 days) change only the hold-out.

Files

File	What it holds
`backtest.ts`	The backtest.
`bootstrap.py`	95 percent bootstrap intervals, resampling independent paths.
`charts.py`	Charts from the CSV results.
`results/cases.csv`	One row per case (12 528 rows), every input and outcome.
`results/summary_headline.csv`	Headline slices.
`results/summary_by_persona*.csv`, `summary_by_history.csv`, `summary_by_persona_history*.csv`	Breakdowns.
`results/summary_by_lead_time.csv`	By how many days until money actually ran out.
`results/summary_by_scenario*.csv`	By balance scenario.
`results/perturbation*.csv`	Shocks the generator never produces.
`results/state_confusion.csv`	Shown state against hindsight state.
`results/calibration_points.csv`	Signed errors and band position per observed case.
`results/generator_diversity.csv`	Distinct random paths the shipped generator produces across cut-offs.
`results/bootstrap_ci.csv`	Confidence intervals.
`results/console.txt`	Console output of the last run.
`charts/*.png`	Error distribution, calibration, error by history, state confusion, perturbation, band position.

Column meanings in `cases.csv`

Day indices count from the cut-off (day 0 is the cut-off day, the engine's "today"). 180 means beyond the engine horizon.

- `fcCentral`, `fcBandFrom`, `fcBandTo`: engine central zero date and band (heavy path to light path).
- `actualRunway`: first day the balance minus actual outflows (spend and bills, no income) is zero or below; -1 if not within 30 days.
- `err`: `fcCentral − actualRunway`. Positive means the forecast was too late (too optimistic).
- `baselineErr`: same for the naive baseline, balance divided by the mean daily outflow of the last 28 days.
- `bandCovered`: 1 inside the band, 0 outside, empty when undecidable (actual not within 30 days and band reaches past day 29).
- `ranOutWithIncome`: first day the real balance, including real income after day 0, is zero or below.
- `nmActual`: first day from day 1 with income at least the engine's significance threshold (10 percent of typical monthly income from the history).

- `hindState`: Short if the outflow-only balance reached zero before `nmActual`; Enough if at least 20 percent of the outflows of that period was left; Tight otherwise. `hindState10` and `hindState30` use 10 and 30 percent.
- `realShort`: 1 when the real balance with all real income hit zero before the next money.

5.3 Edge cases (credit risk stress test)

Source: `docs/review/tools/credit-risk/results.md` and `stress.ts`, run against the real engine.

Id	Scenario	State	Headline	Safe to spend (R)	Versus usual spend	Safe to lend (R)	Observation
S01	Baseline: weekly Friday pay, steady spend, rent on 1st, 90 days	enough	Enough until 2026-10-30	300	257%	690	Reference case.
S02	Zero income for 90 days, R4000 balance	short	Short from 2026-11-01	50	43%	0	No income at all, yet next money is assumed on 2026-11-19 and the state is short. Safe to spend spreads the balance over 30 days.
S02b	Zero income for 90 days, R9000 balance	enough	Enough until 2026-12-01	215	184%	2670	With no income and rent due, the app says enough and offers R2670 as safe to lend.
S03	Overdrawn: balance -R800	short	Short from 2026-10-20	0	0%	0	State short, safe to spend R0. Overdraft depth is not shown and the help plan target is tight.
S04	Windfall R40000 twenty days ago on top of weekly R2000 pay	short	Short from 2026-10-30	0	0%	0	Monthly income is inflated so the 10% threshold rises above the weekly pay. Pattern becomes unknown, next money 2026-11-19, amount 0.
S04b	Loan disbursement R15000 twenty days ago, counted as income	enough	Enough until 2026-10-30	300	257%	690	A loan is treated as income. Next money high amount R2000; the repayment is unknown to the engine.
S05	Missed bill: rent paid Aug and Sep, October rent unpaid (arrears)	enough	Enough until 2026-10-30	300	257%	690	Next rent assumed 2026-11-01. The unpaid October rent (R2500 owed now) is invisible; bills before next money R0.
S05b	Missed bill mid history: rent Aug and Oct, September skipped	enough	Enough until 2026-10-30	300	257%	680	Rent no longer meets the 24 to 36 day rule, so it becomes everyday spend: spend, spend. No November rent is projected; heavy daily spend 128.
S06	Month-end spike: R600 a day extra from the 25th, today 24 Oct	enough	Enough until 2026-11-16	425	333%	350	Central rate 128 a day ignores the known spike; over the next 7 days true spend is about R4370. Engine headline 2026-11-16; a realistic path is below zero by about R1260 before the 31st.
S07-5	Thin data: 5 days of history	short	Short from 2026-11-01	50	41%	0	Band width 0 days at low confidence. Pattern unknown.
S07-7	Thin data: 7 days of history	short	Short from 2026-11-02	50	44%	0	Band width 0 days at low confidence. Pattern unknown.
S07-14	Thin data: 14 days of history	enough	Enough until 2026-11-02	500	444%	1150	Band width 0 days at low confidence. Pattern interval.
S07-20	Thin data: 20 days of history	enough	Enough until 2026-11-02	375	337%	1040	Band width 1 days at low confidence. Pattern weekday.
S08	Outliers: two R9000 one-off spends in the last 60 days	tight	Enough until 2026-11-01	750	590%	2610	Heavy daily spend 1393 versus baseline; one-offs are treated as habitual and drive state.
S08b	Outlier in thin data: R6000 repair with 20 days of history	tight	Enough until 2026-11-14	750	648%	2650	Central daily spend 116: one repair dominates the median of 14 rolling windows.
S09	Holiday: no income for the last 21 days, weekly pattern before	enough	Enough until 2026-11-01	375	321%	990	Engine still expects pay on 2026-10-23 (pattern weekday). Nothing warns that three paydays were missed.
S10	Bill rises 30%: rent R2000, R2000, then R2600	enough	Enough until 2026-11-13	750	609%	1480	Rent rows classified as spend, spend, spend. Projected rent: none.
S10b	Bill rises 15%: rent R2000, R2000, then R2300	enough	Enough until 2026-11-01	750	643%	2490	Projected rent R2000: the median of the last three lags the new R2300.
S11	Income stopped 35 days ago (weekly pay before that)	enough	Enough until 2026-11-01	625	536%	1990	Five missed paydays, still pattern weekday with next money 2026-10-23, amount 2000.
S11b	Lumpy deposits every 7 to 9 days, none in the last 40 days	enough	Enough until 2026-11-01	830	712%	2110	Overdue rule restarts the window from tomorrow: next money 2026-10-21 to 2026-10-23.
S12	Payday is today (Tuesday), payout not yet in balance	short	Short from 2026-10-23	50	43%	0	Next money pushed a week to 2026-10-27; state short on a normal payday morning.
S12b	Payday is today, payout already in balance (R400 + R2000)	enough	Enough until 2026-11-01	300	257%	1380	With the pay in the balance the state is enough.
S13	Rent R3000 due 25 Oct, the day after the latest next money (24 Oct)	enough	Enough until 2026-10-25	400	343%	1090	Safe to spend R400 empties the account by the 24th, while rent of R3000 is due on the 25th and weekly pay is only about R2000.
P-thabo	Persona thabo as seeded	tight	Enough until 2026-10-27	295	100%	320	Safe to spend is 100% of usual spend. Following it every day leaves R32 on 2026-10-27. Small daily income ignored by the headline: R107 a day.
P-lerato	Persona lerato as seeded	enough	Enough until 2026-11-01	330	274%	280	Safe to spend is 274% of usual spend. Following it every day leaves R18 on 2026-10-25. Small daily income ignored by the headline: R170 a day.
P-ayanda	Persona ayanda as seeded	short	Short from 2026-10-23	310	50%	0	Safe to spend is 50% of usual spend. Following it every day leaves R17 on 2026-10-26. Small daily income ignored by the headline: R57 a day.

Condensed to the columns that carry the finding. Full table with balances, patterns, spend rates and bills: docs/review/tools/credit-risk/results.md and results.csv.

5.4 What synthetic data does and does not prove

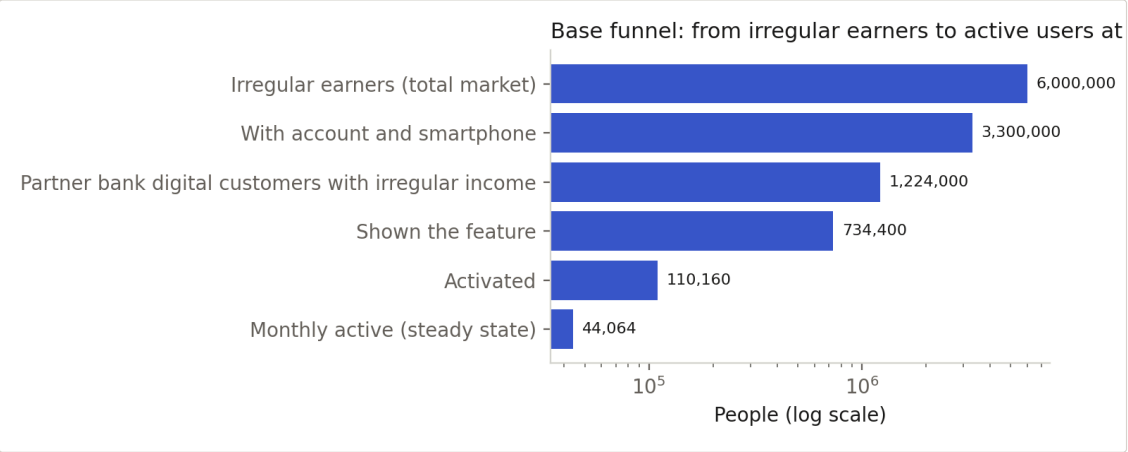
It shows the engine behaves as designed, is stable across seeds and weekdays, degrades predictably with short histories and beats a naive baseline when its own assumptions hold. It does not show that real irregular earners spend, earn or pay bills the way the generator assumes; that the range band is calibrated on real data; or that the states would be right for real people. That needs consented real statements, a held-out period and the same metrics, before any accuracy claim is made in the pitch.

6 Financial model

The model is `docs/review/model.xlsx`: an Inputs sheet with 120 inputs (each with low, base and high values and a source URL with access date or the label Assumption), a Model sheet computing every line for three scenarios side by side, a 24 month Projection, Sensitivity tables and Metrics. Conservative takes the unfavourable end of every range, Ambitious the favourable end. All 1 474 formulas evaluate without errors. Of the 116 inputs prepared by the commercial reviewer, 69 are sourced, 46 are Assumptions and one (FNB's unpaid debit order fee) could not be sourced. Bank outcomes are modelled possibilities, not facts.

6.1 Market sizing

Stats SA's QLFS for Q2 2026 counts 3.6 million people in informal sector employment and 1.1 million in private households. The total market of adults with irregular or informal income is an Assumption of 6.0 million (range 4.7 to 9.0 million). With an account and a smartphone, the serviceable market is 3 300 000 in the base case. One FNB-sized partner (8.16 million digitally active customers, BusinessDay, 16 September 2026) yields about 44 064 monthly active users at steady state in the base case (1 867 conservative, 330 480 ambitious).



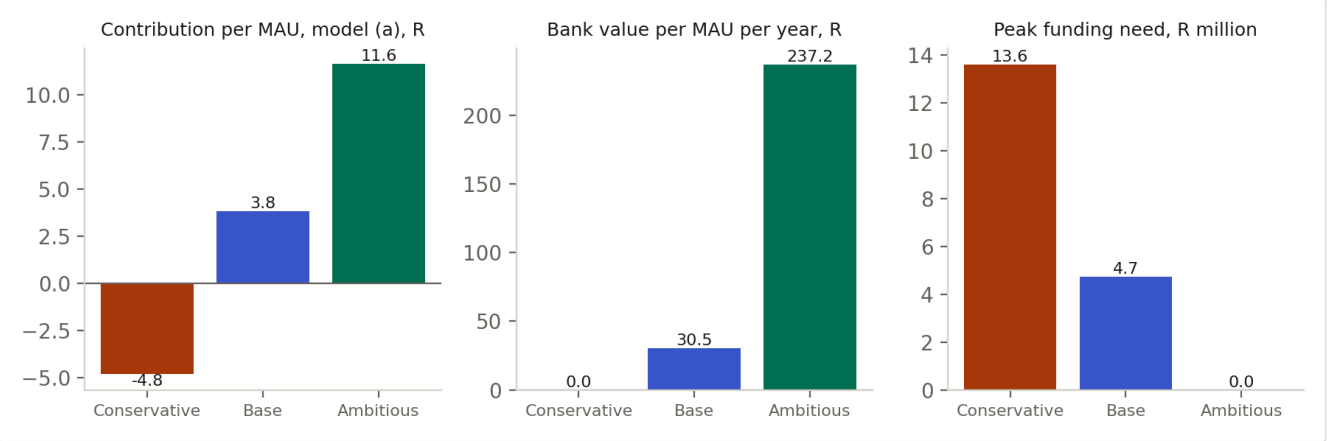
Base funnel from irregular earners to monthly active users at one bank (log scale).

6.2 Unit economics and business models

Output	Conservative	Base	Ambitious
Variable cost per MAU per month, embedded	R6.84	R1.19	R0.36
Fixed cost per month	R600 430	R310 403	R180 393
(a) Bank licence: contribution per MAU per month	-R4.84	R3.81	R11.64
(a) MAU needed to break even	never	81 374	15 496
(b) Revenue share: contribution per MAU per month	-R6.84	-R0.68	R5.57
(c) Consumer subscription: contribution per MAU per month	-R30.53	-R2.37	R1.15
(c) Consumer: lifetime value to acquisition cost	-0.31x	-0.16x	0.36x
(d) Sponsored: contribution per user per month	-R12.07	R1.89	R11.58

Source: model.xlsx, Scenarios sheet. Blank break-even means contribution is zero or negative, so the model never breaks even.

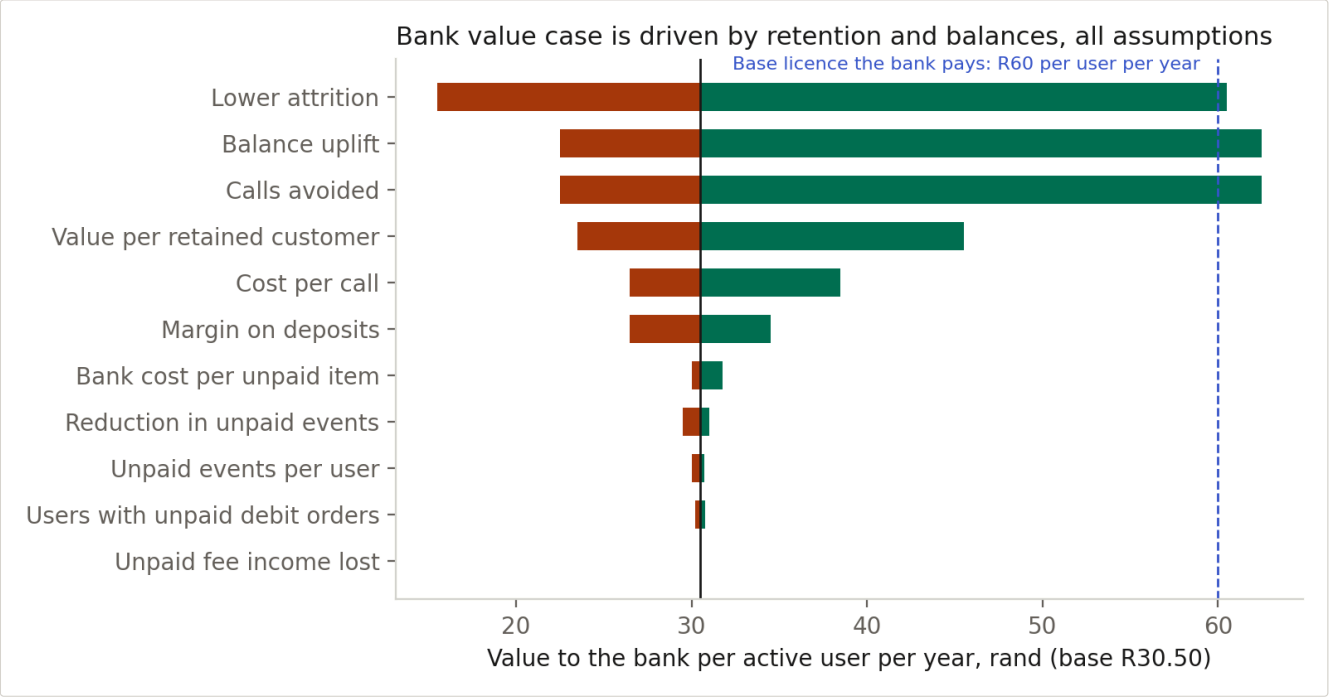
Which model fits. (a) a per-active-user licence to a bank, with a pilot fee and an annual platform minimum, is the only model with positive contribution in the base case and a credible distribution path. (b) revenue share depends on a bank value case that is thin and unproven. (c) a consumer subscription loses money in every scenario because conversion is low and acquisition is expensive for this segment. (d) sponsorship is a useful secondary channel for reach outside banks but does not fund a company on its own. This matches the commercial reviewer (COMM verdict).



Scenario comparison: contribution per MAU (model a), bank value per MAU per year, peak funding need.

6.3 Value to a bank

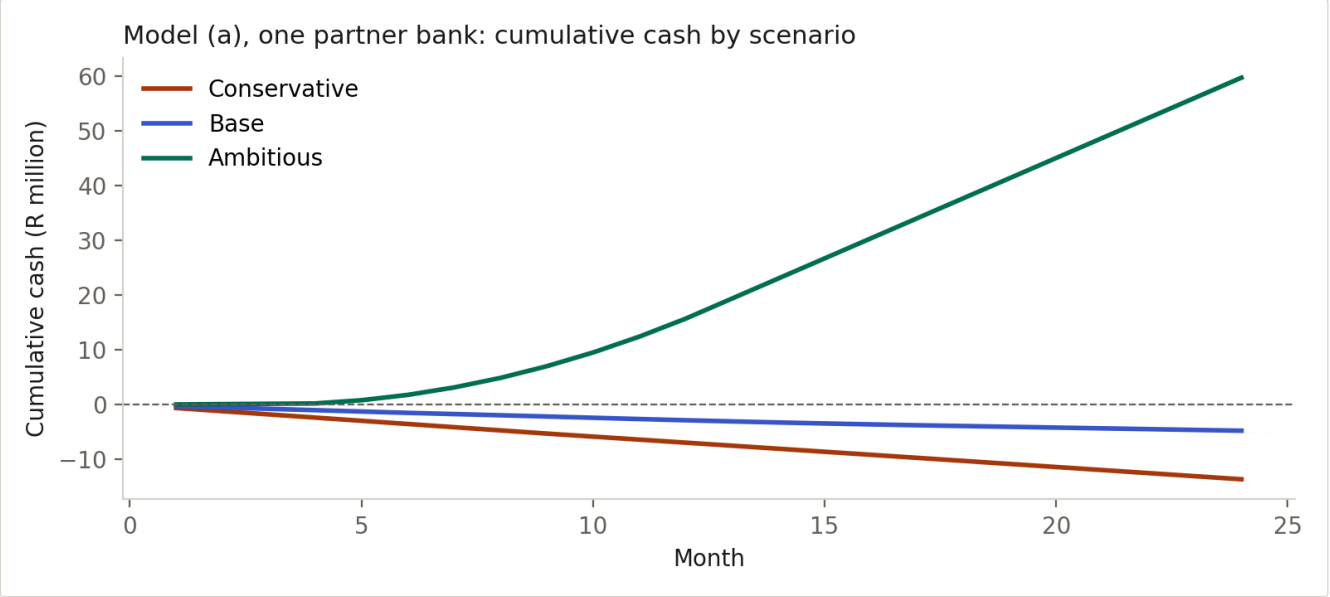
Base value to the bank is R30.50 per active user per year, against a base licence of R60 per user per year: about 0.51 rand of value per rand of licence. The case rests on retention, balances and avoided calls, all Assumptions. Avoided unpaid debit orders also remove fee income from the bank, which offsets part of the operational saving. A pilot with a control group is the only way to establish this.



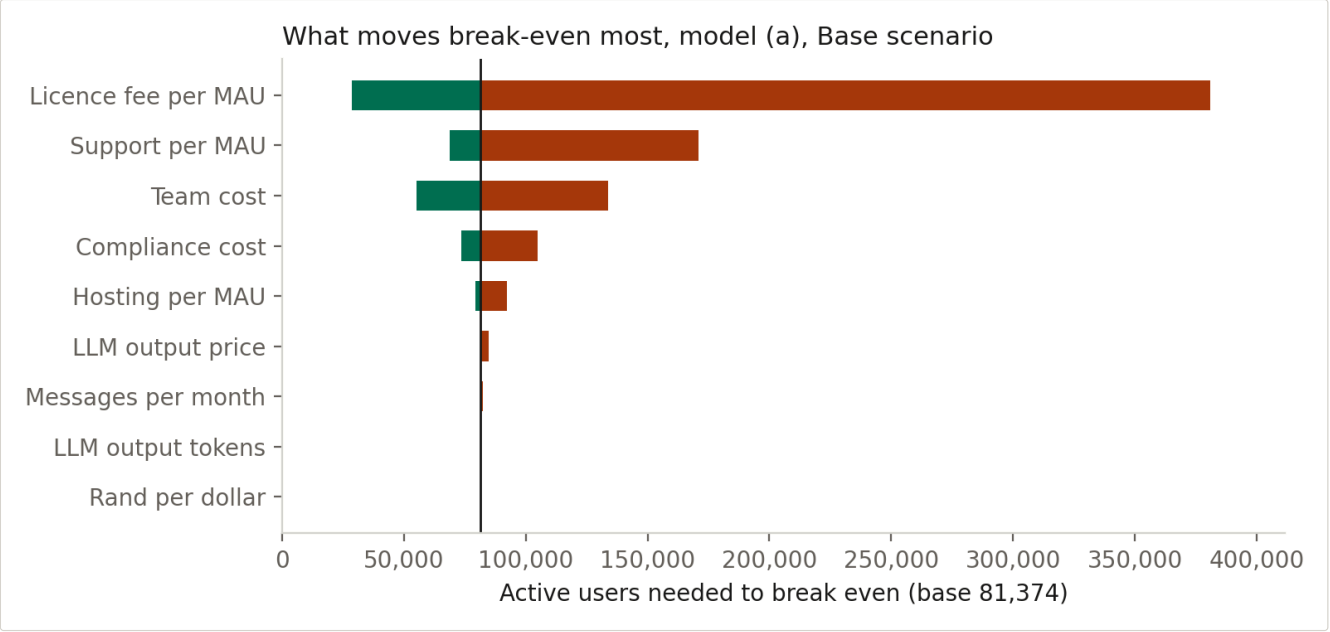
Which bank value assumptions matter most (base case, one input at a time).

6.4 Break-even, scenarios and sensitivity

Model (a), one partner bank	Conservative	Base	Ambitious
First profitable month	beyond 24	beyond 24	1
Peak funding need	R13 598 817	R4 728 272	R0
Cumulative cash at month 24	-R13 598 817	-R4 728 272	R59 734 868



24 month cumulative cash for model (a), by scenario.



What moves break-even most (base case).

The licence fee per user dominates: at R2 a month break-even needs about 381 000 active users, at R12 about 29 000. Support cost per user and the team cost come next. Messaging and LLM costs barely matter when notifications go in-app; paid daily WhatsApp would cost more than the base licence leaves (COMM-03).

7 Impact case and measurement plan

Impact must separate what Tengo produces (outputs) from what changes in people's lives (outcomes). No outcome has been measured.

Type	Metric	Base case value	How to measure
Output	People reached per year (one bank plus a sponsor programme)	54 064	Activated users and sponsor sign-ups
Output	Forecasts viewed, messages opted in	not measured	Consented product analytics
Outcome	Days of cover gained per year (all users)	211 507	Forecast date against actual run-out, before and after
Outcome	Short episodes avoided per year	Assumption: 0.5 per user	Balance hitting zero before next money, pilot against control
Outcome	Fees avoided by users per year	R1 321 920	Unpaid debit order and late fees, pilot against control
Outcome	Confidence that money lasts	Assumption: +5 points	Survey item below, before and after, with a comparison group

Proposed survey item. "How sure are you that your money will last until your next money?" on a five point scale from "not at all" to "completely". The baseline proxy from the World Bank Findex 2025 is that 30.8 percent of South African adults could easily raise emergency money in 30 days, 18.3 percent among the poorest 40 percent (model-inputs.md).

7.1 Measurement plan

1. Before any pilot: the six-person comprehension test (behavioural chapter) to check people read the date, range and states as intended.
2. Pilot design: a randomised or matched control group at the partner bank, 3 to 6 months, with outcomes defined in advance.
3. Track the product metrics in model.xlsx (Metrics sheet): activation, weekly use, day 7 and day 30 retention, imports completed, messages opted in, field forecast accuracy and state transitions acted on.
4. Report harms as well as benefits: users who followed "safe to spend" and ran short, anxiety reported with Short, any late payment linked to a suggestion.

8 Cross-cutting themes and conflicts

8.1 Themes raised by several reviewers

1. **Home contradicts itself.** Raised by product, UX, accessibility, credit risk, behavioural, legal, investor and the judges (PROD-03, UX-01 to UX-04, A11Y-03, RISK-04, BEH-04, LEGAL-08, INV-03). The most cited problem and the cheapest to fix.
2. **The engine misses pattern breaks.** Stopped or absent income, payday morning, stale imported data, bills just after payday (RISK-01 to RISK-03, RISK-10, TECH-01, DATA-02).
3. **Advice-like or harmful outputs.** Late-payment suggestions, "safe" wording, lending framed as safe (LEGAL-01, LEGAL-02, RISK-09, RISK-11, BEH-03).
4. **Claims that are not yet true.** Eleven languages, retention, data minimisation, security ticks (INV-05, L10N-01, SEC-01 to SEC-03).
5. **The money people live on is invisible.** Cash, wallets, stokvels and family transfers (BEH-02, BEH-07, COMM-04, INV-10, PROD-01).
6. **No evidence yet.** No users, synthetic validation, no bank value proof (INV-01, DATA-01, COMM-05).

8.2 Conflicts and how they were resolved

Conflict	Resolution
Offline: the accessibility reviewer reports a blank page offline after a first visit (A11Y-02); the app's own Playwright test passes an offline reload.	Both are true. A synthesis test (`docs/review/tools/synthesis/offline-matrix.csv`) shows offline works only after a second load, because the service worker caches files fetched after it takes control. The app's test reloads before going offline, which hides the gap. A11Y-02 stands.
Novelty: the product lead and commercial lead find no South African bank offering a date-first view for irregular income; the investor says FNB has shown funds after debit orders since 2018.	Compatible. FNB's feature (BusinessTech, 11 June 2018, cited by the investor) shows money after known payments, not a run-out date with inferred next money. The insight is distinct in framing and segment but not defensible as technology (COMM-01, INV-02).
Severity of legal risk: the legal advisor sees low risk for the demo; the credit risk analyst rates the same outputs as High harm.	Both stand: they answer different questions. As a demo with fictional data the legal exposure is low; for real users the harm risk is high. The roadmap treats the fixes as before-submission items because judges will read them as product judgement.
Translation route: the build and earlier guidance assumed Google Cloud Translation covers all 11 languages; the localisation reviewer says it does not support Tshivenda.	Verified on Google's supported-languages page (last updated 2026-10-13, accessed 2026-10-20): Tshivenda is absent; the other nine non-English languages are supported by the standard NMT model. Tshivenda needs a human translator (L10N-02).
Stopped income: the credit risk analyst reports Enough after 35 days without pay.	Confirmed in the analyst's saved run (`tools/credit-risk/results.md`, case S11: Enough with R1 990 set aside after five missed paydays).
Scores: reviewers range from 4 to 6, with technical roles higher and case-making roles lower.	Not reconciled into one number on purpose. The spread reflects a strong build and a weak case, which is the main message of this report.

9 Risk register

Prepared 2026-10-20 from the 13 reviewer files and `findings.csv`. Likelihood and impact are scored 1 (low) to 5 (high); the rating is likelihood times impact. "Before submission" risks affect the hackathon; "before pilot" risks affect any use with real people or bank data. Owners are suggested roles, not named people.

#	Risk	Stage	Likelihood	Impact	Rating	Evidence (finding ids)	Suggested owner	Mitigation
R1	Judges see contradictory numbers on Home (Short beside "R310 safe to spend", chart says Enough, "Enough until" on a Tight state) and stop trusting the forecast	Before submission	5	5	25	PROD-03, UX-01, UX-02, UX-03, UX-04, A11Y-03, RISK-04, INV-03, BEH-04, LEGAL-08	Founder (product)	One reconciled message per state: "Tight until", a daily budget that is zero or marked "not advised" when Short, chart readout labelled as "if money lands as usual", one next-money date
R2	A user follows "safe to spend" or "safe to lend" and is left with nothing on payday or short of rent	Before pilot	4	5	20	RISK-02, BEH-01, BEH-03, RISK-11, LEGAL-02	Founder with a credit risk adviser	Rename to a daily budget with a visible buffer for the next large bill; keep a reserve; show the disclaimer beside the figure; test comprehension with users
R3	The engine says Enough when income has stopped or never existed, giving false reassurance	Before pilot	4	5	20	RISK-01, RISK-03, DATA-02, TECH-01	Engineering	Treat overdue income as late, not coming; cap confidence when the last deposit is older than the usual gap; age imported data; add these cases to the unit tests
R4	"What helps most" tells someone to pay a car instalment, insurance or rent late, causing penalties, a lapsed policy or credit damage	Before pilot	3	5	15	LEGAL-01, RISK-09, LEGAL-06, BEH-05	Founder with legal review	Exclude credit, insurance and rent from bill-move suggestions; never suggest cutting business stock; never suggest borrowing
R5	Privacy screens are read as real compliance or security claims (retention, minimisation, encryption ticks)	Before submission	4	4	16	SEC-01, SEC-02, SEC-03, SEC-05	Founder	Reword every claim to "in this demo" or "in a real service we would"; remove green ticks; label retention as not enforced in the demo
R6	The "11 languages" claim is challenged when every language shows English	Before submission	5	3	15	L10N-01, INV-05, COMM-08, A11Y-01, L10N-05	Founder	Say "built for 11 languages, English today"; stop setting `lang` to a language the text is not in; translate two languages with human checks before submission if time allows
R7	Tshivenda cannot be machine translated with the current script, and a full run leaves locale files inconsistent	Before submission	5	2	10	L10N-02	Engineering	Exclude `ve` from the script and save status per language; commission a human Tshivenda translation and confirm "mutengo"
R8	No evidence of demand or behaviour change; validation is circular on synthetic data	Before pilot	5	4	20	INV-01, DATA-01, INV-06, BEH-08	Founder	Run the 6 person user test in `behavioural.md`; backtest on consented real statements before any accuracy claim
R9	A bank builds the date view itself, removing the reason to license Tengo	Later	3	5	15	COMM-01, INV-02, PROD-02	Founder (commercial)	Compete on design for irregular earners, multilingual messaging and evidence from pilots; pitch as a partner capability, not a moat
R10	Single-bank licence economics do not break even (base case needs about 81 000 active users against about 44 000)	Later	4	4	16	COMM-02, COMM-03; model.xlsx Scenarios sheet	Founder (commercial)	Pilot fee and platform minimum; push notifications instead of paid daily WhatsApp; second bank or sponsor channel
R11	Daily paid messaging costs more than the licence fee leaves	Later	4	3	12	COMM-03, PROD-05	Engineering and commercial	In-app push by default; WhatsApp only on state changes; batch LLM generation or templates
R12	Real data stored in plaintext in the browser, with no auth, backend or security headers	Before pilot	5	5	25	SEC-04, TECH-03, TECH-12, SEC-09	Engineering	Do not touch real data until authenticated backend, encryption, RLS, headers and a data protection impact assessment exist
R13	Messaging consent defaults on and lacks POPIA direct marketing elements	Before pilot	4	3	12	LEGAL-03, SEC-06, UX-09	Founder with legal review	Messages off by default; separate consent per channel; sender identity and STOP in every message; keep consent records
R14	Cross-border transfer when adding a hosted database, an LLM or WhatsApp	Before pilot	4	3	12	SEC-10, LEGAL-07	Engineering with legal review	Choose South African or adequate hosting; operator agreements; minimise data sent to messaging and LLM providers
R15	Cash income, wallets, stokvels and family transfers are invisible, so forecasts are wrong for the target users	Before pilot	5	4	20	BEH-02, BEH-07, COMM-04, INV-10, PROD-01	Founder (product)	Manual cash entries and recurring informal obligations; multi-account view in a bank partnership
R16	Forecast quality is invisible in production (no observability, no stored forecasts to score)	Before pilot	4	4	16	TECH-02, DATA-09	Engineering	Privacy-safe error reporting and forecast accuracy logging, with consent
R17	Offline use fails on a first visit, which matters during loadshedding and on low data	Before submission	4	2	8	A11Y-02, TECH-05	Engineering	Precache the app shell and built assets at install
R18	Shared phones expose financial data	Before pilot	3	4	12	A11Y-06, BEH-06	Engineering	App lock, discreet message previews, quick exit
R19	Advice and credit perimeter unclear (FAIS, NCA) once linked to bank products	Before pilot	3	4	12	LEGAL-05, PROD-04, INV-09	Founder with legal counsel	Written regulatory position; keep outputs factual; bank partner's compliance sign-off
R20	Dependency advisories and a large single bundle on low-end phones	Before pilot	3	2	6	TECH-06, TECH-08, INV-12, A11Y-14	Engineering	Upgrade react-router and vitest; lazy-load locales and pages

Highest ratings

Six risks are rated 20 or more: R1 (contradictory Home) and R12 (no backend security for real data) at 25, then R2 and R3 (safe figures and false reassurance), R8 (no evidence) and R15 (invisible cash) at 20. R1 is the only one of these that can be fixed in hours before submission. The others decide whether a pilot is safe.

10 Prioritised roadmap

Prepared 2026-10-20 from `findings.csv` (all 13 reviewers) and the judge panel's criterion scores in `judges.md`. Every item links to finding ids. Effort: S is up to half a day, M is one to three days, L is one to three weeks. Hours are estimates for one developer familiar with the code.

1. Fix before hackathon submission

Ranked by expected judging score lift per hour. The judge panel (`judges.md`) scored 39 of 70 (average 5.6) and estimates that its top five fixes, about three working days, lift the entry to roughly 7. The "Lift" column splits that estimate across items: it is an **Assumption** anchored to the panel's statement and to how many reviewers raised each issue, so read it as a ranking aid, not a promise. Hours come from the effort the panel gives for each change.

Rank	Item	Judge id	Underlying findings	Hours	Criteria raised	Lift (points of 70)	Lift per hour
1	Honest languages claim ("built for 11, English today") and `lang="en"` while translations are pending	JUDGE-02	INV-05, COMM-08, L10N-01, L10N-05, A11Y-01, L10N-13	1	Presentation, Impact	1.5	1.50
2	One rule drives Home: "Tight until" for Tight, daily budget shown as not advised when Short, chart readout prefixed "If money lands as usual", one next-money date	JUDGE-01	UX-01, UX-02, UX-03, UX-04, PROD-03, INV-03, BEH-04, LEGAL-08, RISK-04, A11Y-03, RISK-12	4	Product, Usability, Presentation	3	0.75
3	Put the backtest in the pitch with its synthetic caveat and stated band coverage	JUDGE-05	DATA-01, DATA-02, COMM-01	2	Intelligence, Presentation	1	0.50
4	Three-minute demo script on the strongest screens (see `judge-qa.md`)	JUDGE-12	walkthrough items 5 and 10	2	Presentation	1	0.50
5	First-impression fixes: empty Plan, messages off by default, neutral icons and conditional privacy copy, distinct light-theme Tight and Short	JUDGE-10	UX-06, UX-09, LEGAL-03, SEC-06, SEC-01, SEC-02, UX-05	2	Usability, Presentation	0.75	0.38
6	One-page business case: bank-embedded licence, pilot fee and floor, in-app push, what the bank has versus what Tengo adds, data contract, pilot KPIs	JUDGE-06	COMM-11, INV-07, COMM-07, PROD-02, PROD-06, COMM-02, COMM-03	4	Sustainability	1.5	0.38
7	Remove harmful outputs: never move credit, insurance or rent, never suggest cutting stock or borrowing, rename "Safe to lend", disclaimer beside the safe figures	JUDGE-03	LEGAL-01, LEGAL-02, LEGAL-06, RISK-09, RISK-11, BEH-01, BEH-03, BEH-05, PROD-04, INV-09	8	Impact, Product	2	0.25
8	Notice when the pattern breaks: paused-income rule, a no-income state, payday-today rule, ageing of imported data; unit tests for each	JUDGE-04	RISK-01, RISK-03, RISK-10, TECH-01, DATA-05	8	Intelligence, Impact	2	0.25
9	Offline after the first visit: precache the build at install, offline fallback, test without the online reload	JUDGE-09	A11Y-02, TECH-05	2.5	Usability, Impact	0.5	0.20
10	Confirm the name with two Tshivenda speakers before telling the name story	JUDGE-11	L10N-12, INV-11, COMM-10	2	Presentation	0.25	0.13
11	Cash in hand and quick manual entries that feed the forecast	JUDGE-08	BEH-02, COMM-04, INV-10, BEH-07, PROD-07	12	Creativity, Impact	1.5	0.13
12	Six-person comprehension test after the fixes, quoted in the pitch	JUDGE-07	INV-01, COMM-05, BEH-08	16	Impact, Product	2	0.13

Items 1 to 8 take about 31 hours (four working days) and carry most of the expected lift. If only one day is available, do items 1, 2, 5 and the copy parts of 7 (about 9 hours).

2. Fix before any real pilot

Item	Findings	Effort
Authenticated backend with encryption at rest and in transit, row-level security, security headers; no real data in localStorage	SEC-04, TECH-03, TECH-12, SEC-09	L
Data protection impact assessment, responsible party, Information Officer, terms of use and complaints route	SEC-08, LEGAL-04, SEC-12	M
Retention and deletion that actually delete (including server copies and backups), consent records that can be proved	SEC-01, SEC-05, SEC-06, SEC-07, SEC-13	M
Operator agreements and cross-border assessment for any hosting, LLM or messaging provider	SEC-10, LEGAL-07, LEGAL-09	M
Written regulatory position on FAIS and NCA with the partner bank's compliance team	LEGAL-05, PROD-04, INV-09	M
Engine: one-offs and windfalls, price rises and skipped bills, month-end spikes, a reserve for the next large bill, background income in the headline	RISK-02, RISK-05, RISK-06, RISK-07, RISK-12, BEH-01	L
Engine: band calibrated near run-out, short-history handling, shock robustness	DATA-03, DATA-04, DATA-05, DATA-06, DATA-10, RISK-08	M
Validation on consented real statements, accuracy regression tests in CI, field accuracy logging	DATA-01, DATA-09, TECH-02, INV-06	L
Cash entries, informal obligations (stokvel, burial society, family transfers) and multiple accounts	BEH-02, BEH-07, COMM-04, INV-10, PROD-01, PROD-07	L
First-language review of all 10 languages with a money glossary; per-string review status; word order out of code	L10N-03, L10N-04, L10N-06, L10N-07, L10N-08, L10N-10, A11Y-09	L
Shared-phone protection: app lock, discreet previews, quick exit; vulnerable users and children policy	A11Y-06, BEH-06, SEC-11	M
Observability with privacy-safe logging; storage error handling; service worker strategy for personal data	TECH-02, TECH-04, TECH-05, SEC-09	M
Bank business case with a measurable outcome and a control group	COMM-05, COMM-02, COMM-11, INV-07	M
Dependency upgrades (react-router, vitest) and bundle splitting	TECH-06, TECH-08, INV-12, A11Y-14, L10N-14	S
UI test coverage for the data layer and failure paths	TECH-07	M

3. Later

Item	Findings	Effort
Low-data and USSD channel; push before paid WhatsApp	PROD-05, COMM-03, COMM-09	L
Final brand and confirmed name; commercial clearance of assets	PROD-08, COMM-10, INV-11, L10N-12	M
Defensibility: design partnerships for irregular earners, evidence from pilots, second bank or sponsor channel	COMM-01, INV-02, PROD-02, COMM-09	L
Remaining UX and accessibility polish (import density, microcopy, truncation, chart scale, command menu)	UX-08, UX-10, UX-11, UX-12, UX-13, A11Y-07, A11Y-08, A11Y-10, A11Y-11, A11Y-12, A11Y-13, A11Y-15, L10N-11	M
Engine performance work (worker thread, memoisation) and removal of the demo latency	TECH-09, TECH-10, TECH-11	S
Strategic use and gaming of the account view; anxiety and avoidance research	BEH-09, BEH-06	M

11 Judge Q&A and demo script

Prepared 2026-10-20 from the 13 reviewer files. Answers are what the founder can honestly say today, based on what exists: a working demo with synthetic data, no users, no bank data, no backend. Finding ids point to the evidence in `findings.csv`. Where an answer depends on a fix from `ROADMAP.md`, it says so.

Twenty likely questions

1. How many real irregular earners have used this? None yet. It is a demo built on three fictional composites. Our next step is a six-person comprehension test with drivers, hairdressers and spaza owners (BEH user-testing plan; INV-01).

2. How accurate is the date? On real data, unknown. On synthetic data the forecast date is off by 2.1 days on average with 90 days of history, against 3.4 days for a naive baseline, and the actual date falls inside our range 72 percent of the time. The test data comes from the same assumptions as the engine, so this is a consistency check, not proof (DATA-01, `backtest/results/summary_headline.csv`).

3. Banks already show money left after debit orders. What is new? Those tools assume a payday. Tengo infers when the next money usually lands from irregular deposits, shows a range rather than a single figure, and turns it into Enough, Tight or Short in plain words. The arithmetic is simple; the design for people without a payday is the contribution (INV-02, PROD-02, COMM-01).

4. Why would a bank not just build it? It could. We pitch Tengo as a tested, explainable layer a bank can switch on for irregular earners quickly, licensed per active user, with the rules, copy and explanations already worked out. The moat is evidence and design for this segment, not code (COMM-01).

5. Where does the data come from? Today, a CSV statement read in the browser. In a pilot, the host bank's own transaction data. We do not rely on open finance, which South Africa has not adopted at scale (PROD-06, INV-04).

6. Who pays? A bank, per active user, with a pilot fee and an annual minimum. Users in this segment should not pay. Our base model needs about 81 000 active users at R5 a month to break even, so one bank alone is not enough at base adoption; a second bank or a sponsor programme closes the gap (COMM-02, `model.xlsx` Scenarios).

7. Ayanda is Short but told R310 is safe to spend, and the chart says Enough. Which is it? If fixed (roadmap item 2): "We fixed that. One rule now drives the screen: when Short, the daily budget is marked not advised, and the chart is labelled as what happens if money lands as usual." If not fixed: "That is a real inconsistency. The three figures answer different questions, and we are merging them so the screen says one thing" (UX-01, UX-02, RISK-04, INV-03).

8. Is "safe to spend" or "safe to lend" financial advice? We present factual estimates, not recommendations, and the reviewers' view is that most of this is likely outside the FAIS meaning of advice. The wording "safe" is too strong, so we are renaming it to describe consequences, and the disclaimer moves next to the figures. Inside a bank it would go through the bank's compliance review. This is not a legal opinion (LEGAL-02, LEGAL-05).

9. Could it tell someone to do something harmful? In the demo, "What helps most" could suggest paying a car instalment later. We are removing bill-move suggestions for credit, insurance and rent, and never suggesting borrowing or cutting business stock (LEGAL-01, RISK-09, roadmap item 7).

10. Your users run on cash and stokvels. How does a bank statement see that? It does not, today. Cash in hand and quick manual entries are the next feature; until then the screen should say it covers this account only (BEH-02, COMM-04, INV-10).

11. What happens if someone's income stops? In the demo the engine still expects it, which is wrong. The fix treats overdue income as late and lowers confidence, so a stopped income cannot show Enough (RISK-01, RISK-03, roadmap item 8).

12. You claim 11 languages. Can I see isiZulu? Not yet. All 11 are wired in, but only English is written. Google's translation API does not support Tshivenda, so that language needs a human translator. Once roadmap item 1 is done, the landing page says "built for 11, English today"; until then, say it out loud before a judge finds it (L10N-01, L10N-02, INV-05).

13. Where is the data stored and who can see it? In this demo, only in the browser on the device. Nothing is sent anywhere. A real service would need an authenticated backend, encryption and a data protection impact assessment before any real data is used (SEC-04).

14. Do your privacy controls meet POPIA? No claim is made. They are labelled "Simulated privacy controls. Not a compliance certification." Some screens describe rules the demo does not enforce, such as 12 month retention; we are rewording those as demo statements (SEC-01, SEC-02).

15. Does it work offline during loadshedding? Yes from the second visit, because the app caches itself after it loads. A first visit followed by going offline shows a blank page today; precaching at install fixes that (A11Y-02, roadmap item 9).

16. Is it accessible? Automated WCAG 2.2 checks find almost nothing across 15 pages in both themes, it reflows at 320 pixels and respects reduced motion. Manual review found real gaps: focus can hide behind the tab bar on phones and the page language was set wrongly for untranslated languages (A11Y-01, A11Y-04).

17. How much does it cost to run per user? About R1.19 a month per active user inside a bank app in the base case, mostly support; paid daily WhatsApp messages would cost more than the licence leaves, so we use in-app notifications (`model.xlsx` Model sheet; COMM-03).

18. What does a bank get out of it? Unproven. Our base assumptions give about R30 of value per active user a year against a R60 licence, driven by retention and balances, which are assumptions. A pilot with a control group would measure unpaid debit orders, balances and retention (`model.xlsx` Sensitivity; COMM-05).

19. What would make you stop? If users read the date as a deadline and make worse decisions, if real-data accuracy is far below the stated range, or if no bank or funder will host it (investor deal breakers).

20. What does the name mean? "Mutengo" is understood by the founder to mean value in Tshivenda. One reviewer notes it may mean price; we are confirming with first-language speakers before telling the story (L10N-12).

Three-minute demo script

Assumes roadmap items 1 to 8 are done. Timing in brackets.

1. **Hook (0:00 to 0:20).** "A balance tells you what you have. Tengo tells you until when." One sentence on who it is for: people paid by the trip, the client or the till, not on the 25th.
2. **Home for Thabo (0:20 to 0:55).** Open Home as Thabo, Tight. Point to the date, the likely range and the next money. "He is covered on a normal week, not on a heavy one." Open "Why this date" and show the inputs in plain words.
3. **The hard case, Ayanda (0:55 to 1:30).** Switch to Ayanda, Short. Show that the screen says one consistent thing: Short from a date, two days before her next money, the daily budget marked not advised, and "What helps most" offering a spending change, not a late bill.
4. **Plan and the family loan (1:30 to 2:00).** Open Plan, enter that her cousin asks for R300. Show the days of cover lost and the state change. "Lending is normal here; Tengo shows the cost before she says yes."
5. **Trust (2:00 to 2:30).** Privacy centre: consent per purpose, see and delete my data, "simulated" label visible. One line on offline after first load and data cost. Name the honest limits: synthetic data, English today, bank account only.

6. **Evidence and model (2:30 to 3:00).** One slide: backtest MAE 2.1 days, 72 percent in range, on synthetic data with the caveat. Business: bank licence per active user with a pilot fee; next step a six-person test and a bank pilot with a control group.

Handling weak spots if they come up. If items are not fixed in time, do not demo Ayanda's Home (use Thabo only), avoid the language selector, and say plainly that the safe figures are being renamed. Never claim real users, real accuracy or compliance.

12 Appendices

A. Full findings table

Id	Severity	Effort	Finding	Evidence	Recommendation
PROD-01	High	M	Engine ignores overdraft, debit orders, holds and multiple accounts	src/engine/forecast.ts:196-210 zero is the floor; src/engine/classify.ts:16 bills inferred only; FNB overdraft page; Standard Bank Future Payments	Define a bank data contract: available balance, facility limit shown separately, scheduled debit orders as known bills, account set
PROD-02	High	S	Overlap with existing bank budgeting and forecast tools not addressed	ITWeb 2025 nav» Money 5m users; Standard Bank Budget Manager and Future Payments; Capitec categorisation; lerato-money-dark-1280.png	Narrow bank pitch to the income-aware run-out date; de-emphasise categorisation; add a "bank has versus Tengo adds" slide
PROD-03	High	S	Headline signals conflict (Enough until with Tight; Short with R310 safe to spend)	src/components/forecast/DateHero.tsx:9; thabo-home-dark-1280.png; ayanda-home-dark-390.png; persona-snapshot.json	Neutral "Lasts until" label; safe to spend obeys state; chart readout states its income assumption clearly
PROD-04	Medium	S	Credit-adjacent and advice-like copy would fail bank compliance review	src/i18n/locales/en.json:523 Safe to lend; help.billToMove Car instalment; en.json:702	Descriptive outcomes, no "safe to lend", never suggest delaying a credit repayment, estimate-not-advice line
PROD-05	Medium	M	No USSD or low-data channel; WhatsApp carries financial figures	fact-sheet Form (web PWA only); thabo-inbox-dark-390.png; FNB USSD *120*321#	In-app card first, one-line USSD answer, SMS under 160 characters, WhatsApp subject to bank policy
PROD-06	Medium	L	No realistic external data path; pilot depends on a bank sponsor	src/lib/csv.ts CSV only; en.json:472 no bank connection; Moonstone 2026 open finance not adopted	Pilot on internal data with a named sponsor; list prerequisites, success metrics and kill criterion
PROD-07	Medium	M	Segment fit asserted not shown; cash-heavy persona fits bank data worst	src/data/seed/personas.ts synthetic; BRIEF.md section 11 lumpy cash takings	Pilot platform and gig earners paid into account first; interviews per segment; cash entry later
PROD-08	Low	M	Standalone onboarding, English-only locales and placeholder brand do not fit a bank app	walkthrough.md item 1; thabo-consent-dark-390.png; src/i18n/status.json; README Brand and type	One-screen in-app opt-in on bank consent; bank design system; first-language review of key locales
TECH-01	High	S	Imported or edited data never ages, so the forecast drifts towards Enough	localRepository.ts:87; forecast.ts:36-41,66; engine-bench.out.txt (Ayanda Short to Enough after 30 days, confidence high)	Store an asOf date, run the engine from it, show data age, lower confidence after 3 days and hide the state after 14
TECH-02	High	M	No observability: errors, performance and forecast quality are invisible	ErrorBoundary.tsx:8,17; App.tsx:22; Import.tsx:55-60; no error reporting found	Error reporting with personal data scrubbed, route-level boundaries, unhandledrejection handler, web vitals, forecast accuracy events
TECH-03	High	L	The repository contract is not ready for a real backend	repository.ts:75-95; personas.ts:3; localRepository.ts:16,57-69,146; PersonaProvider.tsx:112-117; Me.tsx:26	User-scoped, paged, incremental API with concurrency control, server-side append-only audit, shared engine package
TECH-04	Medium	M	localStorage writes are unguarded and the storage is fragile	localRepository.ts:61-69; Import.tsx:57-58; MDN 5 MiB limit and Safari 7-day eviction	Catch quota errors, move to IndexedDB, call storage.persist, sync across tabs, sync to a server before a pilot
TECH-05	Medium	S	The service worker caching strategy is naive	sw.js:4,27-35,40-52; main.tsx:16-19; flows.spec.ts:131-140; vercel.json	Build precache manifest, cache versioned by build hash, cache only ok responses, stale-while-revalidate, exclude assets from the rewrite
TECH-06	Medium	S	One 696 KB bundle with every locale loaded at start	audit.md (695 602 B, 141 864 B gzip); i18n/index.ts:5; App.tsx:8-18; 104 KB gzip of locales compressed one at a time	Load locales and routes on demand, prefetch through the service worker, bundle budget in CI
TECH-07	Medium	M	Test strategy leaves the UI, data layer and failure paths unproven	coverage.txt 19.95 percent; screens.spec.ts:96-98; playwright.config.ts:13 Chromium only; no CI folder	CI pipeline, WebKit and low-end profiles, visual baselines, provider and Import tests, engine property tests
TECH-08	Medium	M	Known advisories in shipped and build dependencies	audit.md: react-router GHSA-wrjc-x8rr-h8h6 moderate; tinypool critical and braces high (dev only)	Upgrade react-router to 7.18 or later, vitest 5, tailwind 4 or overrides; audit gate and Renovate
TECH-09	Low	S	Engine runs on the main thread and is repeated many times per render	help.ts:6,29-39; HelpCard.tsx:17; Plan.tsx:65-73,190; helpPlan 10 to 40 ms on laptop	Debounce inputs, cache the base forecast, closed-form extraNeeded, Web Worker if needed
TECH-10	Low	S	The fake 140 ms latency ships to production and is applied unevenly	data/index.ts:5; localRepository.ts:79-81,129-163; perf-home.json	Development-only latency flag; test loading and error states with Playwright route delays
TECH-11	Low	S	"Today" is fixed when the app opens	PersonaProvider.tsx:36	Recompute today on visibilitychange and at midnight, then reload
TECH-12	Low	S	No security or caching headers in the hosting config	vercel.json has only a rewrite; inline script at index.html:11	Add CSP with script hash, frame-ancestors, Referrer-Policy, Permissions-Policy, immutable asset caching
SEC-01	High	S	Retention rule displayed as fact but never enforced	src/pages/privacy/retention.ts:6-11; en.json:234; README.md:94; deletion-probe.ts shows 2024 record kept	Implement purge or reword as future; server job covering backups in production
SEC-02	High	S	On-screen wording reads as real security and compliance claims	Overview.tsx:105-109 green ticks; en.json:205,207,288,327,330	Neutral planned icons, conditional wording, Demo label per card, banner on all sub-pages
SEC-03	High	M	"Never uses names or account numbers" false for real statements	en.json:188-201; csv.ts:133,163 store description verbatim; sample-statement.csv:2,7 "Mrs K"	Scrub descriptions on import, tokenise merchants, fix copy, add tests
SEC-04	High	L	No auth, plaintext financial data in localStorage, no security headers	localRepository.ts:15,67-69; Picker.tsx:40-41; vercel.json; index.html:11-22; audit.md:45-47	Server storage with auth and app lock, CSP with hash, HSTS and headers, upgrade react-router
SEC-05	Medium	M	Deletion is local only and the receipt proves nothing	localRepository.ts:37-40,165-180; Delete.tsx:21; probe output shows receipt with personald	Server deletion workflow incl. backups and operators, signed receipt, no id in URL, explain downloads
SEC-06	Medium	M	Consent bundled, messages pre-ticked, not provable	Consent.tsx:27,86; localRepository.ts:16,129-135,146	Contract basis for core, opt-in off by default, append-only versioned consent events
SEC-07	Medium	M	Audit log stores descriptions incl. pre-correction values; not tamper-evident	TransactionEditor.tsx:41; localRepository.ts:146; PersonaProvider.tsx:77-81	Ids and field names only, server-side append-only log, no personal data in any logs
SEC-08	Medium	M	Collection notice misses section 18 items; Information Officer placeholder	Consent.tsx:14-20; en.json:146-157; Overview.tsx:150-154	Layered notice with all s18 items, named entity, registered Information Officer, PAIA manual
SEC-09	Medium	S	Service worker would cache personal API responses	public/sw.js:30,38-49; README.md:95	Allow-list static paths, network-only for the rest, clear caches on logout and delete, no-store

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SEC-10	Medium	M	No plan for operators and cross-border transfer	en.json:238; README.md:95-96; Inbox WhatsApp preview	Operator register and s21 contracts, s72 assessment, SA region, de-identified LLM inputs, minimal message content
SEC-11	Medium	M	Children and vulnerable users not considered	No age gate in src/pages; Picker.tsx no lock	18+ gate or bank-verified adult, app lock, discreet mode, abuse risk assessment
SEC-12	Medium	M	No breach response behind the sample notice	src/pages/privacy/Breach.tsx template only	Incident plan, eServices SCN1 process, operator clauses, tabletop exercise
SEC-13	Low	S	Withdrawn consent does not stop processing	PersonaProvider.tsx:83-86; AppGate.tsx:35	Skip forecast when core consent off, offer deletion on paused screen
SEC-14	Low	S	Export and import edge cases (CSV injection, unprotected files, no size cap)	csv.ts:189-199; MyData.tsx:28-38; Import.tsx:99; localRepository.ts:67-69	Neutralise formula prefixes, warn on export, cap import size, handle quota errors
LEGAL-01	High	M	What helps most tells users to pay credit instalments and insurance premiums late	src/engine/help.ts:62 sorts by amount only; Thabo told to pay Car instalment after due date (thabo-home-light-768.png, persona-snapshot.json); FAIS s1(1)(d); NCA s129	Never suggest moving credit or insurance payments; classify bills; reword as ask-first; add unit test
LEGAL-02	High	S	Safe is an unqualified promise and the disclaimer is hidden	en.json:359,523; Ayanda Short shows Safe R310; why.notAdvice only at WhyDrawer.tsx:145 below the fold; CPA s22(2), s41(1)(b)	Rename to estimate wording; qualify Safe to lend at Tight; persistent not-advice line on Home, Plan, Help card
LEGAL-03	High	S	Daily messages default on and lack POPIA s69 elements	Consent.tsx:27 useState(true); templates.ts and en.json:664-684 have no STOP or sender contact; POPIA s1 consent, s69; Meta opt-in; WASPA	Default off; separate named opt-in; Reply STOP and sender details in every message; log consent version
LEGAL-04	Medium	M	No responsible party, terms of use or complaints route	Placeholder Information Officer en.json:239-244; no terms keys; POPIA s18(1); ECTA s43(1)	Publish operator identity, terms, privacy notice, registered Information Officer, complaints process, or use bank's
LEGAL-05	Medium	S	Regulatory perimeter not documented; FAIS position depends on product links	FAIS s1(1) advice and financial product incl. deposits; s1(3)(a); s1(4); set aside copy en.json:702; COFI introduced 17 Apr 2026	Attorney-reviewed perimeter memo; never name products outside a licensed FSP; bank to classify as factual info or automated advice
LEGAL-06	Medium	S	Bring in more can steer people to costly or informal credit	en.json:699; HelpCard.tsx:45	Reword to earn or receive with a borrowing caution; neutral link to NCR consumer help
LEGAL-07	Medium	S	Cross-border transfer and sensitive inference in WhatsApp messages	Short state text en.json:668-669 via Meta; POPIA s72(1)	Name Meta in opt-in and notice; rely on explicit consent; SMS alternative; discreet template
LEGAL-08	Low	S	State label contradicts itself on Home	Enough until next to Tight chip (thabo-home-light-768.png; en.json:346); CPA s22(2), s41(1)(b)	Use Lasts until in the hero; reserve Enough for the chip
LEGAL-09	Low	S	Future bank use could trigger POPIA s71 and purpose creep	Not for marketing or credit scoring en.json:144; POPIA s71	Contractual purpose limit for partners; s71 assessment before any decision use; human review route
RISK-01	High	S	Stopped or paused income still treated as coming	forecast.ts:144-166,178-182; results S09, S11, S11b: Enough, high confidence, safe to lend R990 to R2110	Detect missed paydays, mark pattern paused, cap state at Tight, lend R0, add tests
RISK-02	High	M	Safe figures leave R0 on payday and state ignores bills just after payday	forecast.ts:217-221,294-296,357-367; results S13 Enough and lend R1090 with R3000 rent next day; heavy left after lending R0 to R10	Check cover one cycle ahead, keep a buffer, cap lending at balance after a heavy week
RISK-03	High	S	No income can show Enough and Short refers to a non-existent next money	forecast.ts:28,107-119; forecast.test.ts:116-122; results S02, S02b (Enough, lend R2670), S07-5, S07-7	Show "No regular income found", runway only, lend R0, user-entered income
RISK-04	High	S	Safe to spend contradicts Short and is not a safe amount	persona-snapshot.json Ayanda Short with R310 (50% of usual, mostly stock); Lerato R330 (274% of usual); en.json messages.sms.short; WhyDrawer.tsx:145	State-specific wording, show usual spend, drop "safe" in Short, disclaimer on Home
RISK-05	High	M	One-offs, windfalls and loans distort rates and income detection	forecast.ts:66-82,128-131; classify.ts:66; results S04 (windfall makes pattern unknown), S04b, S08 (heavy R1393/day, cut R645/day)	Winsorise spend, robust income threshold, tag loans and transfers as not income
RISK-06	Medium	M	Bill detection breaks on price rise or skipped month; arrears invisible	classify.ts:16-22,44-54,88; results S05, S05b, S10, S10b	Allow a missed cycle, use latest amount, ask about missed bills, weekly cycles, user-editable bills
RISK-07	Medium	M	Month-end spikes averaged away	forecast.ts:59-88; README.md:319; results S06 headline 16 Nov vs runout about 28 Oct	Day-of-month uplift with 60+ days of history, or a month-end warning
RISK-08	Medium	S	Confidence and band do not reflect real uncertainty	forecast.ts:30-34,83-86; results S07-5, S07-7, S07-14 zero-width band, S11 high confidence	Multi-factor confidence, minimum band, hide lending at low confidence
RISK-09	Medium	S	What helps most can suggest deferring rent or credit and cutting stock	help.ts:56-74; en.json help.moveBill; results S02, P-thabo, P-ayanda, S08	Exclude essential and credit bills unless flexible, warn about fees, cut household spend only, cap cut
RISK-10	Medium	S	Payday morning shows Short and pushes next money a week	forecast.ts:103,149; results S12 Short vs S12b Enough	Treat today as payday until latest passes, "Payday today, not in yet"
RISK-11	Medium	S	Safe to lend wording implies safety and ignores non-repayment	en.json plan.safeToLend, plan.safeToLendHint; Plan.tsx:204-206,236-238; results P-thabo heavy week R695 short	Rename, add "only lend what you can manage without", warn in Tight
RISK-12	Low	S	Small daily income ignored in headline, skewing states and conflicting with chart	forecast.ts:135,259-262; results P rows R57 to R170 a day ignored; walkthrough item 3	Count half the background rate on the central path or disclose it
DATA-01	High	L	Validation is circular and the evidence base is thin	Generator encodes engine assumptions (src/data/seed/generate.ts:18-20, 26, 40, 72-76); startingBalance uses the engine (generate.ts:117-149); shipped seed gives 7, 7 and 1 distinct paths over 24 dates (backtest/results/generator_diversity.csv); no real-data result exists	Treat all backtest numbers as upper bounds, say "on synthetic data" in any claim, run real-data validation before pilot claims, vary demo seed by date
DATA-02	High	M	Tight and Enough give false reassurance for lumpy earners	Seeds 90 day: Tight followed by short 24.4 percent (Ayanda 39.7); Ayanda Enough followed by short 11.6 percent, 24 percent at demo balance; constant P80 rate (src/engine/forecast.ts:81) and one day income slack (forecast.ts:156); copy en.json:365-366	Set state thresholds from a target error rate via simulation, model lumpy outflows as events, make Tight copy name the risk, track P(short) per state
DATA-03	High	M	Band too narrow near run-out and too wide far out	Run-out within 6 days: 53.5 percent coverage, 27.5 percent before heavy edge; 14 to 29 days: 87 to 88 percent (backtest/results/summary_by_lead_time.csv, charts/calibration.png); constant percentile rate (forecast.ts:81-82, 196-213)	Build band from simulated paths and take date quantiles, state a nominal coverage, track coverage by lead time
DATA-04	High	M	Short histories hide bills and lose to a naive baseline	Import accepts 14 days (src/lib/csv.ts:6-7); bills need 2 occurrences 24 to 36 days apart (src/engine/classify.ts:16-21); seeds MAE 5.4 vs baseline 4.2 at 14 days, 3.9 vs 3.3 at 28; Lerato bias +5.5 days	Treat single large debits as provisional monthly bills or ask the user, fall back to mean outflow under 60 days, no Enough under 28 days
DATA-05	Medium	M	Errors grow fast under unmodelled shocks	Missed payout: false reassurance 7.6 to 36.8 percent; prices up 25 percent: 19.9; new R800 debit order: 18.5; R2 000 emergency: 17.4 with coverage 47.5 (backtest/results/perturbation.csv)	Add a missed-payout check to Enough, weight recent weeks more, surface new debit orders after one occurrence
DATA-06	Medium	S	Next money weak for interval income	Ayanda in window 57.5 percent, exact 16.7, MAE 2.1 days; weekday window hard-coded to +1 day (forecast.ts:156); 3 day delay drops window accuracy to 0 percent	Estimate lateness from history, widen interval window, show next money as a range

Id	Severity	Effort	Finding	Evidence	Recommendation
DATA-07	Medium	S	Headline date is a counterfactual users will misread	Runway ignores income (forecast.ts:196-213); date within 30 days in 99.9 percent of cases but real balance hit zero in 20.8 percent; copy "money lasts until" (en.json:666, 680)	Rename to money you have now lasts until, use real shortfall before next money as the primary pilot metric
DATA-08	Medium	S	Short over-warns, especially for Ayanda	Shown Short: 33 percent covered in hindsight, real shortfall 53.1 percent, Ayanda Short right 51.6 percent; background income only used in scrubber (forecast.ts:242-245)	Count a conservative share of background income in the state, track alarm precision
DATA-09	Low	S	No accuracy regression test or field measurement	Unit tests cover rules only (docs/review/prep/checks.md); no stored forecasts to score	Add this backtest to CI with thresholds, log and score every shown forecast in a pilot
DATA-10	Low	S	Day 0 treated inconsistently	Today's income assumed in balance, next money from tomorrow (forecast.ts:106, 124), but a full day's spend subtracted today (forecast.ts:205-209)	Use balance timestamp, include pending same-day credits, subtract only remaining expected spend
UX-01	High	S	Short state sits next to R310 safe to spend and R0 safe to lend	docs/review/screens/ux/ayanda-home-fold-light-390.png; ayanda-plan-light-390.png; ayanda-why-drawer-light-390.png (R623 a day)	Relabel safe to spend as a daily limit when Short, show usual spend beside it, align What helps and Loan check wording
UX-02	High	S	Chart readout says Enough while hero says Short or Tight	src/components/forecast/TimelineScrubber.tsx:110; ayanda-home-dark-1280.png; thabo-home-dark-768.png	Drop the state word from today's readout or label it as the if-income-lands case; show both paths
UX-03	High	S	Headline label says Enough until when state is Tight	src/components/forecast/DateHero.tsx:9; thabo-home-light-1280.png; landing-light-1280.png	Use state-specific labels such as Lasts until (on a normal week) for Tight
UX-04	Medium	S	Three different next-money dates on one screen	src/pages/Home.tsx:56 and 74; docs/review/screens/ux/ayanda-home-fold-light-390.png	Show one next-money date and explain the planning buffer in words
UX-05	Medium	S	Tight and Short nearly identical colour in light theme	src/styles/index.css:47-48 (contrast between them 1.01); picker-light-390.png	Shift light Tight to a distinct amber and re-run contrast checks
UX-06	Medium	S	Plan opens with pre-filled R500 what-if and R300 loan	src/pages/Plan.tsx:59 and 187; thabo-plan-dark-1280.png	Start empty with a neutral prompt; keep quick-amount chips
UX-07	Medium	S	Range band marker unlabelled and headline at edge of likely range	src/components/forecast/RangeBand.tsx:29-48; lerato-home-dark-390.png	Label Your date and Likely range on the track; explain in Why drawer
UX-08	Medium	M	Chart scale driven by optimistic band, line hugs zero, shading below R0	src/components/forecast/TimelineScrubber.tsx:59-60; ayanda-home-light-390.png	Scale to central line and capped band, clip at R0, mark the cover date
UX-09	Medium	S	Daily messages on by default at consent	src/pages/Consent.tsx:27; thabo-consent-dark-390.png	Default to off and offer opt-in after first Home view
UX-10	Low	S	Import step 2 dense and summary tiles ambiguous	thabo-import-sample-light-390.png	Collapse mapping when detection succeeds, add amount to preview, relabel tiles as row counts
UX-11	Low	S	Microcopy and date format inconsistencies	src/i18n/locales/en.json:697 and 704; lerato-home-dark-390.png; thabo-plan-dark-1280.png	Lower-case bill names, reword To reach Tight, reconcile R330 and R280, drop leading zero on days
UX-12	Low	M	Phone first fold cuts safe to spend; tablet uses stretched phone stack; Privacy is an unlabelled icon	docs/review/screens/ux/ayanda-home-fold-light-390.png; thabo-home-dark-768.png; src/layouts/nav.ts	Tighten hero on phones, two columns from 768, label Privacy
UX-13	Low	S	Persona lists truncate role and show fixed start state	lerato-me-light-390.png; src/pages/Picker.tsx	Allow two-line roles and compute chip from current forecast
A11Y-01	High	S	Page language says Tshivenda or Sepedi while text is English (3.1.1)	src/components/Preferences.tsx:12; src/i18n/status.json all pending; results/manual.json lang shows html lang ve/nso/zu with English h1	Set html lang to the language actually rendered (en while pending); mark fallback strings lang=en; add a test
A11Y-02	High	S	Blank page offline after a first visit	public/sw.js precaches only / and icons; results/offline-single-visit.json shows no JS or CSS cached and empty body offline; tests/flows.spec.ts:131 reloads online first	Precache build assets at install, add a plain offline fallback, test offline straight after first visit
A11Y-03	High	S	Chart slider says Enough while headline says Short and hides bills and income	results/manual.json chart.ayanda; TimelineScrubber.tsx:79 valueText; TimelineScrubber.tsx:118 aria-hidden notes	Add the assumption, bills, income and range to aria-valuetext; rename or relabel the per-day state
A11Y-04	Medium	S	Keyboard focus hidden behind fixed bottom tab bar (2.4.11)	results/focus-obscured-app-390.png; results/focus-obscured-app-money-390.png; AppLayout.tsx:166; no scroll-padding in src	Add scroll-padding-top and scroll-padding-bottom below lg; add a Playwright check
A11Y-05	Medium	M	Home too dense in numbers with contradictory signals	docs/review/screens/ayanda-home-light-390.png: Short, R310 safe, chart Enough, range -R1 882 to R21 173	One sentence plus one action first; move detail behind a link; reconcile safe to spend with Short; test with low-numeracy users
A11Y-06	Medium	M	No privacy protection on a shared phone	results/manual.json storage 16.5 KB plaintext; no lock or hide mode; en.json:666 messages include date and state	Optional app lock or biometrics, hide-amounts toggle, discreet lock-screen messages; keep 3.3.8 in mind
A11Y-07	Medium	S	Form field borders and chart band below 3:1 (1.4.11)	border token 1.24:1 dark and 1.26:1 light on surface; band 1.25:1; src/styles/index.css:16,40	Raise input border to 3:1; add non-text pairs to contrast-check script
A11Y-08	Medium	S	Chart data table not keyboard scrollable (2.1.1)	axe scrollable-region-focusable in results/axe-*-*.json chart-table; TimelineScrubber.tsx:224	Add tabIndex 0, role region and label; simplify the table on phones
A11Y-09	Medium	L	Multilingual, low-literacy audience gets English only	src/i18n/status.json; PIRLS 2021 81 percent cannot read for meaning; privacy strings estimate grade 9 to 11	Ship 2 or 3 reviewed languages first; add read-aloud on the date; rewrite privacy copy to grade 5
A11Y-10	Low	S	Import errors not explained or announced (3.3.1, 3.3.3, 4.1.3)	en.json:582 readError; Feedback.tsx:54 Notice not live; Import.tsx:248 aria-invalid without message	Add next step to errors, role alert, visible balance error, keep button enabled
A11Y-11	Low	S	Negative amounts may be read without minus (Assumption)	results/manual.json chart.ayanda.slider "-R384"; src/lib/format.ts:13	Use "minus R384" or "R384 below zero" in spoken text; verify on NVDA and TalkBack
A11Y-12	Low	S	Truncated descriptions and roles lose meaning	ayanda-home-light-390.png "Cash deposit, shop takin..."; results/home-320.png "Spaza s..."	Use two-line clamp instead of truncate
A11Y-13	Low	S	Command menu keyboard-only with unannounced groups	results/manual.json commandMenu.mobileOpenButtonVisible 0; CommandPalette.tsx:197; results/command-menu.json opensFromTextField 1	Use role group with labels; ignore shortcut in non-empty text fields
A11Y-14	Low	S	Low-end Android parses 337 KB of untranslated locale JSON	docs/review/prep/audit.md; perf-home.json cold hero 1516 ms at 4x CPU	Lazy-load the active locale; test on a real entry-level Android Go phone
A11Y-15	Low	M	No plan yet for accessible login, consistent help or testing with disabled users	no auth in app; walkthrough.md is developer-only; 3.2.6 and 3.3.8 apply once real	Design biometric or paste-friendly login, one Help link in the same place on every page, test with TalkBack and low-literacy users
BEH-01	High	S	Safe to spend can license overspending and hides the next big bill	Lerato R330 safe vs R120 usual; following it daily drops Enough to Tight, date 1 Nov to 25 Oct (docs/review/tools/behavioural/frames.ts); rent R2 800 on 1 Nov not shown (lerato-home-dark-1280.png); src/engine/forecast.ts perDay	Cap at the amount that keeps the current state, rename "Spend up to R... a day to stay Enough", always show the next large bill after next money
BEH-02	High	M	Model sees one bank account while users live in cash, wallets and several accounts	No cash ledger, manual entry or transfer rule in src/engine or src/lib; FinScope 2023 94 percent withdraw cash monthly; SARB cash 56 percent of payments by volume; seed deposits cash fares daily (src/data/seed/generate.ts)	Add cash in hand, quick entries, transfer detection and a prompt on large deposits; label the date "bank account only" until cash is added

Id	Severity	Effort	Finding	Evidence	Recommendation
BEH-03	High	M	Safe to lend anchors a default and double counts the same buffer	"Use R320" chip (src/pages/Plan.tsx:204-238); canSetAside reuses safeToLend (src/engine/help.ts:49); lend R280 plus spend R330 drops Lerato to Tight (frames.ts); no repayment option	Reframe as give or lend without dropping a level, show one shared budget, remove or relabel the chip, add expected repayment date, show days lost instead of "no amount is safe"
BEH-04	Medium	S	Conflicting signals undermine comprehension and trust	"Enough until" with Tight chip (src/components/forecast/DateHero.tsx:9); hero Short vs chart Enough (ayanda-home-light-390.png); two daily targets R310 and about R373 for Ayanda; headline at band edge for Lerato (persona-snapshot.json)	State-neutral label, one daily number per state, explicit hero and chart reconciliation line, test range comprehension
BEH-05	Medium	M	Business inputs treated as household spending to cut	Ayanda top spend stock R15 137 and advice to spend R250 less a day (persona-snapshot.json); Thabo told to move car instalment (walkthrough.md item 10)	Add a business stock and fuel group excluded from cuts, timing-based advice, exclude vehicle finance and funeral cover from bill moves
BEH-06	Medium	S	Daily bad news risks avoidance, anxiety and exposure on shared phones	Daily 07:00 Short message (ayanda-inbox-light-390.png, en.json:666-682); ostrich effect (Karlsson et al. 2009); nudge side effects (Medina 2021)	Change-based messages, neutral lock-screen preview, action-first copy naming a specific cost, track opt-out by state
BEH-07	Medium	M	Stokvels, burial societies, family transfers and informal credit absent	No such entries in src/data/seed/generate.ts; FinScope 2023 informal savings 30 percent, burial society 29 percent; stokvel payout can pass the 10 percent next money threshold (src/engine/forecast.ts:24)	Add these flows to a persona, let users tag deposits as one-off or stokvel payout and contributions as commitments
BEH-08	Low	S	Date-first behavioural claim is unproven	No published runway vs balance experiment found; README.md:5 implies it; nearest evidence is adjacent (Hershfield et al., Karlan et al.)	Present as a hypothesis and test it in task T2 of the user study
BEH-09	Low	S	Account-level view can be gamed or used strategically	State depends only on the imported account balance (src/engine/forecast.ts forecast); Assumption on frequency	Treat as a design input, ask about showing or hiding the screen in testing, keep shared messages free of amounts
L10N-01	High	M	Ten of 11 languages are English	All 10 locale files byte-identical to en.json; status.json all pending; translated-from.json empty	Claim "built for 11, N live"; get speakers to write the money keys for 2 to 3 Gauteng languages and mark them reviewed
L10N-02	High	S	Tshivenda not in Cloud Translation; a full run would leave files inconsistent	Cloud Translation language page lists no Venda; translate.mjs:16,76,123,127-128; i18n-sync.mjs:33; README estimate 155k vs about 171.6k chars	Human-translate ve; save status after each language; pre-check supported targets; fix README figure
L10N-03	High	M	MT money messages without glossary, context or in-message caveat	translate.mjs:74 bare strings, v2 no glossary; 156 strings of 12 chars or fewer incl. state words; notice only at LanguageSelector.tsx:53-57; Google and Stellenbosch quality caveats	Block MT on money keys until reviewed; speaker term base; v3 glossary and context; bilingual notice in app shell
L10N-04	Medium	M	Review status is per language, not per string	translate.mjs:124 keeps reviewed after new MT strings; LanguageSelector.tsx:56 claims speaker-checked	Track reviewed status per key; show reviewed only when all keys are reviewed
L10N-05	Medium	S	html lang declares a language the text is not in	Preferences.tsx:12 sets lang=ve on English content; Inbox.tsx:117; WCAG 3.1.1 and 3.1.2	Use lang=en while pending; mark English fallback strings lang=en
L10N-06	Medium	M	Word order and agreement fixed in code	WhyDrawer.tsx:105, Import.tsx:260, privacy/Review.tsx:22 concatenate; {{state}} and English date interpolated; DateHero.tsx:9 shows Enough until for Tight	One key per full sentence; month and day names in locales; nest nav labels; test bold on WhatsApp
L10N-07	Medium	S	SMS limit check only works for English and GSM	MessageWriter.ts:28 160; Inbox.tsx:129 counts length; templates.test.ts English only; ê ë ô š ḓ not GSM-7 so 70 per SMS	Count segments by encoding; test every language; per-language short SMS
L10N-08	Medium	S	Layout clips expanded text	expansion.l10n.ts at 360px: bottom nav truncates Money and Inbox (AppLayout.tsx:171,185); StateChip nowrap squeezes Me rows; channel control stacks; shots/pseudo-ve-ibm-inbox-360.png	Two-line nav labels; shrinkable chips; wrapping segmented control; keep pseudo-locale in CI
L10N-09	Medium	S	Quality gates do not check translations	i18n-check.mjs checks keys and exclamation marks only; fonts-check passes only because all text is English	Check placeholder and bold parity, dashes, share identical to English; Tshivenda font fixture
L10N-10	Low	S	Plural rules need briefing and tests	Intl.PluralRules zu and nso map 0 to one, so i18next renders the _one form for 0; no non-English plural tests	Brief translators; test plural keys for 0, 1, 2 in all languages
L10N-11	Low	S	Decomposed Tshivenda characters use a fallback font	glyphs.mjs: Inter lacks U+032D and U+0307; no NFC normalisation in translate.mjs or CSV import	Normalise to NFC everywhere; add rule to i18n-check
L10N-12	Low	S	Name origin unconfirmed, may mean price not value	README.md:106-110; en.json:460; only hit is a Shona page glossing mutengo as price (unverified)	Confirm with two Tshivenda speakers before pitch; fix gloss everywhere
L10N-13	Low	S	Demo safety label English in every language	translate.mjs:18 keeps common.demoChip as "Demo, Fictional Data"	Show translated label alongside English
L10N-14	Low	S	All locales bundled eagerly	i18n/index.ts:5 eager glob; 337 KB of locale JSON in a 696 KB bundle	Bundle English only; lazy-load and cache the chosen language
COMM-01	High	M	No defensible moat against the bank building it	src/engine/forecast.ts is simple percentile maths; FNB nav» Money available funds (BusinessTech 2018); Standard Bank Future Payments; FNB 8.16m digitally active (BusinessDay 2026-09-16)	Lead with backtested accuracy, tested multilingual content and a pilot playbook; license the engine; do not claim a tech moat
COMM-02	High	M	Single-bank unit economics do not break even	model-inputs.md funnel gives about 44 000 MAU; R5 per MAU less R1.23 cost gives R166k contribution against R310k fixed; break-even about 82 000 MAU	Pilot fee plus per-MAU fee plus annual platform floor; plan a second bank or white-label within 18 months
COMM-03	High	S	Daily paid messaging costs more than the licence fee	WhatsApp utility USD 0.0095 (ChatMaxima, Meta) x 30 x R16.13 = R4.60; marketing R18.34; SMS R0.24 × 30 = R7.20 (Panacea)	Default to in-app push; WhatsApp on state change, at most weekly; keep templates
COMM-04	High	M	The segment's cash income is invisible to the product	Personas earn cash (en.json personas); TengoRepository has no add method (src/data/repository.ts:75 to 96); stokvels not modelled (Findex 19.6 percent)	Add cash in, cash out and cash on hand entry, plus stokvel contributions
COMM-05	Medium	M	No evidence of outcomes for a bank buyer	Synthetic data only (fact sheet); bank value inputs are all Assumption (model-inputs.md section 5)	Run 5 to 8 user sessions, publish backtest, offer a control-group pilot with four KPIs
COMM-06	Medium	S	A consumer subscription does not fit the segment	Capitec account R7.50 a month; free bank budgeting tools; PayFast 3.2 percent plus R2; finance app day 30 retention 4.2 to 15 percent (weak source)	Keep Tengo free; test a paid tier only for small business owners later
COMM-07	Medium	S	The distribution story conflicts with the product as shown	Landing stat "Bank logins needed"; CSV-only onboarding (src/pages/Import.tsx); open banking start date unconfirmed	Pitch one channel: inside the bank app with bank data and consent; CSV is the demo path
COMM-08	Medium	S	The pitch overclaims languages	Landing "Official languages" stat; all 10 non-English locales are English (fact sheet, src/i18n/status.json, walkthrough item 5)	Show one or two reviewed languages or reword the stat
COMM-09	Low	M	One buyer, one channel	Built for an FNB competition (README line 9); Capitec 15.3m, Absa 5.4m, Nedbank 3.2m digital users exist	Prepare white-label and platform-partner tracks alongside FNB
COMM-10	Low	S	Brand assets are not commercially cleared	Public Stunt personal-use licence and unconfirmed name origin (BRIEF.md sections 1 and 5); placeholder logo (fact sheet)	Clear the font licence, confirm the name, run a trademark search
COMM-11	Low	S	No written business case in the repository	README.md and BRIEF.md have no buyer, pricing or metrics section	Add a one-page business case linked to model.xlsx
INV-01	High	M	Zero evidence of demand or behaviour change	No users or sessions in repo; personas fictional (README.md:11, src/data/seed/personas.ts)	Run 5 to 8 sessions with real irregular earners and quote them
INV-02	High	S	Core insight easy to copy and partly exists	src/engine/forecast.ts is 368 lines of percentiles; FNB My available funds since 2018 (BusinessTech); Pocket Runway	Pitch design, segment and language work as the asset; position as bank licence

Id	Severity	Effort	Finding	Evidence	Recommendation
INV-03	High	S	Headline, safe to spend and chart contradict	ayanda-home-dark-390.png Short plus R310 safe plus chart Enough; forecast.ts:282-296; DateHero.tsx:9 shows Enough until for Tight	One rule per screen; Tight until label; daily cap when Short
INV-04	High	M	No path to real data	CSV only, PDF not built (fact-sheet.md); open finance not finalised (Moonstone, Stitch)	Name bank-embedded data route and write integration spec
INV-05	High	S	Overclaiming 11 languages	landing-light-1280.png stat; src/i18n/status.json all pending; en.json:475 says machine made	Honest stat now; two reviewed languages before submission
INV-06	High	M	Accuracy unproven, demo tuned to states	Seed balance chosen to hit Tight, Enough, Short (src/data/seed/generate.ts, fact-sheet.md)	Backtest on real consented statements and report error and band coverage
INV-07	Medium	S	Who pays is undefined	No pricing in README.md or BRIEF.md; nav Money is free (Engineering News 2022)	Choose B2B2C bank licence and state bank value hypothesis
INV-08	Medium	S	Headline date ignores income	forecast.ts:195-211; Lerato Enough until 1 Nov driven by rent though paid 24 Oct	Test what users think the date means; lead with state when income comes first
INV-09	Medium	S	Advice-like outputs raise harm exposure	safeToLend forecast.ts:357-368; help.ts:58-73 suggests moving car instalment	Exclude credit instalments; reword safe to lend; add estimate not advice line
INV-10	Medium	M	Cash and informal flows invisible	Engine reads bank rows only (forecast.ts:105-191); Ayanda cash takings	Add cash in hand and owed entries; test with cash heavy user
INV-11	Low	S	Placeholder brand and unconfirmed name	BRIEF.md section 1; fact-sheet.md placeholders	Confirm name with Venda speaker; simple wordmark
INV-12	Low	S	Dependency vulnerabilities and one large bundle	audit.md: 2 critical, 5 high, react-router moderate; 696 KB single JS file	Update react-router; split locale files
JUDGE-01	High	S	Make one rule drive the Home screen	ayanda-home-dark-1280.png Short plus R310 plus chart Enough; thabo-home-light-390.png and landing-dark-1280.png Enough until with Tight; UX-01, UX-02, UX-03, UX-04, PROD-03, INV-03, BEH-04, LEGAL-08, RISK-04, A11Y-03	State-specific hero label, daily limit tied to state with usual spend, no state word on chart readout, one next-money date, recapture landing mock
JUDGE-02	High	S	Replace the 11-languages claim and fix page language	landing stat 11 Official languages; locales identical to English (L10N-01); html lang wrong (A11Y-01, L10N-05); L10N-02; COMM-08; INV-05	Honest stat, lang=en while pending, one reviewed Gauteng language if time allows
JUDGE-03	High	M	Remove outputs that could cause harm	help.ts:62 suggests late car instalment; Safe to lend with Use button; shared buffer double counted; LEGAL-01, LEGAL-02, RISK-09, RISK-11, BEH-01, BEH-03, BEH-05, PROD-04, INV-09	Never move credit, insurance or rent; rename safe to lend; exclude stock and fuel from cuts; estimate-not-advice line on Home, Plan, help
JUDGE-04	High	S	Engine does not notice when the pattern breaks	RISK-01 S11 Enough after 35 days without pay; RISK-03; RISK-10; TECH-01 Ayanda Short to Enough after 30 days; DATA-05 missed payout 36.8 percent false reassurance	Paused income rule, no-income state, payday-today rule, asOf date and data age, with unit tests
JUDGE-05	Medium	S	Put the backtest in the pitch with its limits	backtest/results/summary_headline.csv MAE 2.1 vs 3.4, coverage 71.8 percent; DATA-01 circular; DATA-02 Tight short 24.4 percent; COMM-01	One slide or About section with synthetic caveat, stated band coverage, honest Tight copy
JUDGE-06	Medium	S	No business case or single distribution story	COMM-11, INV-07 no pricing; COMM-07 0 Bank logins vs bank embed; PROD-02, PROD-06, COMM-02, COMM-03	One page: bank-embedded licence, fees and floor, in-app push, bank-has vs Tengo-adds, data contract, pilot KPIs
JUDGE-07	Medium	M	No user evidence	INV-01, COMM-05 no sessions; BEH-08 claim untested; BEH user-testing plan	Run BEH tasks T1 and T2 with six real earners after fixes; report and quote
JUDGE-08	Medium	M	Cash income invisible	BEH-02 FinScope 94 percent withdraw cash; COMM-04; INV-10; BEH-07; PROD-07	One-tap cash in hand and quick entries feeding the forecast; bank-only label until used
JUDGE-09	Medium	S	Offline fails on first visit	A11Y-02 blank page after one visit; TECH-05 registration on load and naive cache; tests/flows.spec.ts:131-140 reloads online first	Precache build at install, offline fallback, test offline after first visit
JUDGE-10	Low	S	First-impression issues on Plan, consent, privacy and colour	UX-06 pre-filled Plan; UX-09, LEGAL-03, SEC-06 messages on by default; SEC-01, SEC-02 ticks and retention; UX-05 light Tight vs Short	Empty Plan, messages off, neutral planned icons and conditional copy, amber Tight
JUDGE-11	Low	S	Name meaning and brand assets unconfirmed	L10N-12 mutengo may mean price; INV-11; COMM-10 font licence and placeholder logo	Confirm name with two Tshivenda speakers before using the name story
JUDGE-12	Low	S	No demo script built on the strongest screens	No deck or script in repo; walkthrough items 5 and 10; UX walkthrough	Three-minute script: tagline, reconciled Short, Why drawer, loan check, pattern-break case, backtest, business page

Appendix B: Test and audit outputs

B.1 Check outputs

Tool output is quoted as produced, except that the build warning marker is shown as "(warning)".

Run on 2026-10-20 12:03 at commit d114497. Node v26.4.0, npm 11.17.0.

`npm run build`

```
✓ 2060 modules transformed. rendering chunks... computing gzip size... dist/index.html 1.77 kB │ gzip: 0.80 kB dist/assets/inter-vietnamese-wght-normal-CBcvBZtf.woff2 10.25 kB dist/assets/figtree-latin-ext-wght-normal-DCwSJGxG.woff2 10.28 kB dist/assets/inter-greek-ext-wght-normal-DlzME5K_.woff2 11.23 kB dist/assets/inter-cyrillic-wght-normal-DqGufNeO.woff2 18.75 kB dist/assets/inter-greek-wght-normal-CkhJZR-_.woff2 19.00 kB dist/assets/figtree-latin-wght-normal-D_ZTVpCC.woff2 20.16 kB dist/assets/inter-cyrillic-ext-wght-normal-BOeWTOD4.woff2 25.96 kB dist/assets/inter-latin-wght-normal-Dx4kXJAl.woff2 48.26 kB dist/assets/inter-latin-ext-wght-normal-DO1Apj_S.woff2 85.07 kB dist/assets/index-DGIJMyHN.css 47.58 kB │ gzip: 9.14 kB dist/assets/index-BLX9Vxli.js 695.60 kB │ gzip: 141.85 kB (warning) Some chunks are larger than 500 kB after minification. Consider: - Using dynamic import() to code-split the application - Use build.rollupOptions.output.manualChunks to improve chunking: https://rollupjs.org/configuration-options/#output-manualchunks - Adjust chunk size limit for this warning via build.chunkSizeWarningLimit. ✓ built in 1.76s exit code: 0
```

`npm run test`

```
> tengo@0.1.0 test
> node scripts/contrast-check.mjs && vitest run

Contrast: 116 of 116 pairs pass across dark and light.

 RUN  v3.2.7 /Users/matamelamakhado/Documents/development/tengo-finance

 ✓ src/lib/format.test.ts (4 tests) 20ms
 ✓ src/lib/csv.test.ts (9 tests) 9ms
 ✓ src/engine/forecast.test.ts (24 tests) 20ms
 ✓ src/engine/help.test.ts (6 tests) 59ms
 ✓ src/data/localRepository.test.ts (5 tests) 95ms
 ✓ src/messages/templates.test.ts (3 tests) 28ms
 ✓ src/engine/classify.test.ts (7 tests) 8ms
 ✓ src/engine/spending.test.ts (2 tests) 4ms
 ✓ src/data/seed/generate.test.ts (6 tests) 1219ms
   ✓ seed data > Thabo starts in tight on any day 541ms
   ✓ seed data > Lerato starts in enough on any day 355ms
   ✓ seed data > Ayanda starts in short on any day 316ms

 Test Files  9 passed (9)
      Tests  66 passed (66)
   Start at  12:04:01
   Duration  1.54s (transform 383ms, setup 0ms, collect 889ms, tests 1.46s, environment 1ms, prepare 535ms)
exit code: 0
```

`npm run lint`

```
> tengo@0.1.0 lint
> eslint . && node scripts/check-dashes.mjs

No em or en dashes found.
exit code: 0
```

`npm run i18n:check`

```
> tengo@0.1.0 i18n:check
> node scripts/i18n-check.mjs

All 11 locales have the same 565 keys.
exit code: 0
```

`npm run fonts:check`

```
> tengo@0.1.0 fonts:check
> node scripts/fonts-check.mjs

Fonts cover every locale string. Inter: 1623 characters, Figtree: 348.
exit code: 0
```

`npm run contrast`

```
> tengo@0.1.0 contrast
> node scripts/contrast-check.mjs

Contrast: 116 of 116 pairs pass across dark and light.
exit code: 0
```

B.2 Playwright

```
  221 passed (5.1m)
exit code: 0
```

B.3 Unit coverage

```
File               | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
All files          |   19.95 |    82.18 |   63.79 |   19.95 |
 src               |       0 |        0 |       0 |       0 |
 src/components    |       0 |    28.57 |   28.57 |       0 |
 src/data          |   45.75 |     75.6 |   77.27 |   45.75 |
 src/data/seed     |   98.02 |    98.48 |     100 |   98.02 |
 src/engine        |   97.55 |    91.96 |   93.75 |   97.55 |
 src/i18n          |   82.05 |      100 |       0 |   82.05 |
 src/layouts       |       0 |    66.66 |   66.66 |       0 |
 src/lib           |   72.44 |    84.42 |   62.96 |   72.44 |
 src/messages      |     100 |    33.33 |     100 |     100 |
 src/pages         |       0 |        0 |       0 |       0 |
 src/pages/privacy |       0 |     37.5 |    37.5 |       0 |
 src/store         |       0 |        0 |       0 |       0 |
```

B.4 Audit, bundle and licences

Run on 2026-10-20 at commit d114497.

npm audit (all dependencies)

```
fix available via `npm audit fix --force`
Will install vitest@5.0.5, which is a breaking change
node_modules/@vitest/mocker
  vitest  0.0.95 - 4.1.10
  Depends on vulnerable versions of @vitest/mocker
  Depends on vulnerable versions of tinypool
  node_modules/vitest
braces  *
Severity: high
braces vulnerable to stack-exhaustion denial of service through deeply nested patterns - https://github.com/advisories/GHSA-vfj7-8cjw-p6xm
fix available via `npm audit fix --force`
Will install tailwindcss@4.3.3, which is a breaking change
node_modules/braces
  chokidar  2.0.0 - 3.6.0
  Depends on vulnerable versions of braces
  node_modules/chokidar
  tailwindcss  <=0.0.0-oxide-insiders.ff2c25f || 0.5.0 - 3.4.19
  Depends on vulnerable versions of chokidar
  Depends on vulnerable versions of fast-glob
  Depends on vulnerable versions of micromatch
  Depends on vulnerable versions of postcss-nested
  Depends on vulnerable versions of postcss-selector-parser
  node_modules/tailwindcss
micromatch  >=0.2.0
Depends on vulnerable versions of braces
node_modules/micromatch
fast-glob  *
Depends on vulnerable versions of micromatch
node_modules/fast-glob
postcss-selector-parser  <7.1.6
Severity:
```

moderate PostCSS: Quadratic complexity in flat selector parsing allows CPU exhaustion – https://github.com/advisories/GHSA-rj75-hqrm-r3gf fix available via `npm audit fix --force` Will install tailwindcss@4.3.3, which is a breaking change node_modules/postcss-selector-parser postcss-nested 2.0.3 - 6.2.0 Depends on vulnerable versions of postcss-selector-parser node_modules/postcss-nested react-router 6.0.0 - 7.17.0 Severity: moderate React Router: Open redirect via backslash in <Link> and useNavigate (CVE-2025-68470 bypass) – https://github.com/advisories/GHSA-wrjc-x8rr-h8h6 React Router: Arbitrary Constructor Injection via deserializeErrors() in React Router SSR Hydration – https://github.com/advisories/GHSA-337j-9hxr-rhxg fix available via `npm audit fix --force` Will install react-router-dom@7.18.4, which is a breaking change node_modules/react-router react-router-dom 6.0.0-alpha.0 - 7.17.0 Depends on vulnerable versions of react-router node_modules/react-router-dom tinypool <=2.1.1 Severity: critical Tinypool: Prototype Pollution gadget in worker options leads to Remote Code Execution – https://github.com/advisories/GHSA-5gmw-xhrv-c9v3 Tinypool: Prototype Pollution Gadget to RCE in run() options – https://github.com/advisories/GHSA-85c8-ppgw-ccpr fix available via `npm audit fix --force` Will install vitest@5.0.5, which is a breaking change node_modules/tinypool 12 vulnerabilities (5 moderate, 5 high, 2 critical) To address issues that do not require attention, run: npm audit fix To address all issues (including breaking changes), run: npm audit fix --force

npm audit --omit=dev (shipped to users)

```
# npm audit report

react-router 6.0.0 - 7.17.0 Severity: moderate React Router: Open redirect via backslash in and useNavigate (CVE-2025-68470 bypass) - https://github.com/advisories/GHSA-wrjc-x8rr-h8h6 React Router: Arbitrary Constructor Injection via deserializeErrors() in React Router SSR Hydration - https://github.com/advisories/GHSA-337j-9hxr-rhxg fix available via `npm audit fix --force` Will install react-router-dom@7.18.4, which is a breaking change node_modules/react-router react-router-dom 6.0.0-alpha.0 - 7.17.0 Depends on vulnerable versions of react-router node_modules/react-router-dom 2 moderate severity vulnerabilities To address all issues (including breaking changes), run: npm audit fix --force
```

Production bundle (dist/assets)

File	Raw bytes	Gzip bytes
figtree-latin-ext-wght-normal-DCwSJGxG.woff2	10280	10348
figtree-latin-wght-normal-D_ZTVpCC.woff2	20156	20168
index-BLX9Vxli.js	695602	141864
index-DGIJMyHN.css	47578	9155
inter-cyrillic-ext-wght-normal-BOeWTOD4.woff2	25960	26034
inter-cyrillic-wght-normal-DqGufNeO.woff2	18748	18818
inter-greek-ext-wght-normal-DIzME5K_.woff2	11232	11276
inter-greek-wght-normal-CkhJZR-_.woff2	18996	19063
inter-latin-ext-wght-normal-DO1Apj_S.woff2	85068	85152
inter-latin-wght-normal-Dx4kXJAl.woff2	48256	48288
inter-vietnamese-wght-normal-CBcvBZtf.woff2	10252	10319

Locale files are bundled into the main JS: 336600 bytes of JSON across 11 files.

Licences (production dependencies)

```
├─ MIT: 18
├─ OFL-1.1: 2
├─ ISC: 1
├─ UNLICENSED: 1
└─ Apache-2.0: 1
```

UNLICENSED is the private tengo package itself.

B.5 Home timing under mobile throttling

Run	Date visible	First paint	DOM ready	Transferred
cold	1516 ms	1188 ms	1081 ms	149 KB
warm	627 ms	404 ms	342 ms	1 KB

4x CPU slowdown, 1.6 Mbps down, 750 kbps up, 150 ms latency, 390×844, service worker blocked. Source: docs/review/prep/perf-home.json.

B.6 Offline matrix

```
first,offline,reloadBeforeOffline,swControlled,renderedHeading,bodyTextChars
/,/,true,true,"Money you can plan around.",1510
/,/,false,true,"",0
/,/app,true,true,"Enough until Tuesday 27 OctoberTue 27 Oc",1396
/app,/app,false,true,"",0
/app,/app,true,true,"Enough until Tuesday 27 OctoberTue 27 Oc",1396
```

Appendix C: Lighthouse results

Page	Performance	Accessibility	Best practices	SEO	FCP	LCP	TBT	CLS
landing	94	100	100	91	2.2 s	2.4 s	160 ms	0
picker	96	100	100	91	2.2 s	2.3 s	10 ms	0

Lighthouse 12.8.2, mobile form factor, simulated throttling (150 ms RTT, about 1.6 Mbps, 4x CPU). Reports: docs/review/prep/lighthouse/.

Appendix D: Licence summary

Production dependencies: MIT 18, SIL Open Font License 1.1 2 (Inter and Figtree), ISC 1, Apache 2.0 1, UNLICENSED 1 (the private Tengo package itself). No copyleft licences ship to users. Source: docs/review/prep/audit.md.

Appendix E: Source list

168 distinct URLs cited across the reviewer files and model inputs, all accessed on 2026-10-20 unless the citing file states otherwise.

URL	Cited by	Accessed
https://acts.co.za/news/blog/2025/09/cofi-bill-status-update-2025	Legal and regulatory advisor	2026-10-20
https://africaglobalfunds.com/shop/product/finmark-trust-cenfri-and-mastercard-foundation-to-launch-data-facility-for-service-providers	Model inputs	2026-10-20
https://appoftheyear.co.za/faqs/	Hackathon judge panel	2026-10-20
https://appoftheyear.co.za/wp-content/uploads/2025/10/2025-Hackathon-programme.pdf	Hackathon judge panel	2026-10-20
https://bis.org/review/r240910a.htm	Behavioural economist and financial inclusion researcher	2026-10-20
https://blog.google/intl/en-in/24-new-languages/	Localisation and language reviewer	2026-10-20
https://blog.google/products/translate/google-translate-new-languages-2024/	Localisation and language reviewer	2026-10-20
https://briefly.co.za/business-economy/money/254111-sarb-raises-repo-rate-725-what-means-bills-sa/	Model inputs	2026-10-20
https://bulksms.com/resources/regulations/south-africa-sms-regulations.htm	Legal and regulatory advisor	2026-10-20
https://business.columbia.edu/faculty/research/poverty-and-economic-decision-making-evidence-changes-financial-resources-payday	Behavioural economist and financial inclusion researcher	2026-10-20
https://businesstech.co.za/news/?p=136339	Model inputs	2026-10-20
https://businesstech.co.za/news/banking/250773/fnb-launches-new-nav-money-tool/	Investor and sceptic, Product strategist and commercial lead	2026-10-20
https://businesstech.co.za/news/business-opinion/869804/one-silver-lining-for-unemployment-in-south-africa/	Investor and sceptic	2026-10-20
https://businesstech.co.za/news/industry-news/843969/introducing-standard-bank-money-reels-see-what-your-money-habits-say-about-you/	Product strategist and commercial lead	2026-10-20
https://businesstech.co.za/news/wealth/862257/south-africas-middle-class-is-in-serious-trouble-2/	Model inputs	2026-10-20
https://capeargus.co.za/personal-finance/financial-planning/2024-04-02-stokvels-and-societys-success-is-all-about-relationships-support-and-understanding	Behavioural economist and financial inclusion researcher	2026-10-20
https://census.statssa.gov.za/assets/documents/2022/P03014_Census_2022_Statistical_Release.pdf	Localisation and language reviewer	2026-10-20
https://chatmaxima.com/whatsapp-api-pricing/south-africa/	Model inputs, Product strategist and commercial lead	2026-10-20
https://claude.com/pricing	Model inputs, Product strategist and commercial lead	2026-10-20
https://clevertap.com/wp-content/uploads/2025/11/BFSI-Benchmark-Report-Retention.pdf	Model inputs	2026-10-20
https://connectingafrica.com/financial-inclusion/vodacom-launches-vodapay-super-app-	Product strategist and commercial lead	2026-10-20
https://dailyinvestor.com/banking/112501/billionaire-patrice-motsepe-backed-bank-hits-12-million-clients/	Model inputs, Product strategist and commercial lead	2026-10-20
https://datareportal.com/reports/digital-2026-south-africa	Model inputs, Product strategist and commercial lead	2026-10-20
https://developer.mozilla.org/en-US/docs/Web/API/Storage_API/Storage_quotas_and_eviction_criteria	Head of technology and solution architect	2026-10-20
https://developers.facebook.com/docs/whatsapp/pricing	Model inputs, Product strategist and commercial lead	2026-10-20
https://developers.facebook.com/documentation/business-messaging/whatsapp/getting-opt-in	Legal and regulatory advisor	2026-10-20
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https://docs.cloud.google.com/translate/docs/languages	Localisation and language reviewer	2026-10-20
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https://eservices.inforegulator.org.za/compromises/default.aspx	Security and privacy lead	2026-10-20
https://fastcompany.co.za/tech/2025-08-07-are-you-prepared-for-spotifys-new-subscription-prices-in-south-africa/	Model inputs	2026-10-20
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https://help.netcash.co.za/?p=9799	Model inputs	2026-10-20
https://ideas.repec.org/a/oup/rfinst/v34y2021i5p2580-2607..html	Behavioural economist and financial inclusion researcher	2026-10-20
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https://inforegulator.org.za/?p=7427	Legal and regulatory advisor	2026-10-20
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https://www.capitecbank.co.za/blog/articles/best-way-to-bank/how-to-use-our-app-to-manage-your-money/	Product strategist and commercial lead	2026-10-20
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https://www.capitecbank.co.za/globalassets/pages/documents-library/transact/personal-banking-fees-2026.pdf	Model inputs, Product strategist and commercial lead	2026-10-20
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https://www.cars.co.za/motoring-news/what-happens-when-you-miss-a-car-finance-payment-in-south-africa/355033/	Legal and regulatory advisor	2026-10-20
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https://www.clydeco.com/en/insights/2026/07/the-fsca-has-published-its-three-year-regulation-p	Legal and regulatory advisor	2026-10-20
https://www.cmu.edu/tepper/faculty-and-research/assets/docs/ppaper-61264983153898-ostrich-effect-1.pdf	Behavioural economist and financial inclusion researcher	2026-10-20
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https://www.dailymaverick.co.za/article/2023-05-16-international-study-shows-81-of-grade-4s-in-south-africa-cannot-read-for-meaning/	Accessibility and inclusive design specialist	2026-10-20
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https://www.ewn.co.za/2026/07/23/sarb-keeps-interest-rates-unchanged-surprising-markets-and-analysts	Model inputs	2026-10-20
https://www.ewn.co.za/google-increases-youtube-premium-prices-in-south-africa-by-36	Model inputs	2026-10-20
https://www.fnb.co.za/for-you/borrow/overdraft.html	Retail banking product lead	2026-10-20
https://www.fnb.co.za/ways-to-bank/cellphone-banking-index.html	Retail banking product lead	2026-10-20
https://www.fnb.co.za/ways-to-bank/digital/navMoney.html	Retail banking product lead	2026-10-20
https://www.fscamymoney.co.za/Legislation%20Strategy%20and%20Policy/FINANCIAL%20ADVISORY%20AND%20INTERMEDIARY%20SERVICES%20ACT%2037%20OF%2020.pdf	Legal and regulatory advisor	2026-10-20
https://www.fundsdata.co.za/ebook/2025-2U/files/basic-html/page35.html	Legal and regulatory advisor	2026-10-20
https://www.getpanto.ai/blog/mobile-app-retention-statistics	Model inputs, Product strategist and commercial lead	2026-10-20
https://www.gov.za/documents/financial-advisory-and-intermediary-services-act	Legal and regulatory advisor	2026-10-20
https://www.gov.za/documents/protection-personal-information-act	Security and privacy lead	2026-10-20
https://www.icasa.org.za/uploads/files/2025-26FY-Q2-Bi-annual-Tariff-Analysis-Report-Abridged-Report-For-Publication.pdf	Accessibility and inclusive design specialist	2026-10-20
https://www.informs.org/News-Room/INFORMS-Releases/News-Releases/New-Research-Exposes-the-Mindset-Required-to-Save-Money	Behavioural economist and financial inclusion researcher	2026-10-20
https://www.itweb.co.za/content/DZQ58MVPaJOvzXy2	Investor and sceptic	2026-10-20
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https://www.moonstone.co.za/key-fintech-policy-developments-to-watch-in-2026/	Investor and sceptic, Retail banking product lead	2026-10-20
https://www.moonstone.co.za/updated-popia-compliance-data-breach-reporting-must-be-online/	Security and privacy lead	2026-10-20
https://www.nber.org/papers/w17028	Behavioural economist and financial inclusion researcher	2026-10-20
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Appendix F: Glossary

Term	Meaning
Band	The likely range for the run-out date, from a heavy week (early) to a light week (late).
B2B2C	A business sells to a business (a bank) that serves consumers.
CAC	Customer acquisition cost.
Enough, Tight, Short	Tengo cover states: covered until the latest likely next money; covered only on a normal week; runs out before the usual next money.
FAIS	Financial Advisory and Intermediary Services Act, 2002.
LTV	Lifetime value: contribution per user times expected months as a user.
MAE	Mean absolute error, here in days.
MAU	Monthly active user.
NCA	National Credit Act, 2005.
Next money	The next significant deposit, inferred from past deposits.
POPIA	Protection of Personal Information Act, 2013.
Runway date	The day the balance reaches zero if spending continues at the usual rate and no new income arrives.
Safe to spend today	Balance minus bills before the latest next money, spread over the days until then, rounded down to R5.
Synthetic data	Fictional transactions produced by the app's seed generator.
WCAG	Web Content Accessibility Guidelines.